

TFLite to Verilog streaming CNN compiler for embedded AI applications

andromeda.py
+ conv2d.v
+ concatenate.v
+ replicate.v

Replicate(), Concatenate(), Add()

<https://vision.middlebury.edu/stereo/data/scenes2006/>

Applications

[LAD-RCNN: A Powerful Tool for Livestock Face Detection and Normalization](#)

[EvConv: Fast CNN Inference on Event Camera Inputs For High-Speed Robot Perception](#)

[Sony Event-based Vision Sensor](#)

[CNN-based Methods for Object Recognition with High-Resolution Tactile Sensors](#)

[FusionRCNN: LiDAR-Camera Fusion for Two-stage 3D Object Detection](#)

[Low-complexity CNNs for Acoustic Scene Classification](#)

[Counting Varying Density Crowds Through Density Guided Adaptive Selection CNN and Transformer Estimation](#)

[Agricultural Plantation Classification using Transfer Learning Approach based on CNN](#)

[Predictive Modeling of Charge Levels for Battery Electric Vehicles using CNN EfficientNet and IGTD Algorithm](#)

[A CNN Approach for 5G mmWave Positioning Using Beamformed CSI Measurements](#)

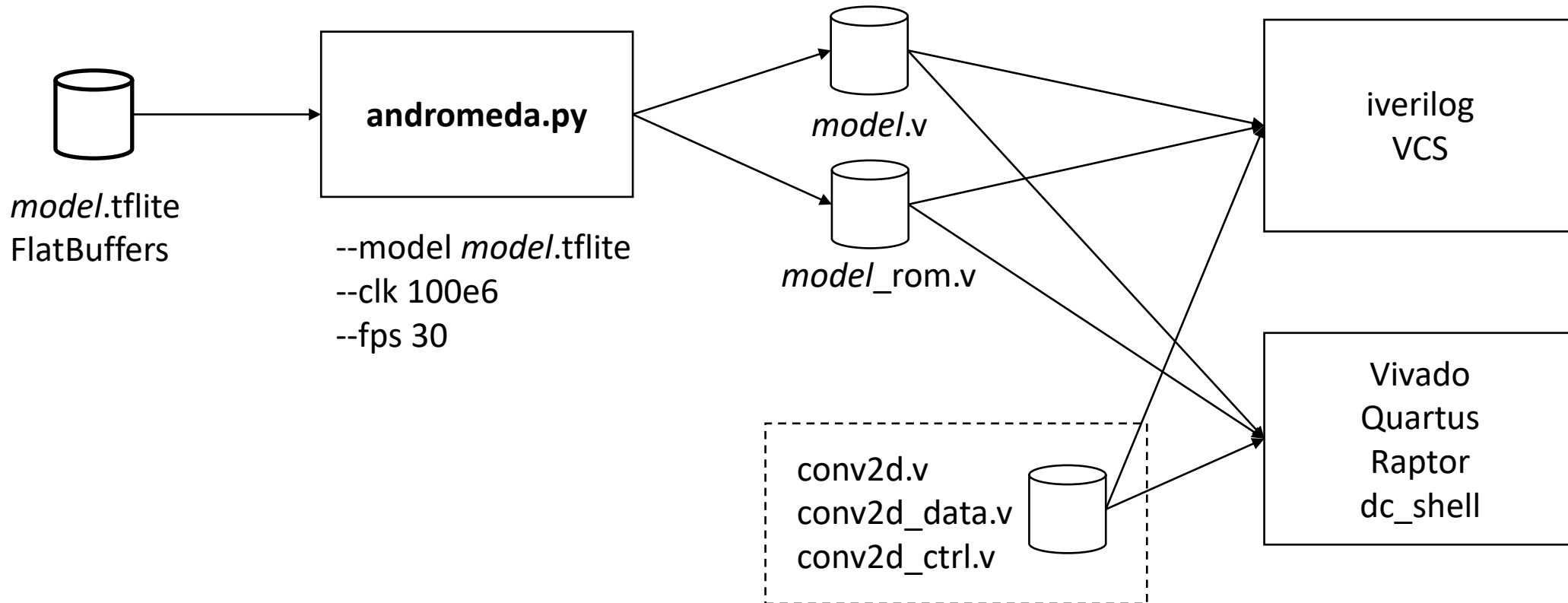
Target applications

1. Prediction / parameter estimation from joint sensor data at ~100K samples/sec
 - e.g. [MLX90640](#) thermal array, piezo, Hall effect
 - app: thermal screening , motor control, equipment monitoring
2. AI audio processing using multiple microphones/speakers
3. Advanced multi-camera vision tasks at 100M+ pixels/s
 - 300+ FPS, 20M+ uncompressed pixel resolution, 10+ cameras, hyperspectral
4. AI radio using RF baseband IQ samples, MIMO antennas

Replace traditional DSP algorithms with trained deep models

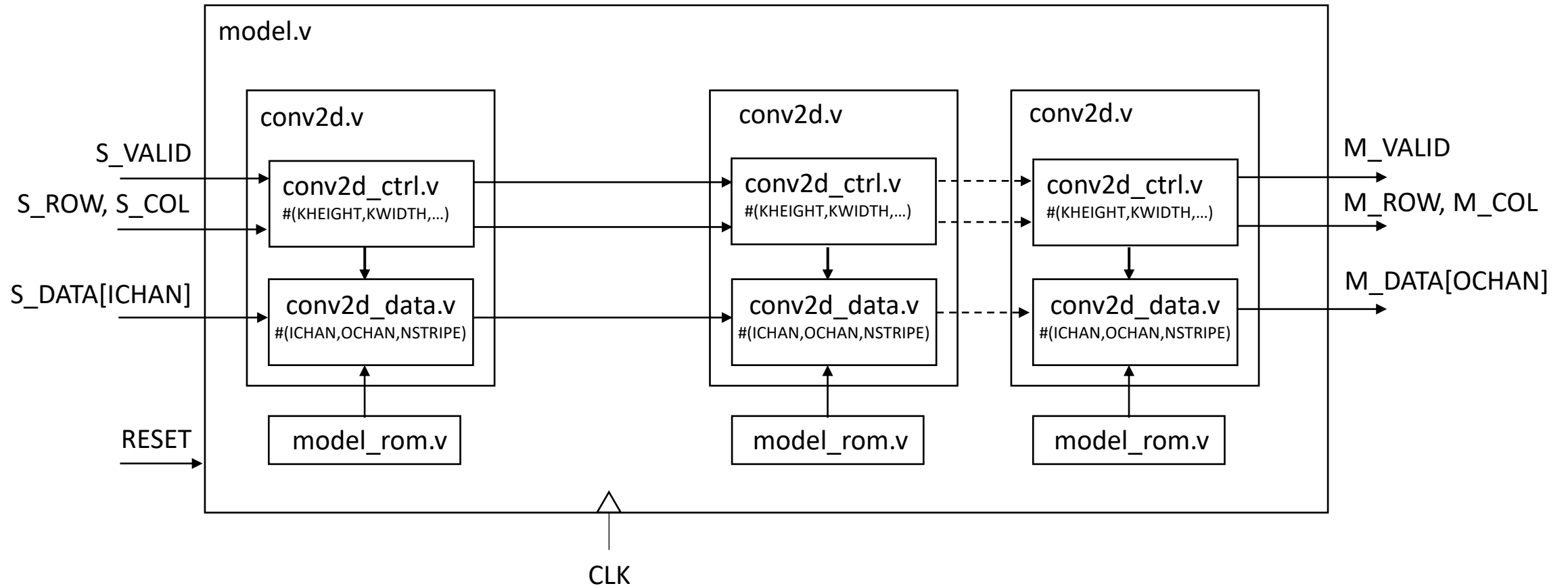
- Performance improvement from using deep model instead of heuristics
- Can learn over time and adapt
- Can automate the implementation of e.g. MATLAB model using imitation learning

Streaming CNN compiler

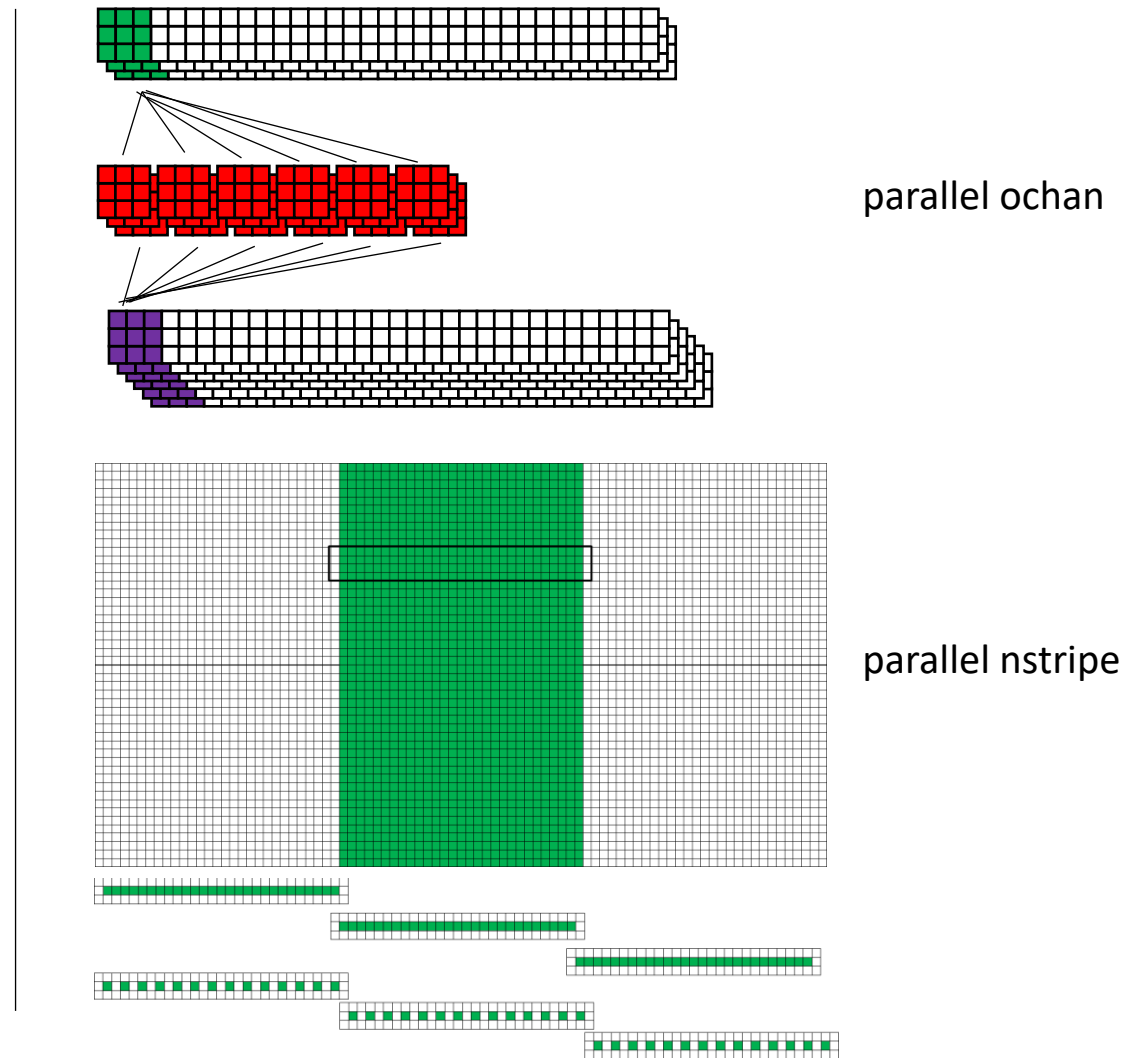
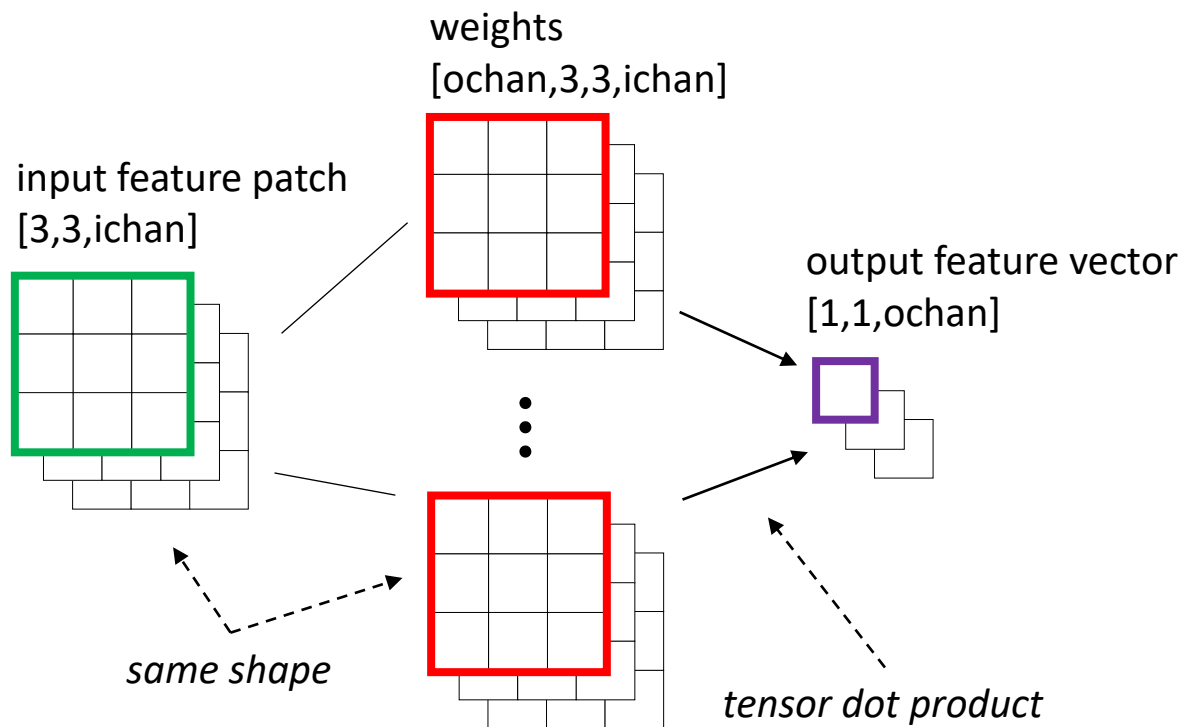


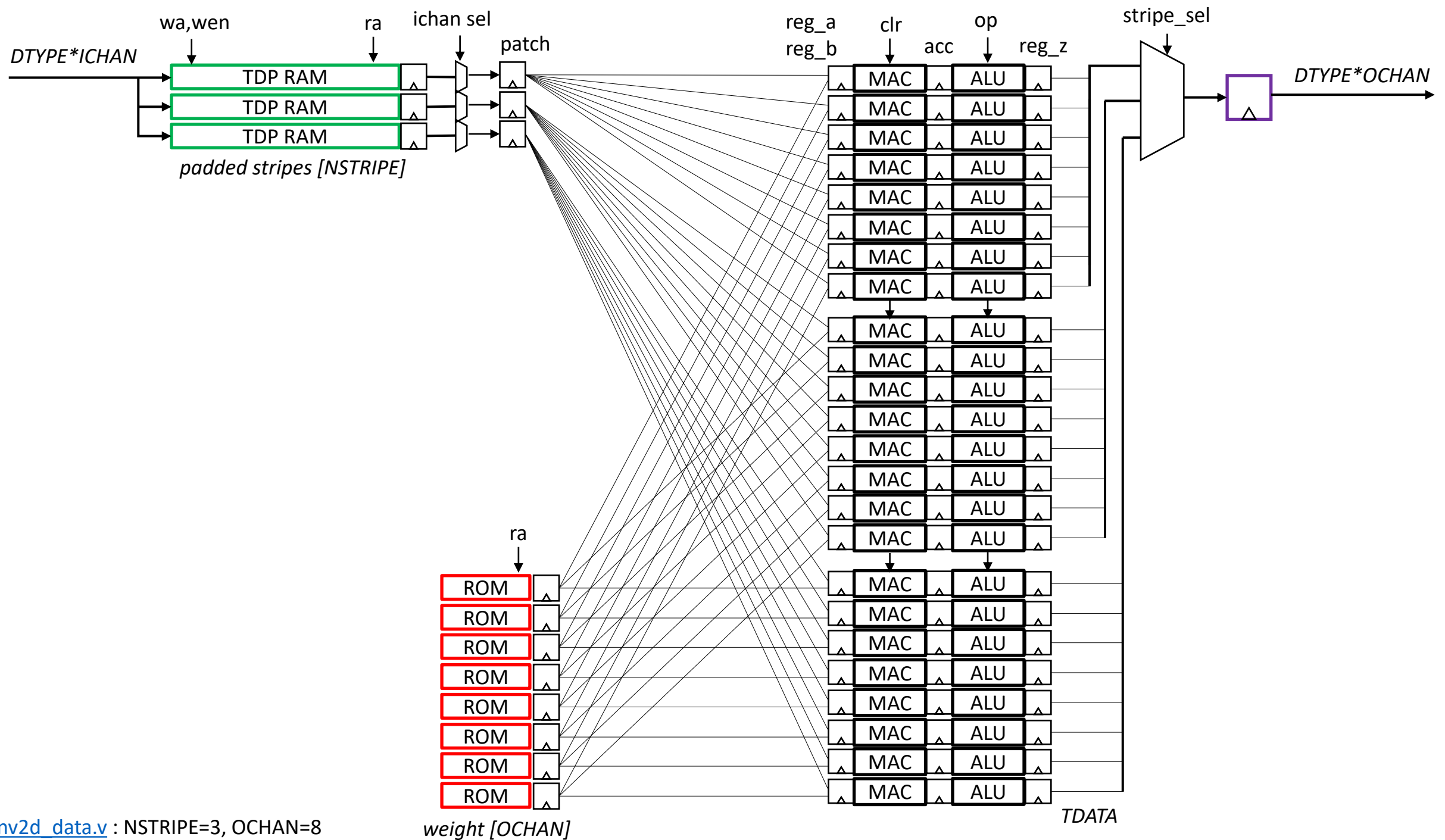
automated tool to compile TFLite model into synthesizable Verilog, using conv2d.v parameterized module
compiler generates parallel code for $OCHAN * NSTRIPE$ MAC operations per layer

Generated Verilog



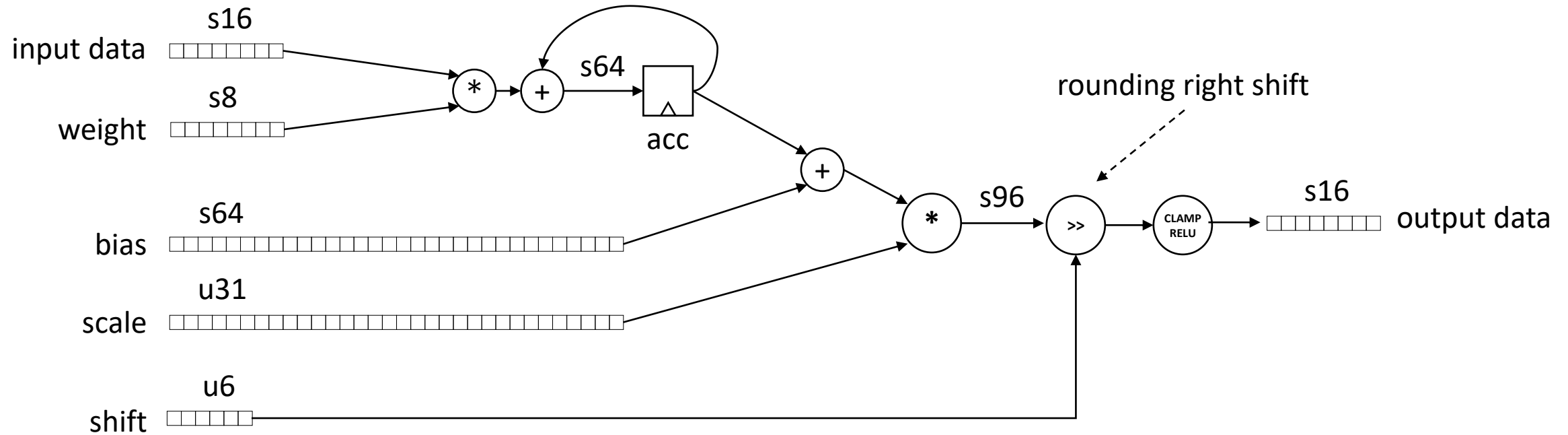
Parallel computation of CNN dot products



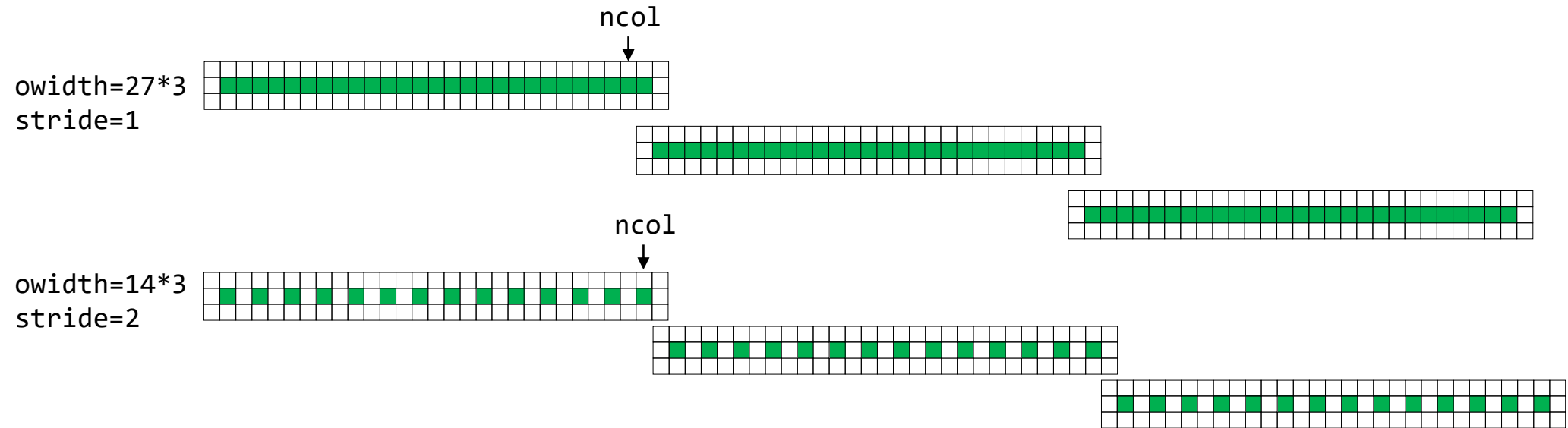


TFLite 16x8 quantization

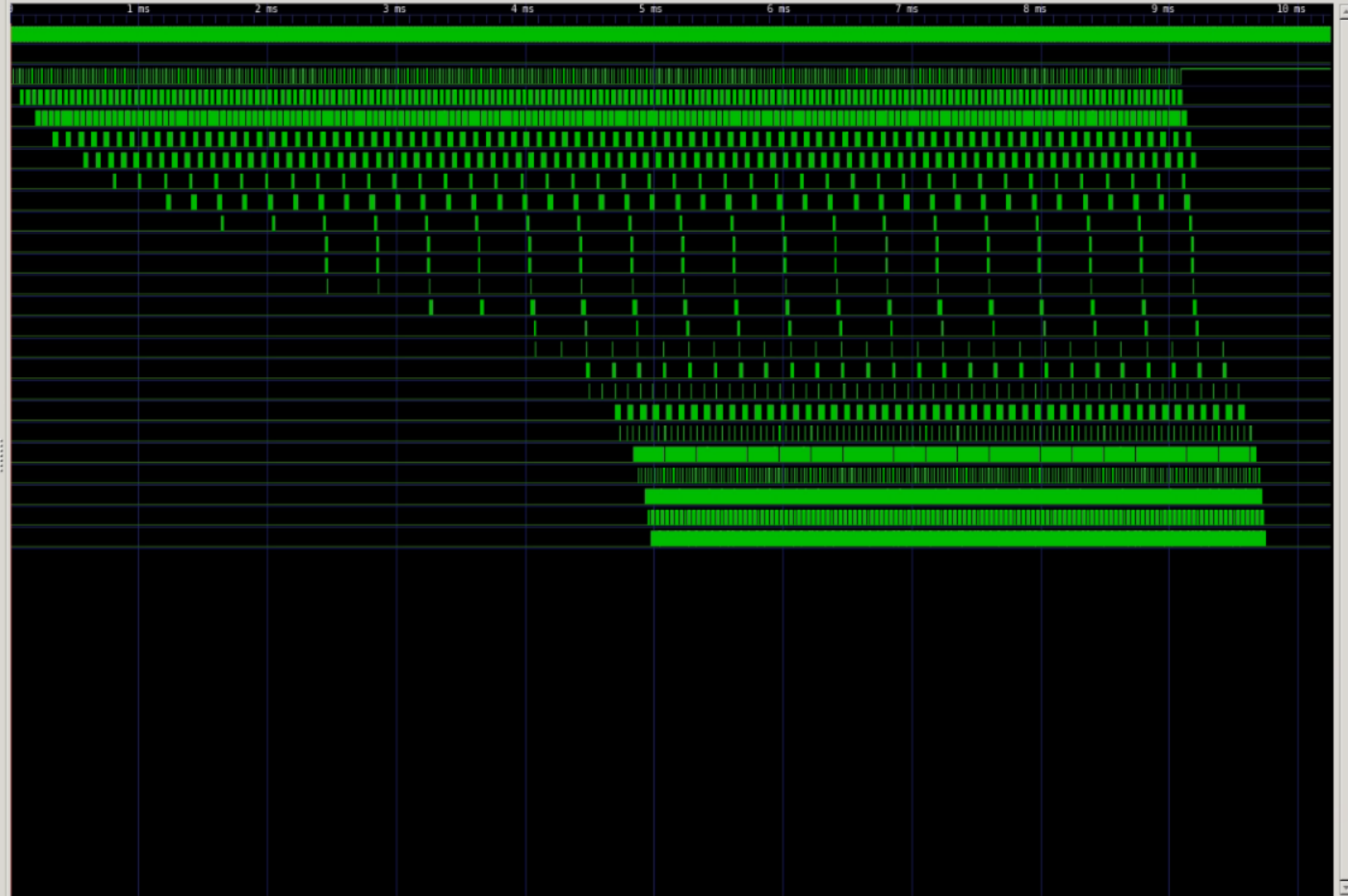
- 1: compute dot product
- 2: add bias
3. multiply by scale
4. shift right with rounding
5. clamp and RELU



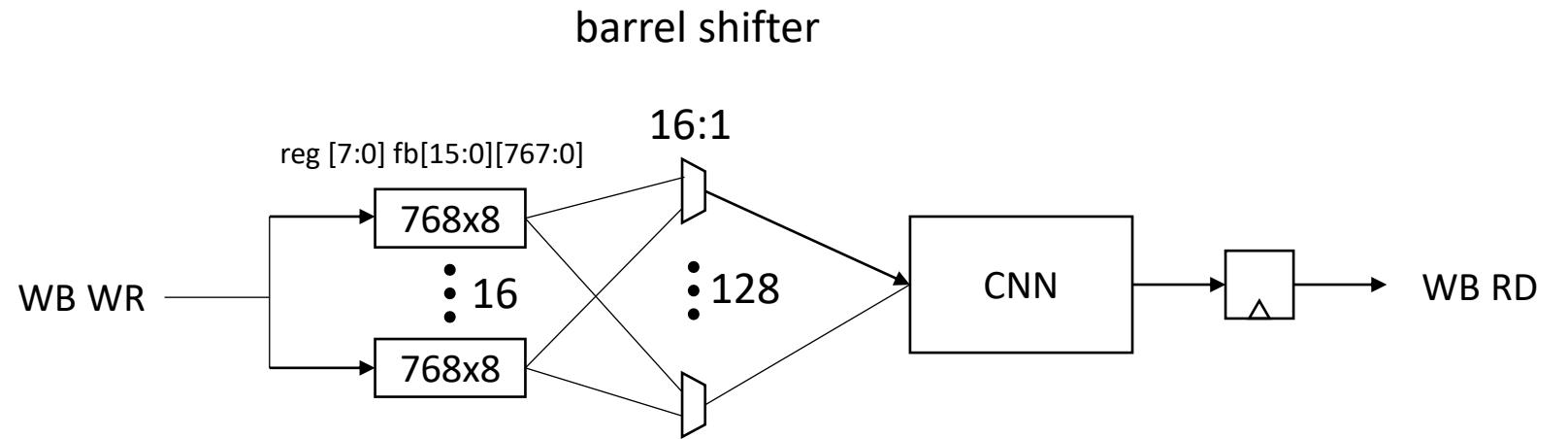
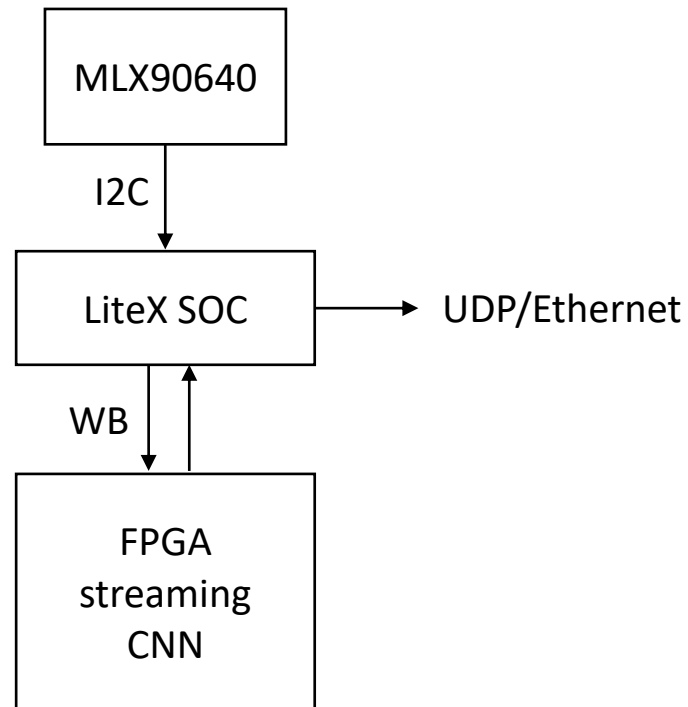
CNN striped parallel execution



Waves



Demo

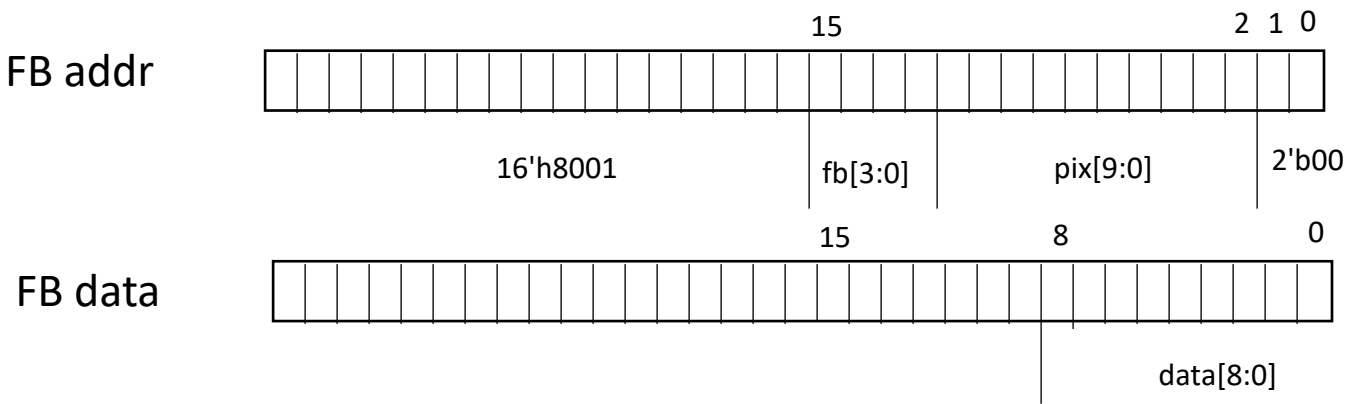


1. read MLX90640 frame using I2C
2. read previous prediction and assert seq++
3. assert reset
4. copy frame to next fb[15:0]
5. increment and write chan[3:0]
6. deassert reset
7. goto 1

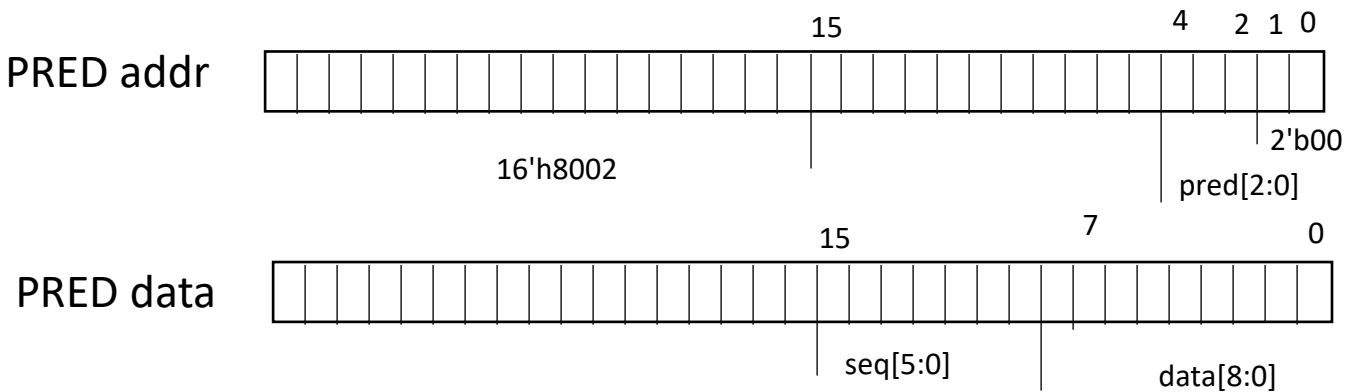
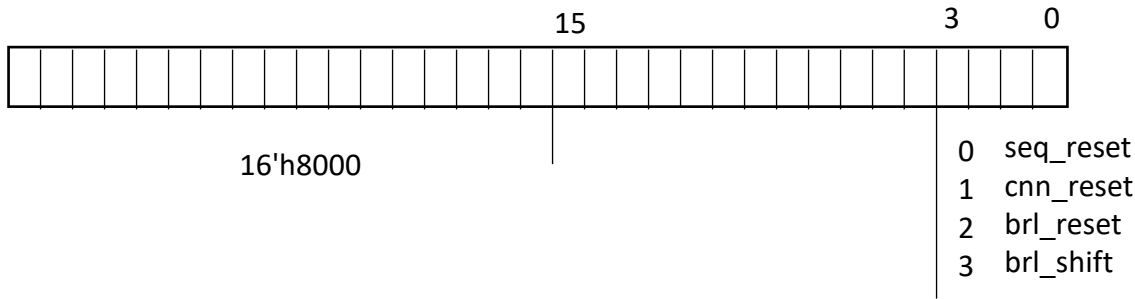
Demo

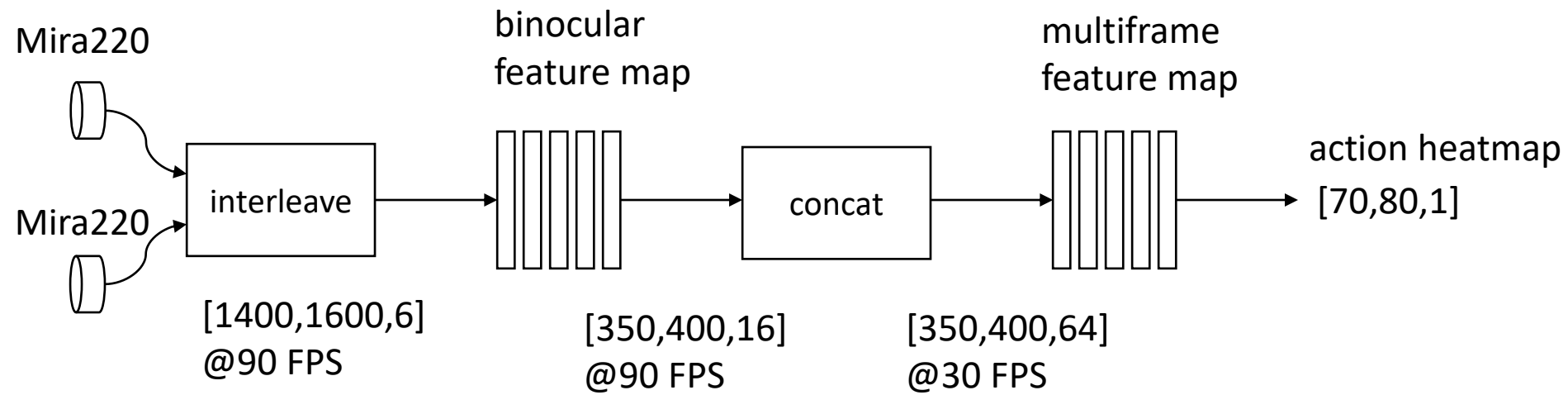
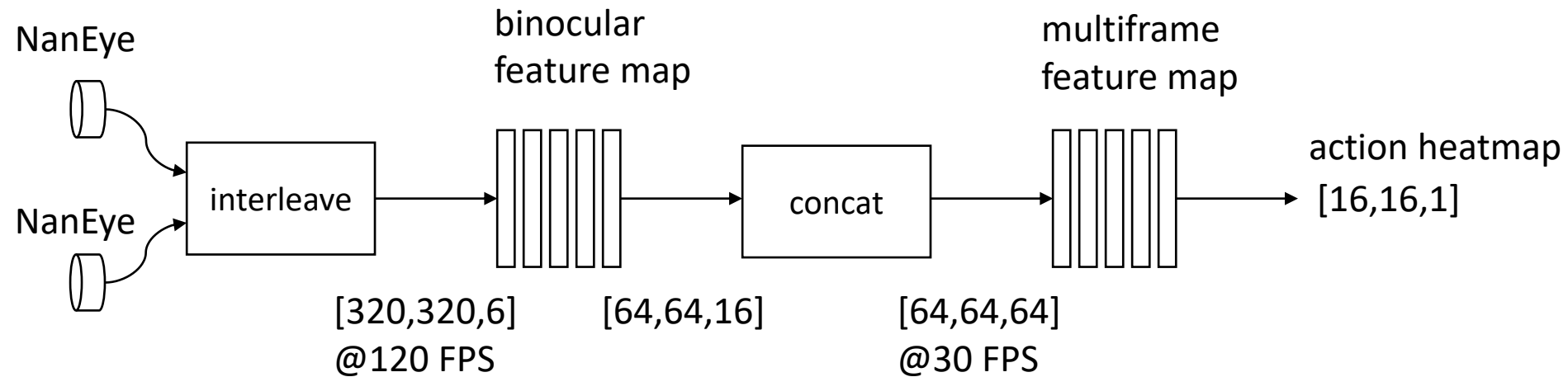
chan0[3:0] selects which physical fb is fed to the CNN input as channel 0

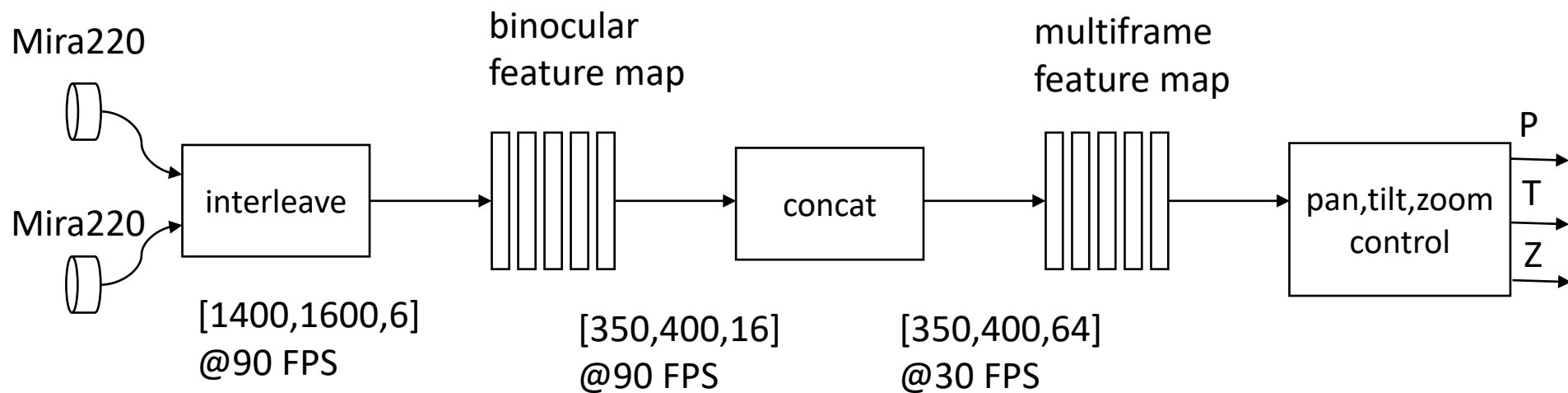
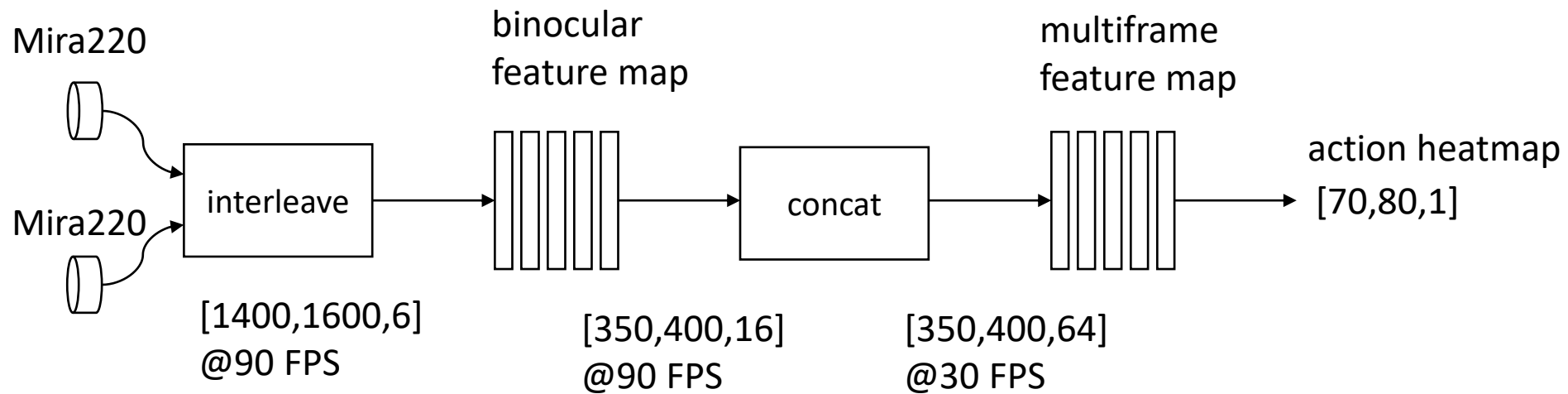
pred seq[5:0] increments every m_tdata_valid



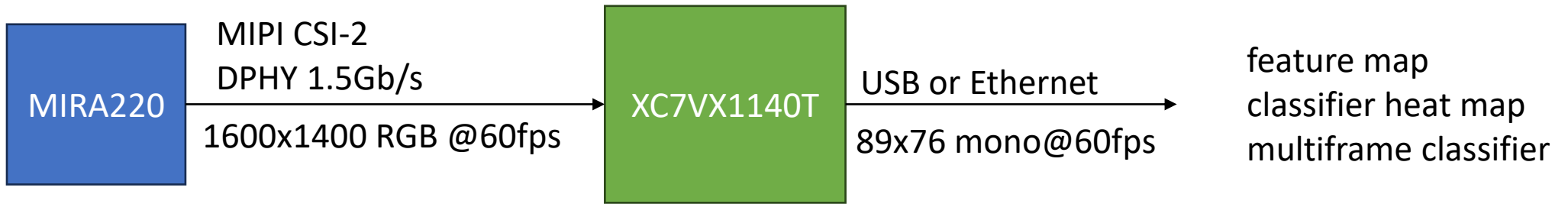
CSR 0x80000000
CSR data







High performance demo



classifier heat map (auto PTZ)

multiframe classifier (action recognition)

stereo vision (RGBD)

hyperspectral imaging, feature map based on sensor fusion

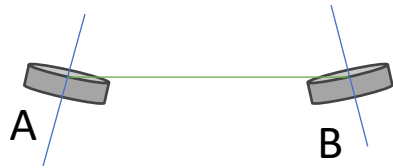
line scan image sensor with infinite scrolling

audio analysis using spectrograms <https://en.wikipedia.org/wiki/Spectrogram>

self supervised visual feature learning <https://arxiv.org/pdf/1902.06162.pdf>

binocular vision demo - self supervised binocular feature map training

https://en.wikipedia.org/wiki/Binocular_vision



~10 deg vergence angle

~10 cm distance between cameras

predict A given B, B given A, sharing weights

self supervised "pixels to feature map" learning with binocular inputs, multi-frame features

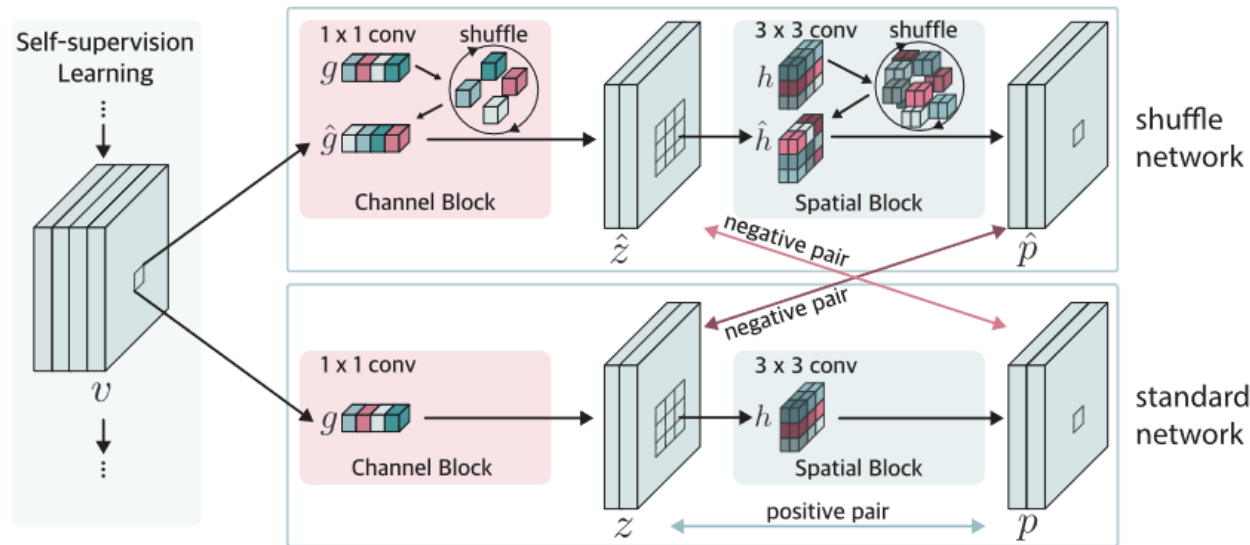


Fig. 2. Overall architecture. Our framework shuffles the convolution kernels to make the network learn spatial information within a small region. We use two networks: standard network and shuffle network. Each network contains two blocks called channel block and spatial block. The channel blocks are attached to the output v which is an arbitrary output of the backbone network. The output v is fed into both channel blocks g and $\hat{g}$ where $\hat{g}$ is randomly shuffled kernel of g . The output of each channel block is then fed into the spatial block h and $\hat{h}$. The outputs of each block of standard network z and p are considered as a positive pair. The outputs between the shuffle network and standard network are considered as negative pairs. To make the pairs asymmetric, $\hat{z}$ and p are considered as one negative pair and z and $\hat{p}$ are considered as the other negative pair for our module.

<https://www.sciencedirect.com/science/article/pii/S0020025522013032>

<https://www.robots.ox.ac.uk/~vgg/publications/2021/Ge21/ge21.pdf>

self supervised egocentric multicamera representations

- predict camera A feature map given camera B feature map
- predict camera B given camera A, share weights
- can extend to more cameras, multiframe feature maps at high frame rate

References

Research

[AoCStream: All-on-Chip CNN Accelerator With Stream-Based Line-Buffer Architecture](#)

<https://github.com/Xilinx/finn>

[A Study on the Intersection of GPU Utilization and CNN Inference](#)

Commercial Edge AI

<https://www.nvidia.com/en-us/autonomous-machines/embedded-systems/>

<https://perceive.io/>

TFLite

https://www.tensorflow.org/lite/api_docs

<https://arxiv.org/pdf/1712.05877.pdf>

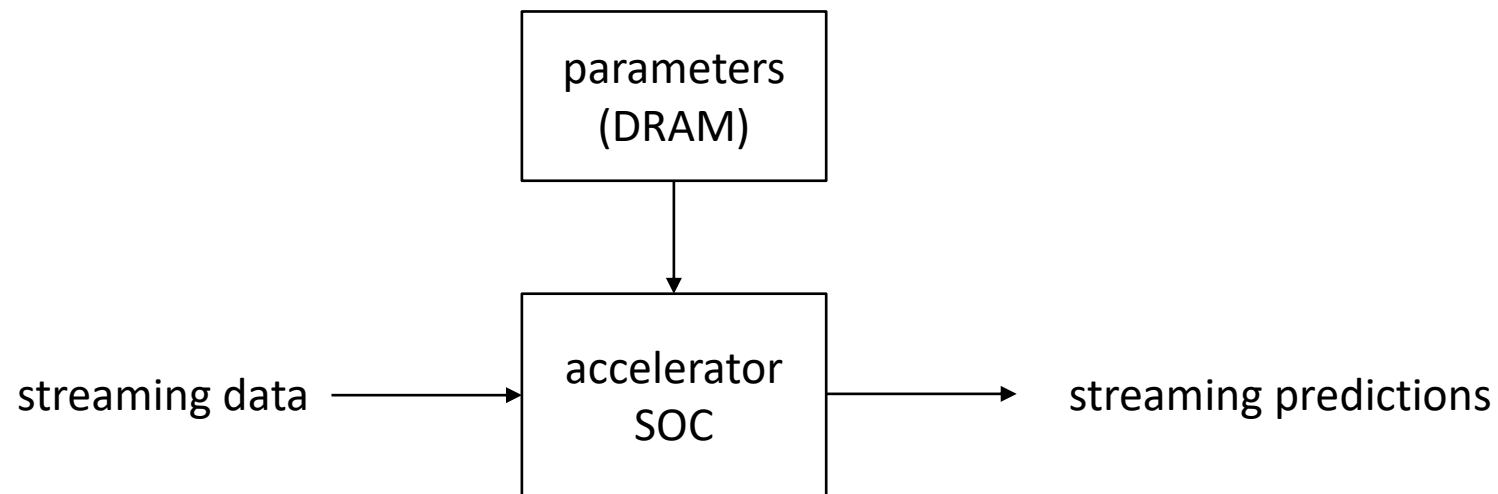
https://www.tensorflow.org/lite/performance/quantization_spec

https://www.tensorflow.org/lite/performance/post_training_quantization

<https://github.com/tensorflow/tensorflow/blob/92f84571edb98bdb31f1e7b1e23f17bf2c1438e9/tensorflow/lite/kernels/internal/reference/conv.h#L101>

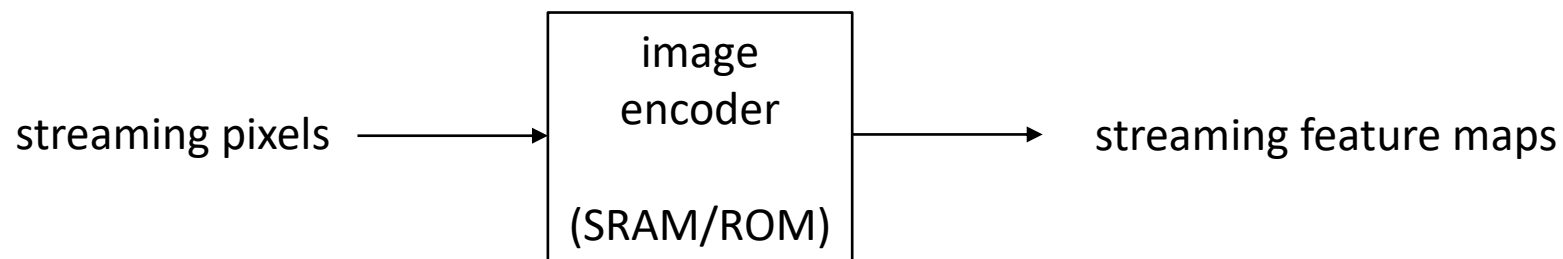
<https://pypi.org/project/tflite/>

Image Encoder Concept

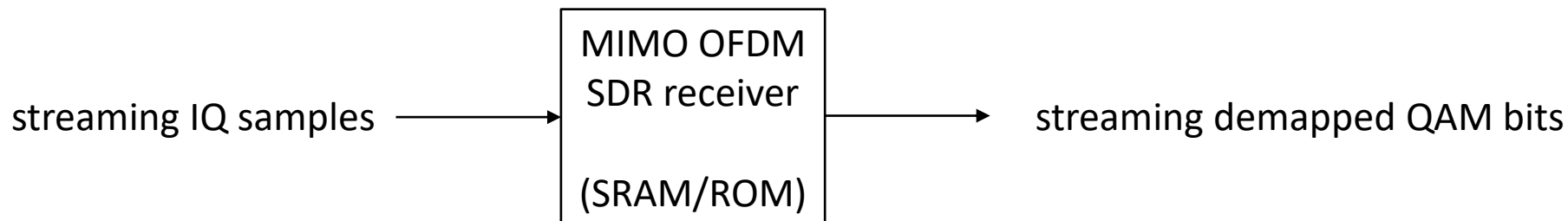


general purpose inference architecture ("compute platform")

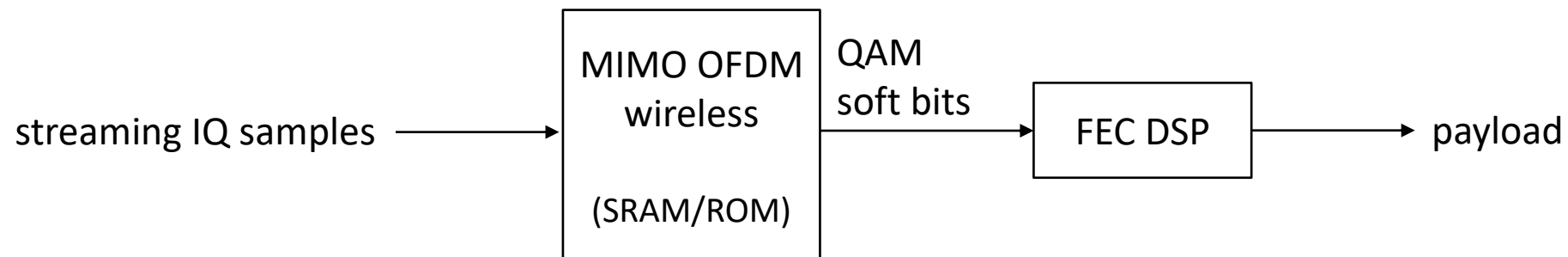
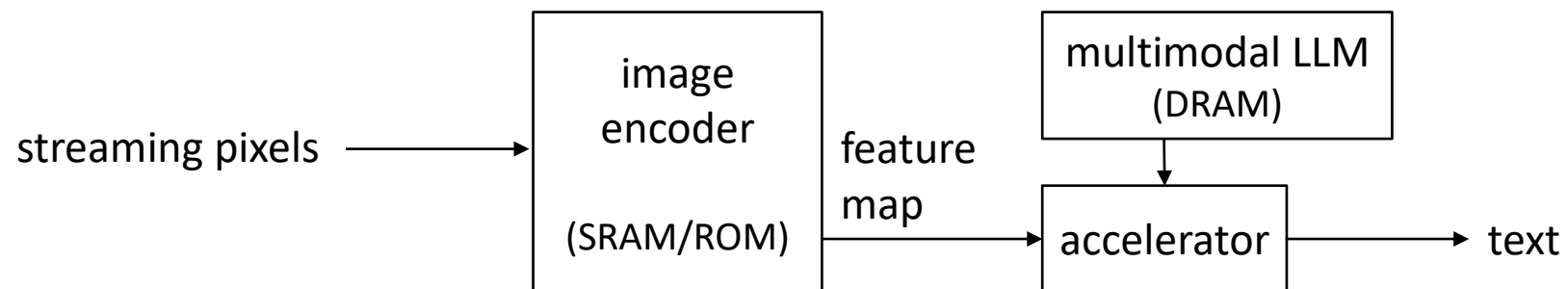
- arbitrary model architecture
- arbitrarily large model size
- DRAM based model parameters



- fixed model architecture
- fixed model size
- SRAM/ROM based parameters



data/task specific inference architectures ("perception components")



Technology Strategy

1. Powerful on-chip models can perform multiple tasks
 - a. for a vision convnet feature map pyramid, 8M weights ~ foundation vision model
 - b. it is feasible to use fixed model arch w/ params on-chip (SRAM or ROM)
 - c. supports image, video, RF and encoder/decoder tasks
2. Use existing algorithms, implemented in Verilog
 - a. high throughput ($>100\text{M}$ samples/s), low latency ($<10\mu\text{s}$), zero jitter, low power ($<2\text{W}$)
 - b. no new algorithms, leverage TFLite framework, correlated with TFLite reference
 - c. parameterized Verilog code with Python compiler

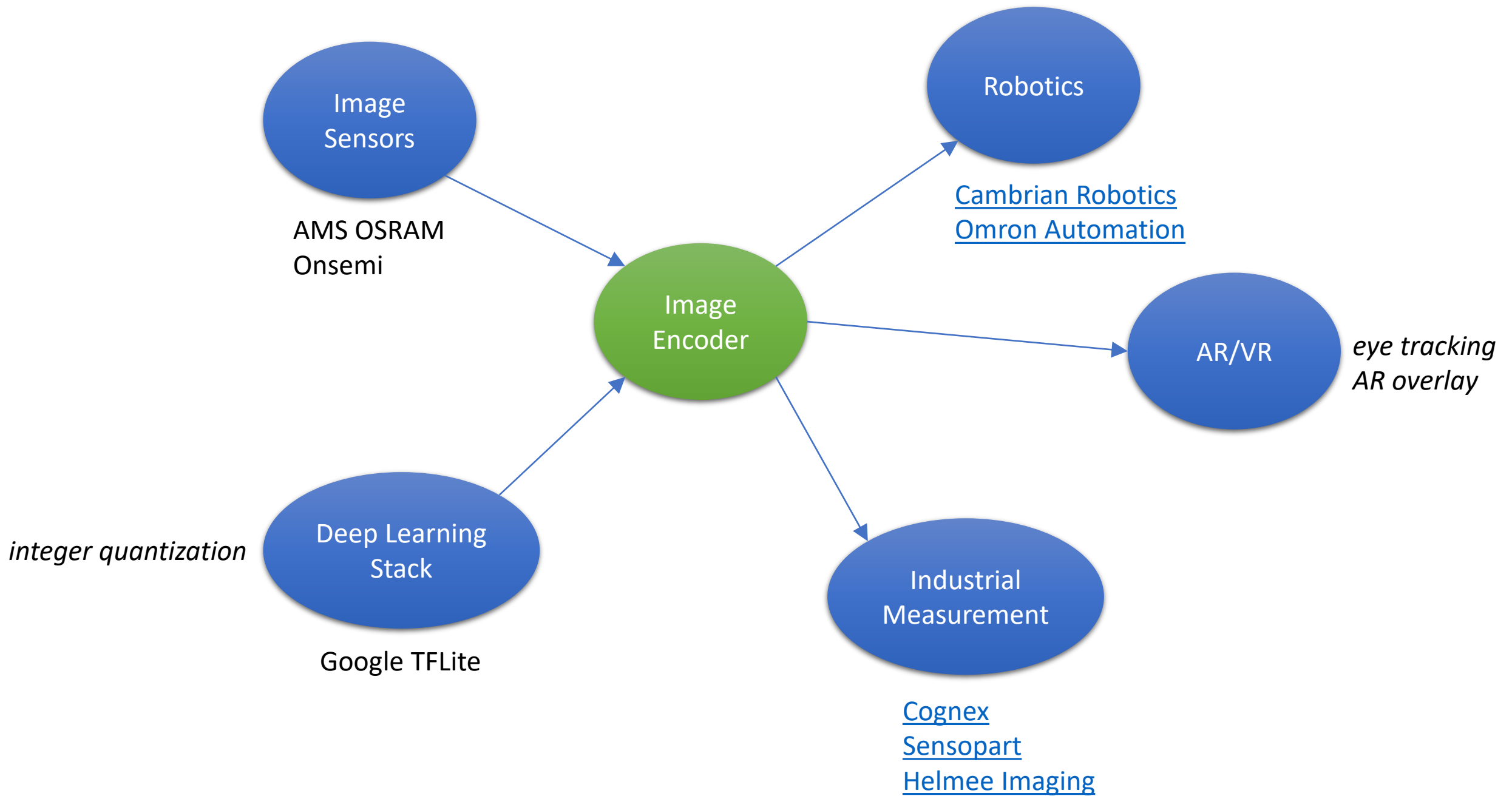
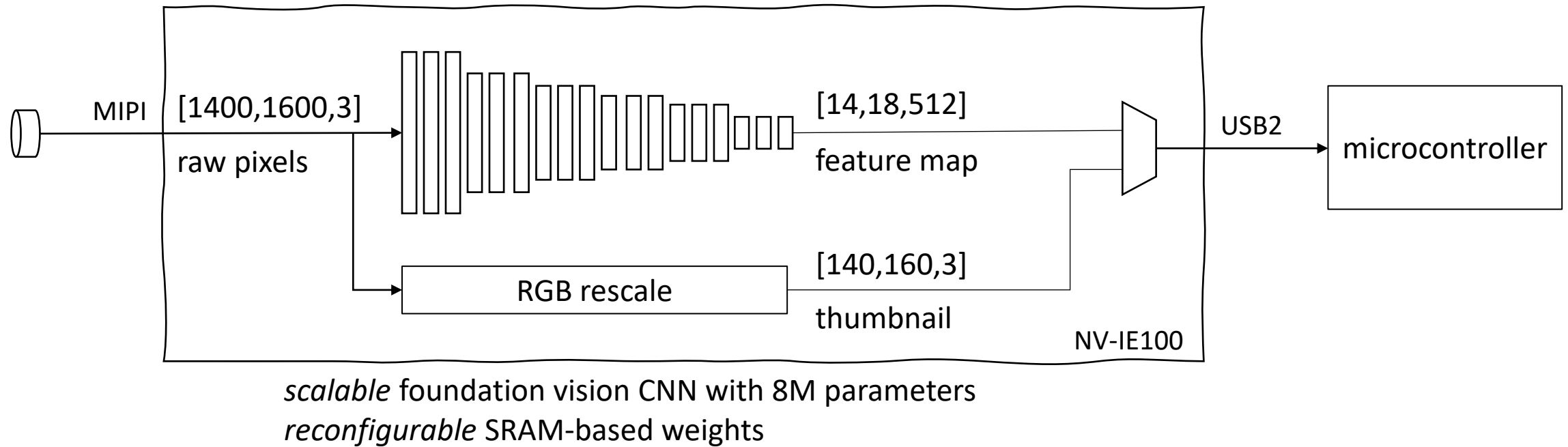


Image Encoder Concept

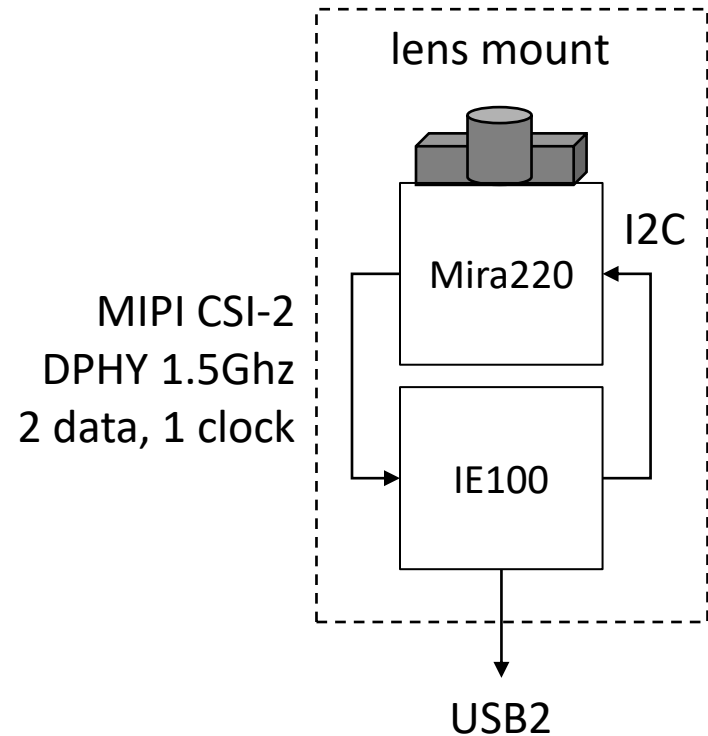


MIPI		USB	USB
1600x1400 RGB pixels		[14,18,512] features	[140,160,3] thumbnail
12 bits/pixel		16 bits/feature	8 bits/pixel
30 frames/s		30 frames/s	30 frames/s
= 2.42 Gb/s	30X compression	= 62 Mb/s	= 16 Mb/s

NV-IE100

500MHz clock
3840 integer MAC units
64Mb SRAM (weights)
20Mb SRAM (rowbuf)

Reference Board Concept



- real time image encoder
- compact form factor, USB powered
- supports a broad range of applications
- low pin count interfaces
- includes USB to I2C for configuration
- memory map provides access to weights

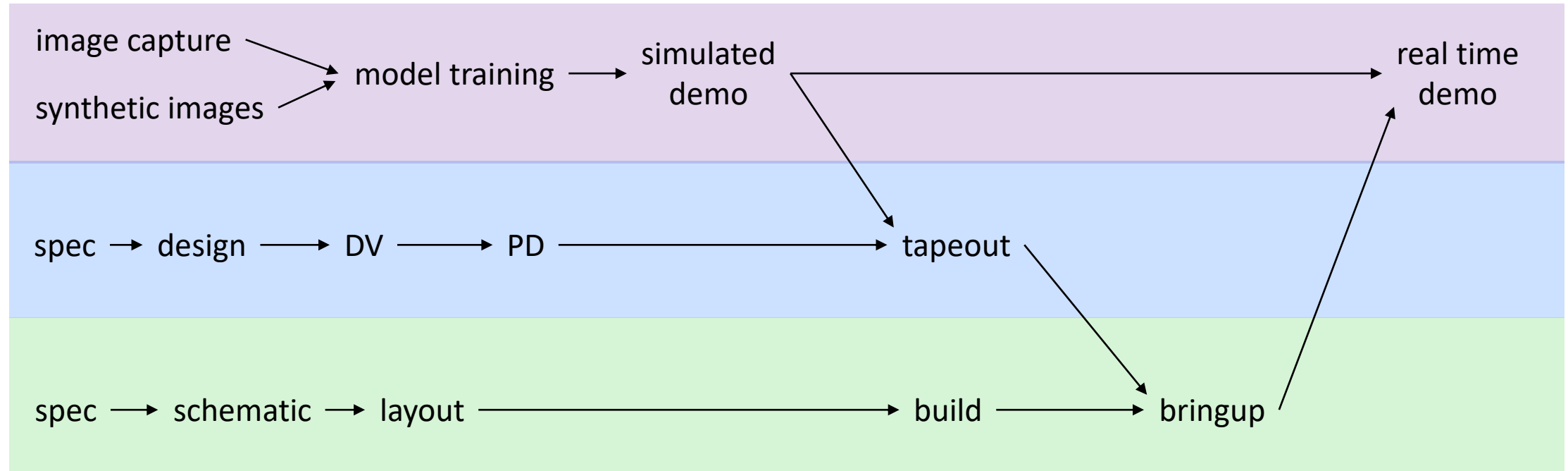
NV-IE USB endpoint	M/S
feature pipe	M
thumbnail pipe	M
I2C bridge	S
NV-IE memory map	S

Execution Plan

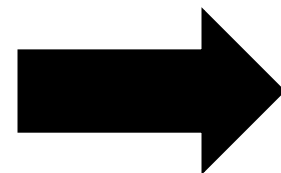
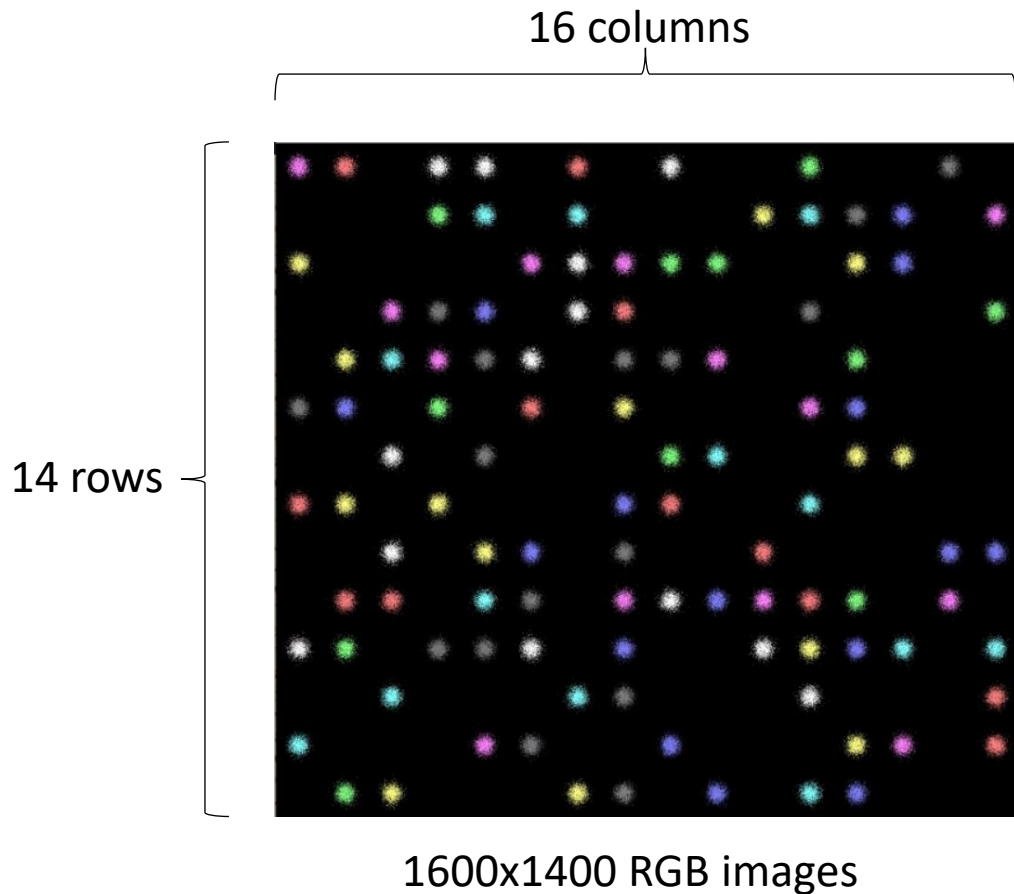
applications

chip

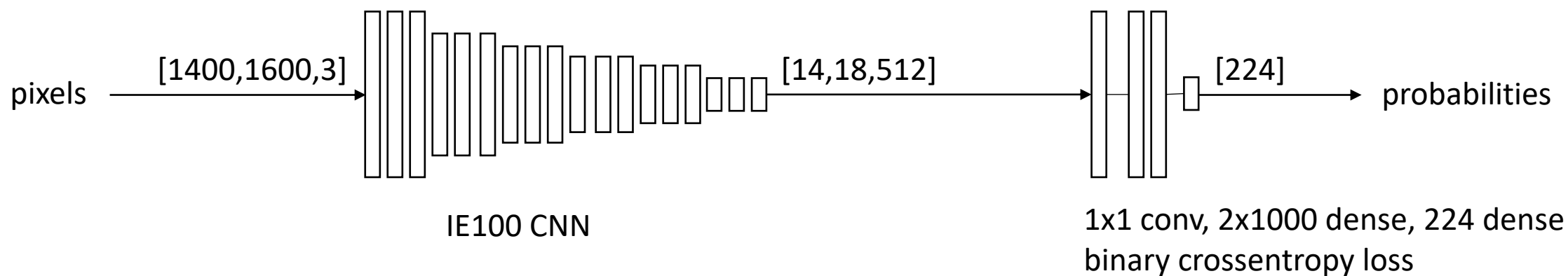
reference board



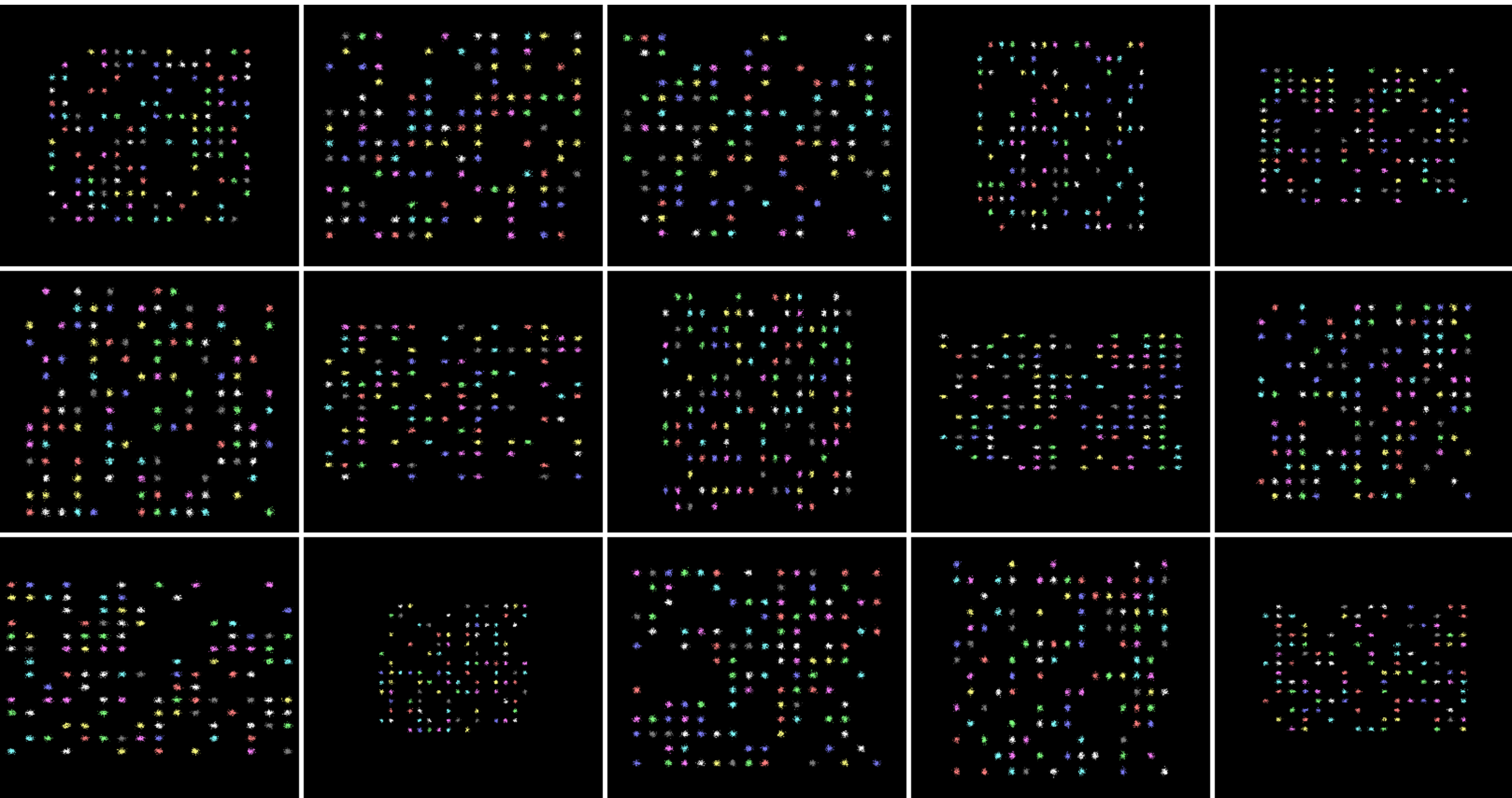
Receptive field unit test



randomly populate 16x14 grid locations
label is '0' if empty, '1' if colored ball
224 binary labels



thumbnail gallery of 1600x1400 RGB input images

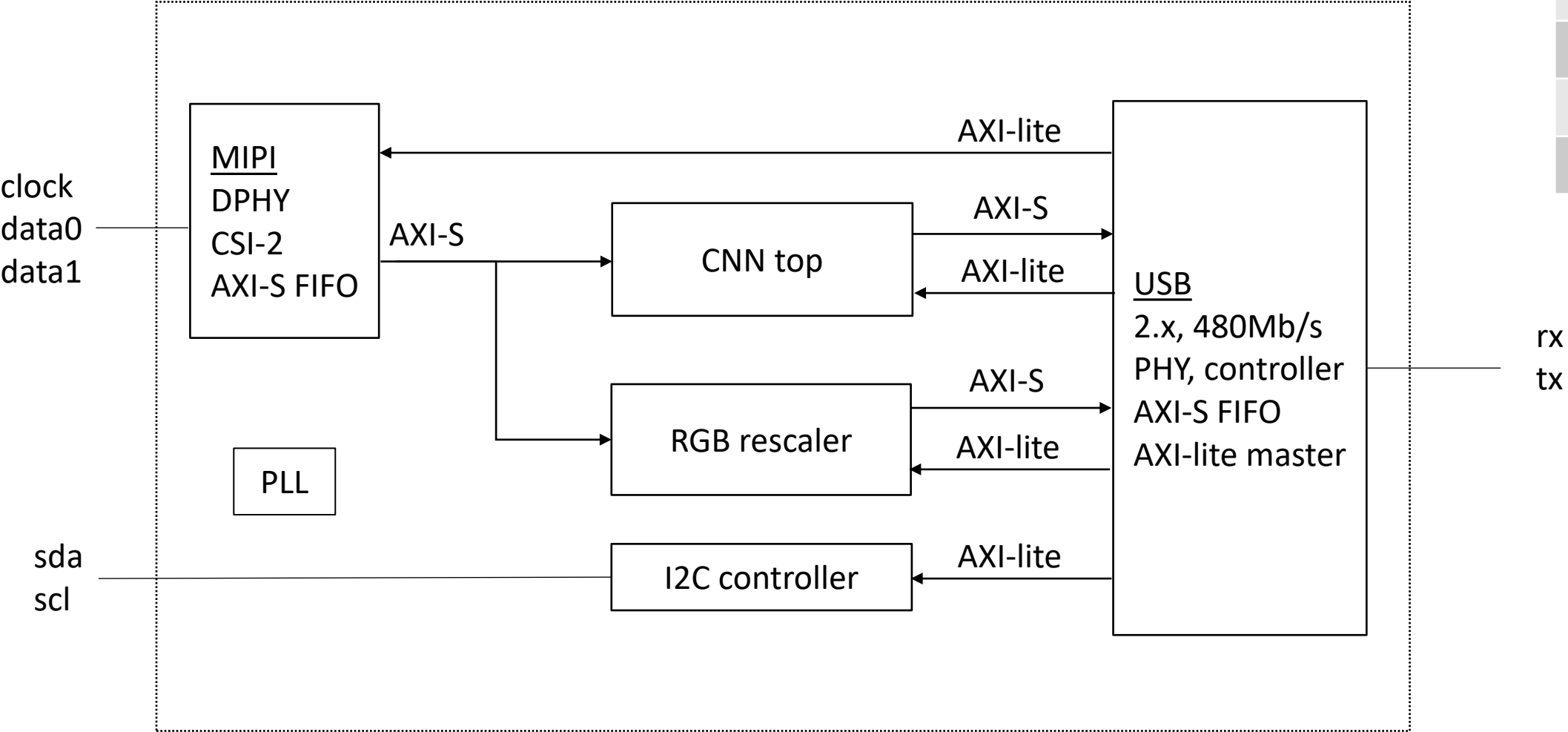


Product Planning

Milestone	Dependencies	Status
Concept Commit	PRD/datasheet die size/power estimate preliminary budget/schedule	needed for joint development proposal
Execution Commit	complete program plan to FCS	
FCS	standard SoC development process	

PRD Feature	Priority	Schedule
MIPI CSI-2, DPHY RX, 1C 2D, 1.5GHz	P1	MVP, fastest TTM
[1400,1600,3] input shape (frame) 12 bits/pixel 30 frames/s	P1	Mira220 RGB
[14,18,512] output shape (feature map)	P1	
[140,160,3] output shape (thumbnail)	P1	
USB 2.0 PHY, controller	P1	

NV-IE100 top level diagram

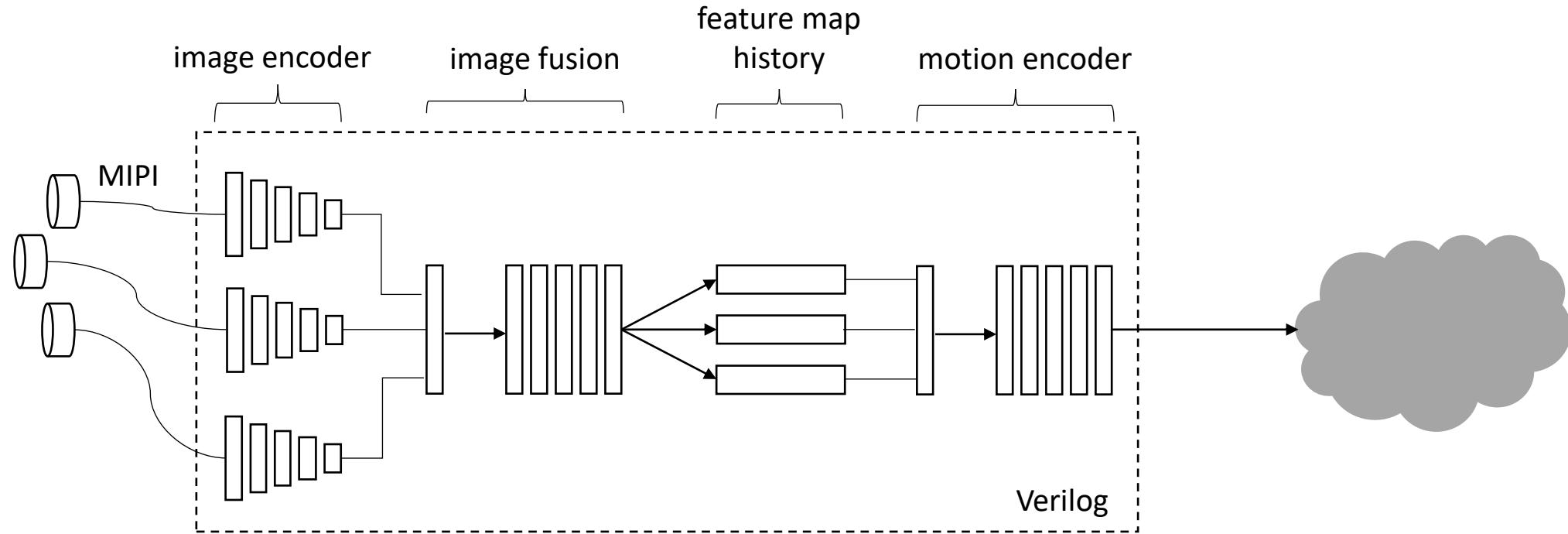


Block	Owner
MIPI	
USB	
RGB	
CNN	
I2C	

Execution Planning

Function	Tasks	Lead
CNN compiler	add level of hierarchy for inferred SRAM add SRAM option to replace weight ROM (SPI config) enhancements and fixes as needed (DAG,TFLite metadata,quantization)	
RTL design	MIPI CSI-2 RX with DPHY USB PHY and controller CNN top input/output interfaces and FIFO RGB image rescaler and FIFO I2C controller for image sensor configuration	
Verification (simulation)	End to end unit tests, pixels to predictions Also correlate predictions with TFLite reference interpreter	
Physical implementation	Standard ASIC flow with 500MHz core clock, 16nm libraries Run initial flat P&R experiment on compiled CNN with 18 layers	
Tools / Automation	Automate SRAM physical memory compiler for arbitrary size CNN instances	
Verification (emulation)	Prototype full chip at 200MHz, partition across multiple FPGAs	
Reference board	Mira220 RGB + NV-IE100, compact fanless form factor, USB powered	
Applications/Models Documentation FAE Support	Mira220+ Opal Kelly Artix US+ data capture and demo Build reference apps/models e.g facial keypoint detection, image super resolution, support HuggingFace model community	

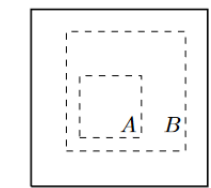
Roadmap



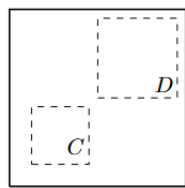
- integrate image fusion and motion encoder into the chip
- output feature maps will contain motion information which reduces the frame rate
- 100M IQ baseband samples/s for wireless sensing and communication e.g. MIMO OFDMA

NeuVision Robotics Demo Application

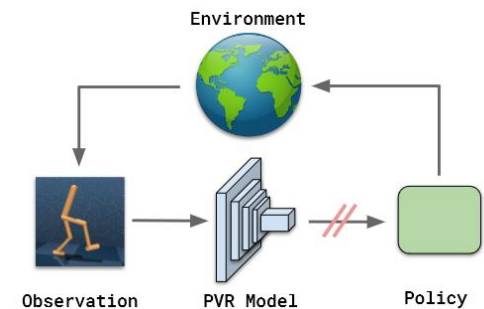
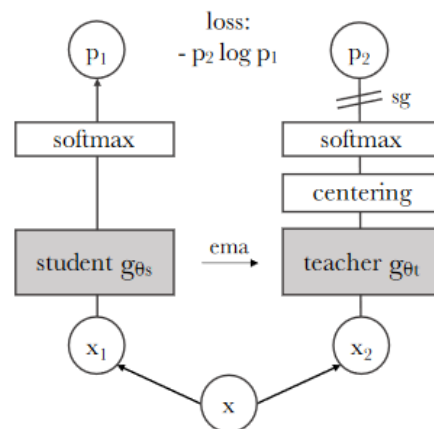
Paper	Year	
A Simple Framework for Contrastive Learning of Visual Representations	2020	random crop and rescale
Bootstrap Your Own Latent A New Approach to Self-Supervised Learning	2020	positive pairs only
Emerging Properties in Self-Supervised Vision Transformers	2021	our baseline method (DINO)
The (Un)Surprising Effectiveness of Pre-Trained Vision Models for Control	2022	robot demo



(a) Global and local views.



(b) Adjacent views.



APPENDIX

Risks

Risk	Sev	Mitigation
negative effects of integer quantization		POR use TFLite signed integer 8b weights, 16b activations Straightforward to switch to 8b/16b floating point as a fallback
no batch normalization or skip connections		POR use RELU activations with reduced learning rate Straightforward to support SELU using 64Kx16 SRAM LUT as a fallback
MIPI burstiness due to vertical blanking		CNN throughput is specified as a row time, not frame time

MIPI SoC platforms exceed the cost of the image sensor

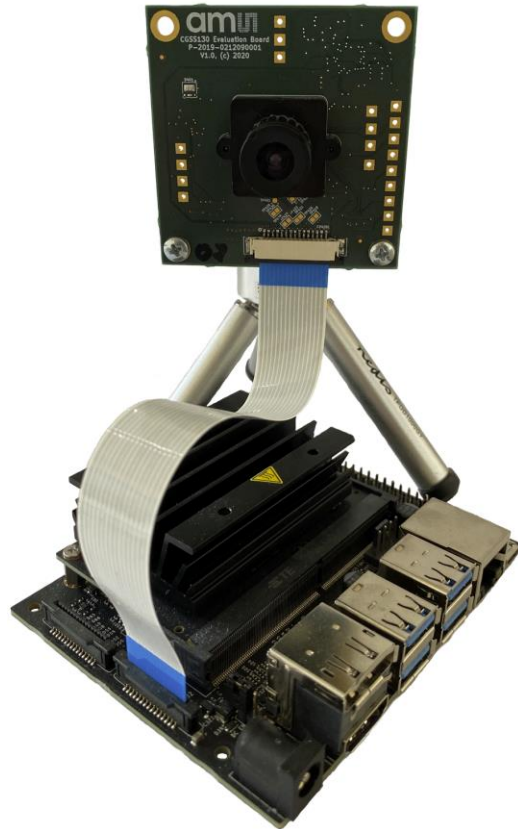


Image Encoder Product Options

NeuVision Product	Model Architecture	Model Weights	Scalability
IE-FPGA	reconfigurable	reconfigurable	<1M weights, <1 TOP, chiplets?
IE-100-RAM	fixed VGG18	reconfigurable (RAM)	8M weights, 2 TOP
IE-100-ROM	fixed VGG18	fixed (ROM)	8M weights, 2 TOP

- Image Encoder performs the initial convolutional pyramid using a VGG-like architecture.
- The proposed 8M weight VGG18 architecture is a strong foundation vision model with broad application.
- The image encoder also performs a valuable compression function.
- Weights are 8 bits, so 64Mb SRAM (weights) + 20Mb SRAM (row buffers) + standard cell logic

proposed VGG-like CNN

uses conv2d layer only

downsample using stride=2

padding=valid

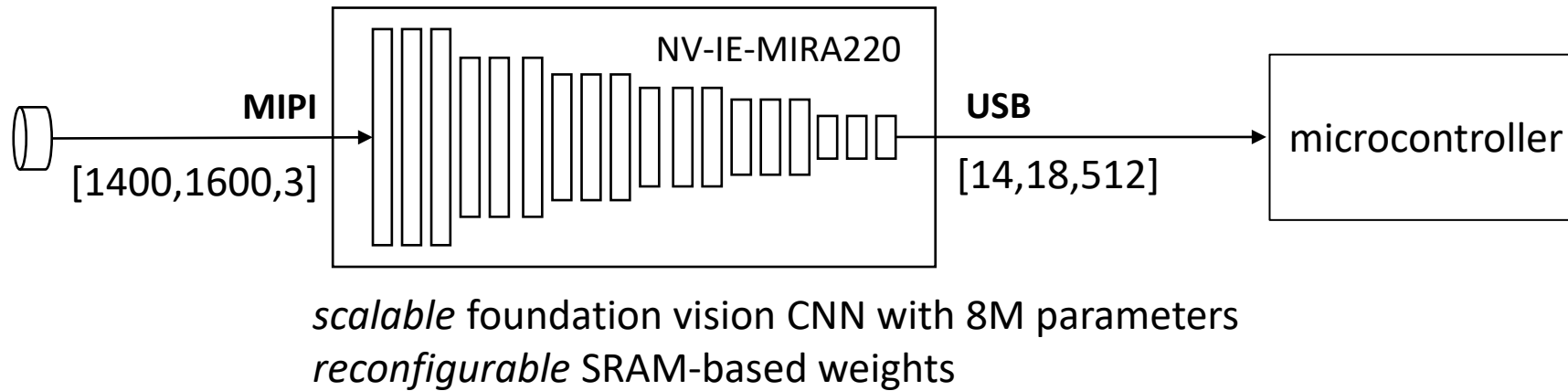
activation nonlinearity=RELU, last layer=None

8 bit integer weights, 16 bit integer activations

Layer (type)	Output Shape	Param #
input_1 (InputLayer)	[(None, 1400, 1600, 3)]	0
conv2d (Conv2D)	(None, 1398, 1598, 16)	448
conv2d_1 (Conv2D)	(None, 698, 798, 16)	2320
conv2d_2 (Conv2D)	(None, 696, 796, 16)	2320
conv2d_3 (Conv2D)	(None, 694, 794, 32)	4640
conv2d_4 (Conv2D)	(None, 346, 396, 32)	9248
conv2d_5 (Conv2D)	(None, 344, 394, 32)	9248
conv2d_6 (Conv2D)	(None, 342, 392, 64)	18496
conv2d_7 (Conv2D)	(None, 170, 195, 64)	36928
conv2d_8 (Conv2D)	(None, 168, 193, 64)	36928
conv2d_9 (Conv2D)	(None, 166, 191, 128)	73856
conv2d_10 (Conv2D)	(None, 82, 95, 128)	147584
conv2d_11 (Conv2D)	(None, 80, 93, 128)	147584
conv2d_12 (Conv2D)	(None, 78, 91, 256)	295168
conv2d_13 (Conv2D)	(None, 38, 45, 256)	590080
conv2d_14 (Conv2D)	(None, 36, 43, 256)	590080
conv2d_15 (Conv2D)	(None, 34, 41, 512)	1180160
conv2d_16 (Conv2D)	(None, 16, 20, 512)	2359808
conv2d_17 (Conv2D)	(None, 14, 18, 512)	2359808
Total params: 7,864,704		
Trainable params: 7,864,704		
Non-trainable params: 0		

Proposed Image Encoder

NV-IE-MIRA220
500MHz clock
7500 integer MAC units
64Mb SRAM (weights)
20Mb SRAM (rowbuf)



MIPI

1600x1400 RGB pixels
12 bits/pixel
110 frames/s
= 8.87 Gb/s

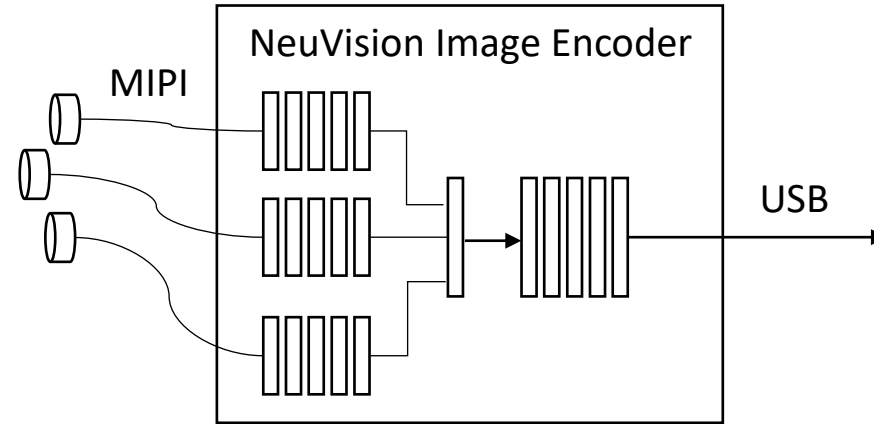
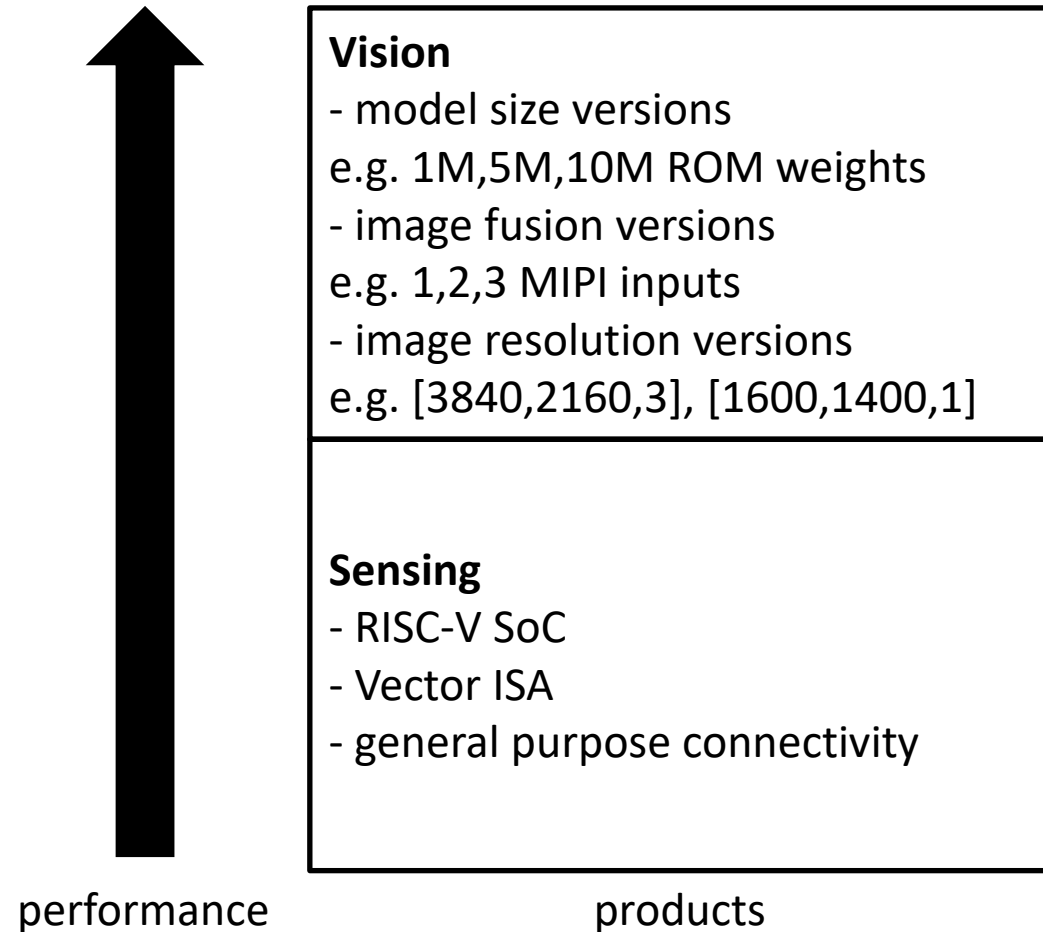
39X compression



USB

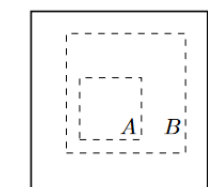
$[14, 18, 512]$ features
16 bits/feature
110 frames/s
= 0.227 Gb/s

Smart Sensor Silicon Segmentation

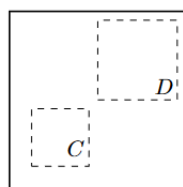


1. collect unlabeled dataset
 - a. synchronized frame capture from multiple image sensors
 - b. synchronized metadata e.g. robot servo angles
2. self supervised pretraining using unlabeled dataset
 - a. BYOL, DINO
3. few shot learning using pretrained vision model
 - a. imitation learning for robotics
 - b. classifier for object detection
 - c. image decoder for object segmentation
 - d. patch embeddings for multimodal LLM
4. data sheet for {100K,1M,5M,20M parameter VGG-like CNN}, {FPGA, ASIC}
 - a. area, ROM area, clock frequency

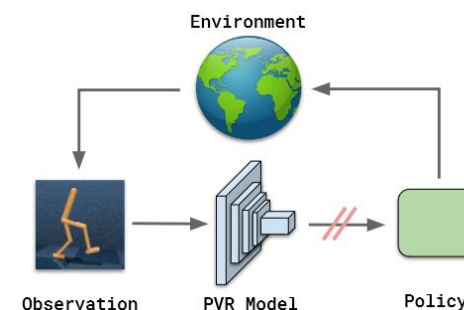
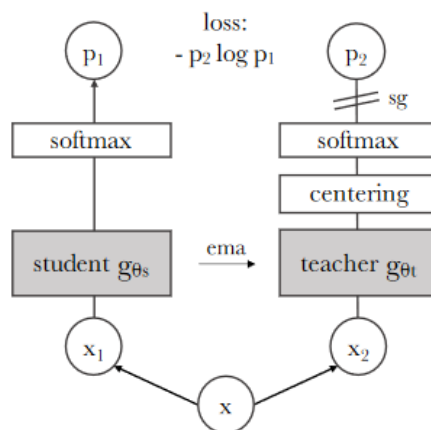
Paper	Year	
A Simple Framework for Contrastive Learning of Visual Representations	2020	random crop and rescale
Bootstrap Your Own Latent A New Approach to Self-Supervised Learning	2020	positive pairs only
Emerging Properties in Self-Supervised Vision Transformers	2021	our baseline method (DINO)
DINOv2: Learning Robust Visual Features without Supervision	2023	
Masked Autoencoders As Spatiotemporal Learners	2022	
The (Un)Surprising Effectiveness of Pre-Trained Vision Models for Control	2022	robot demo



(a) Global and local views.



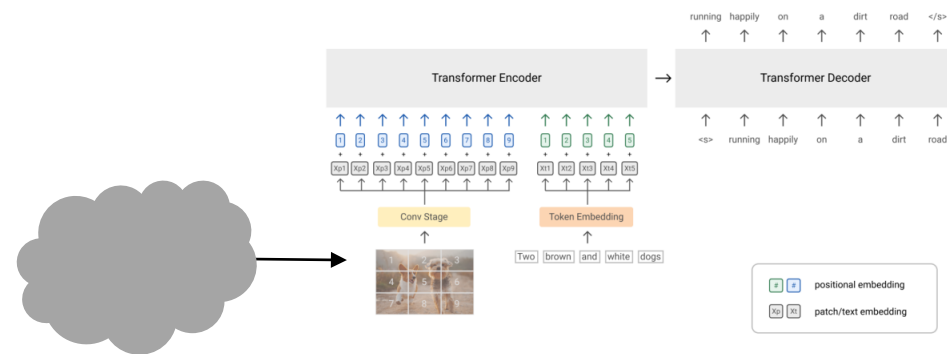
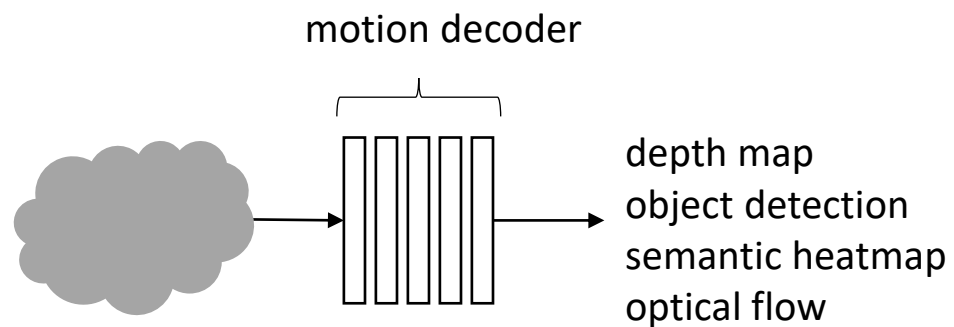
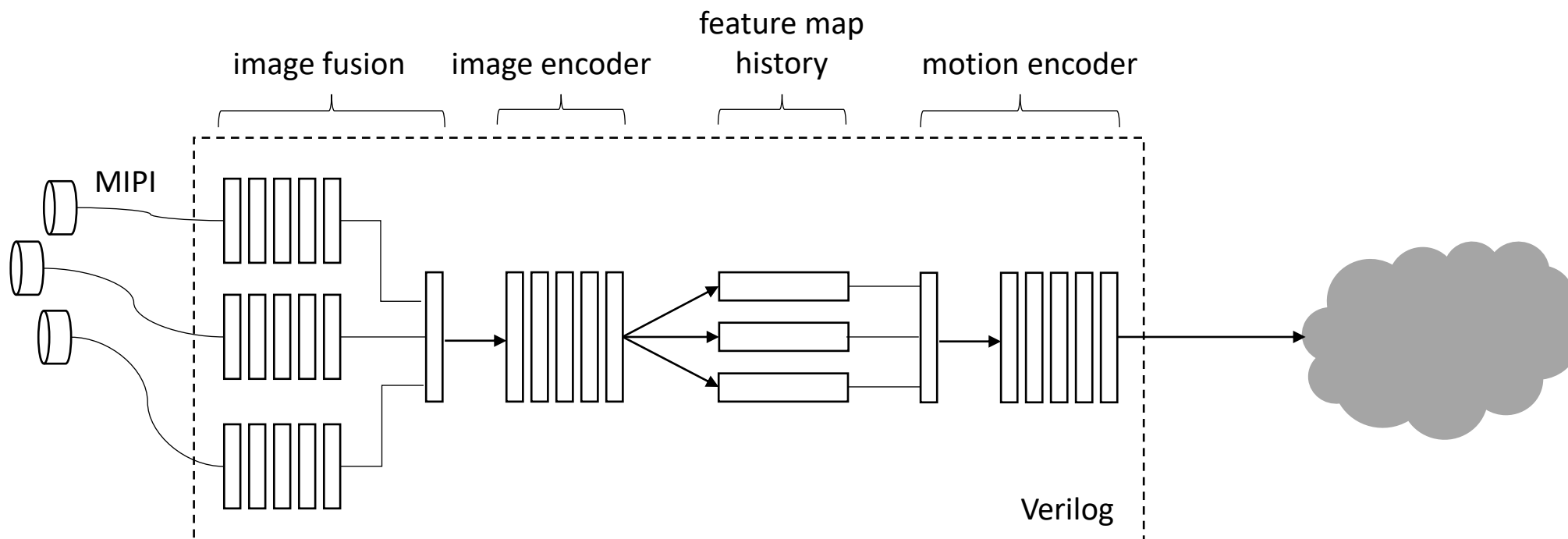
(b) Adjacent views.



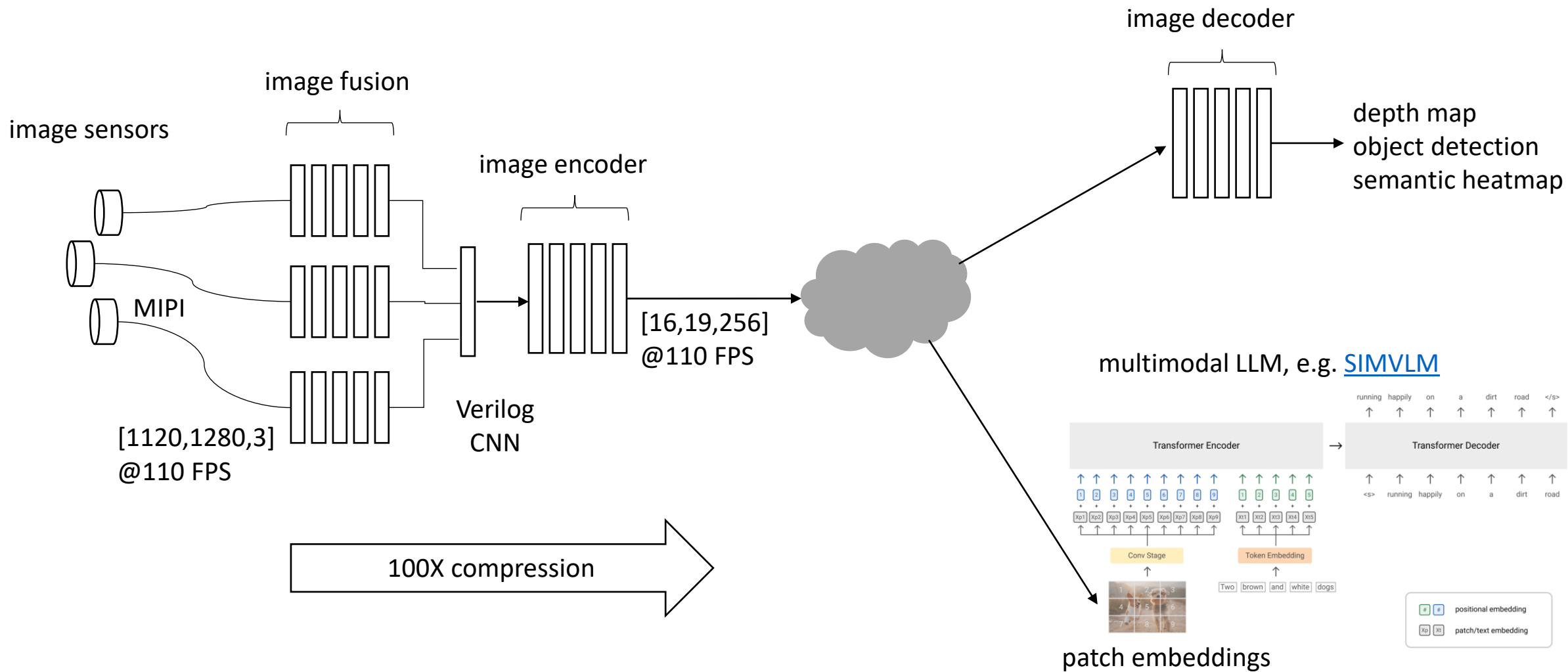
[1120,1280,3]
@110 FPS

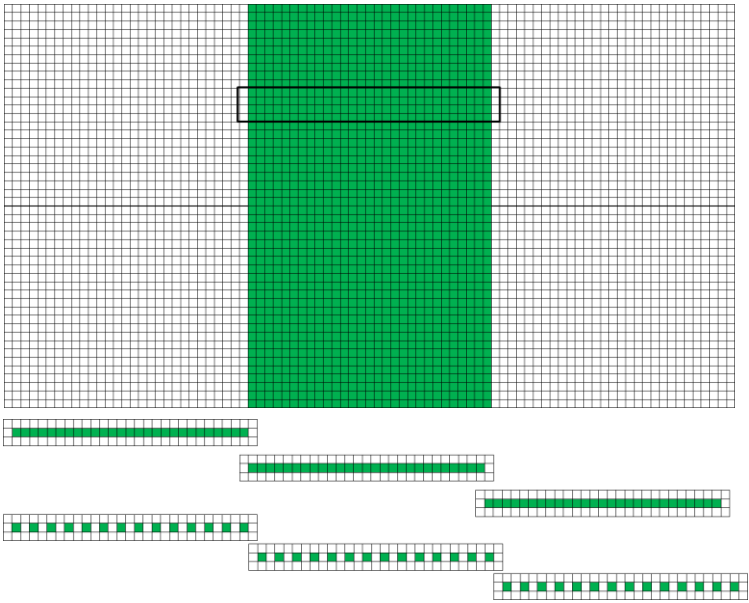
1000X compression

[16,19,256]
@10 FPS

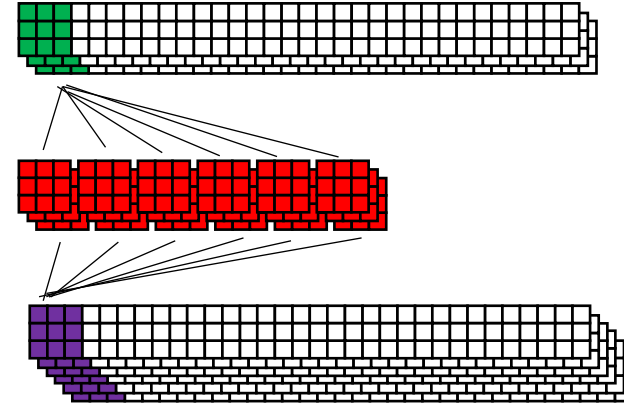


patch embeddings for multimodal LLM, e.g. [SIMVLM](#)

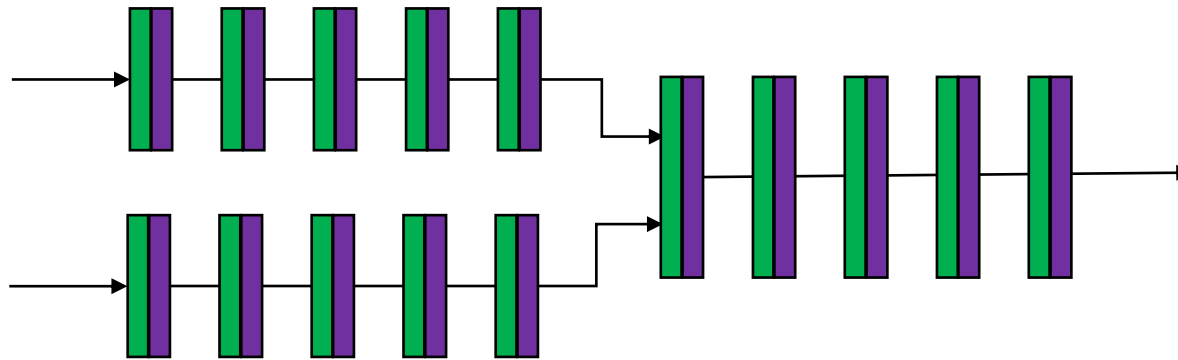




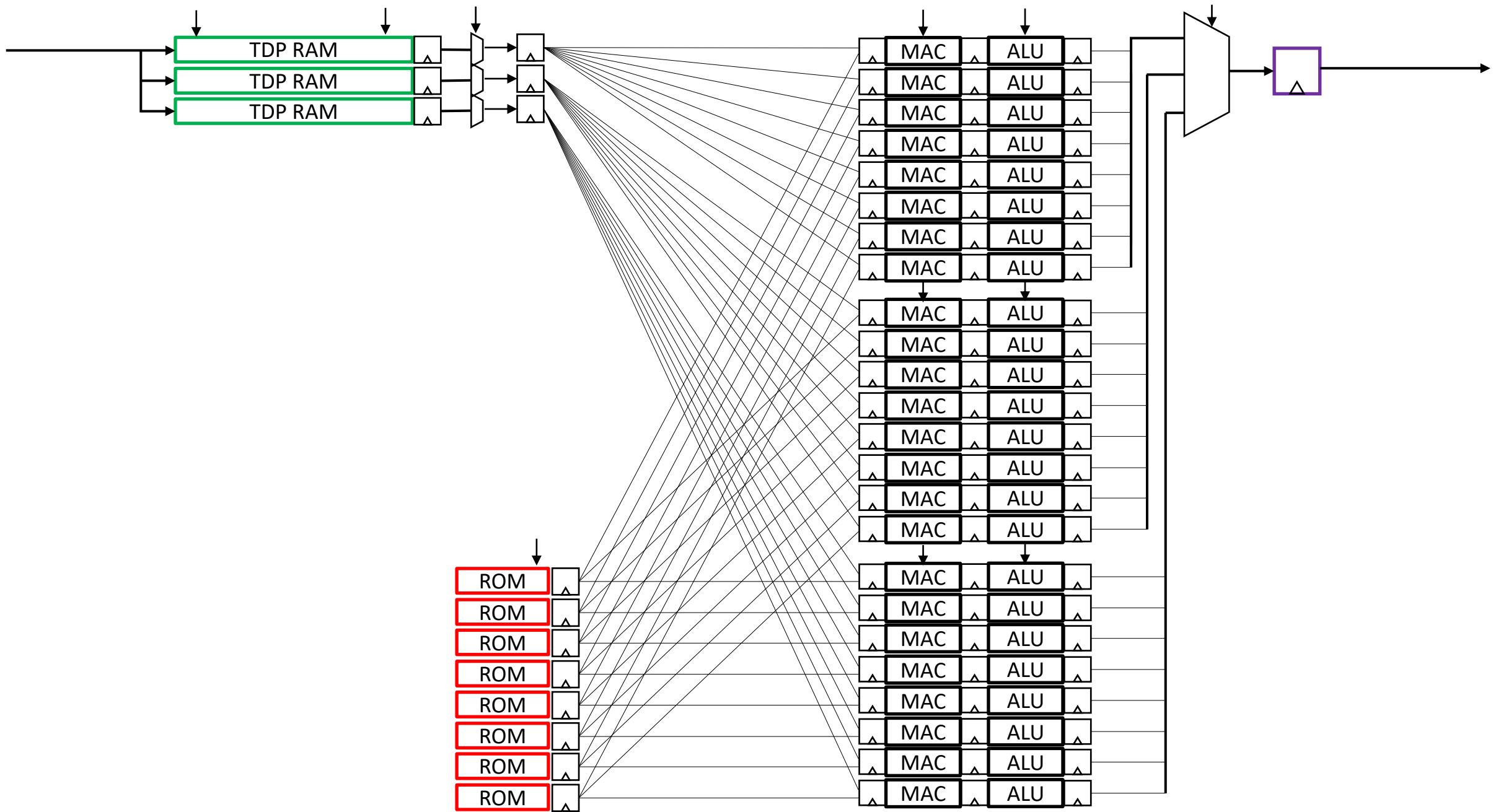
A) parallel vertical stripes

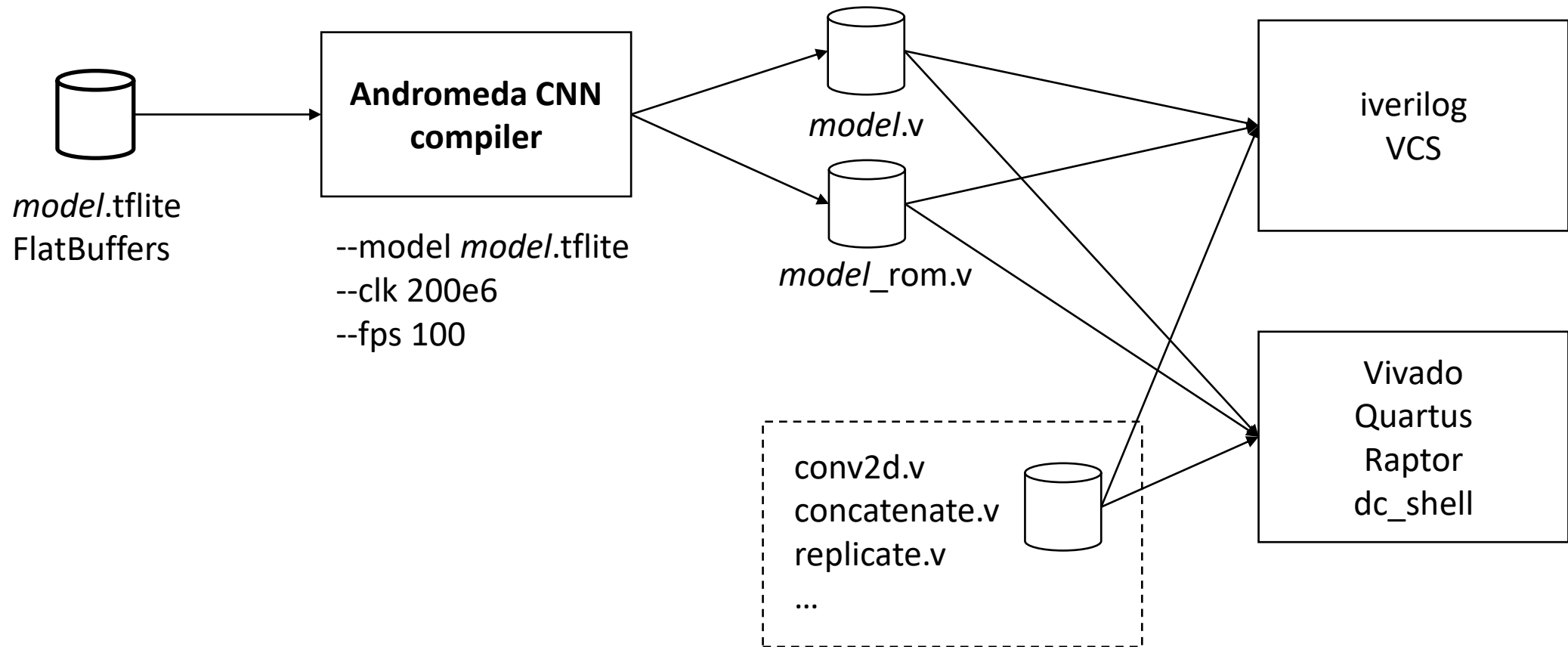


B) parallel output channels

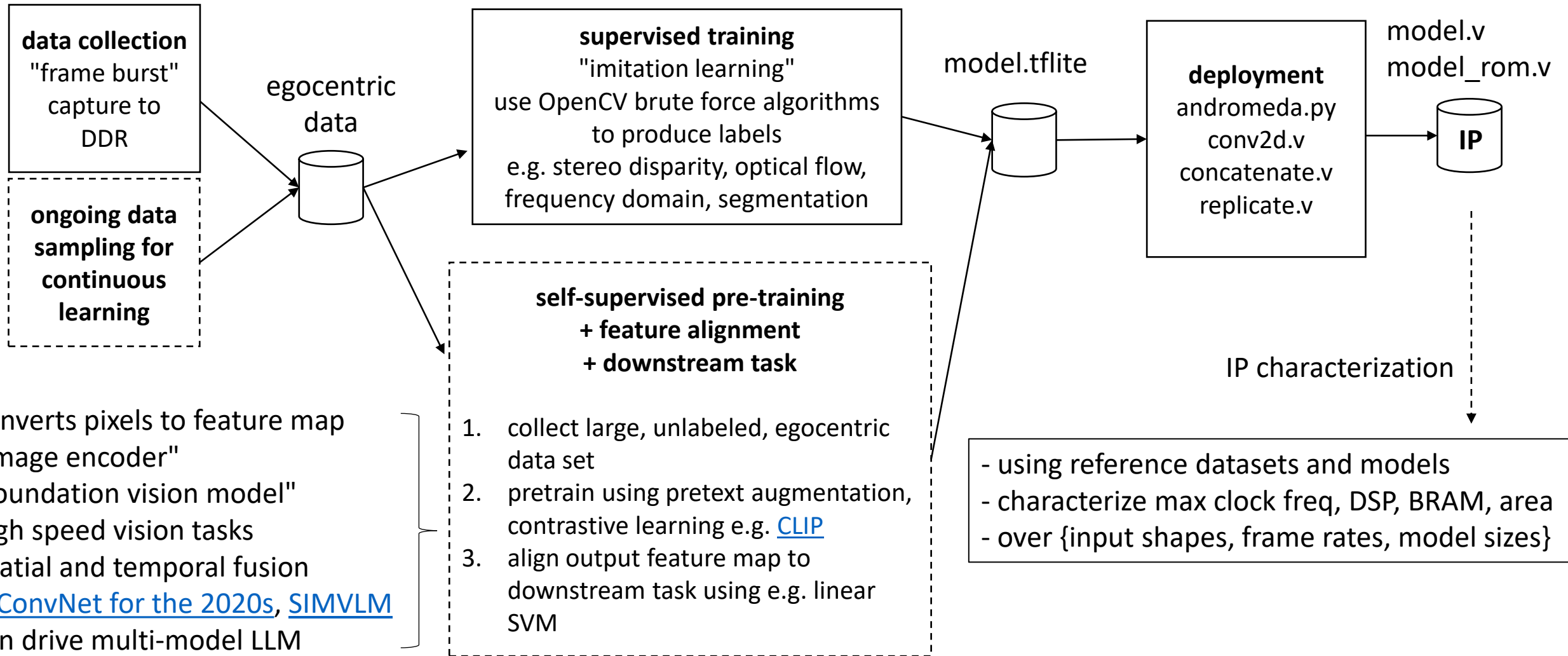


C) parallel layers

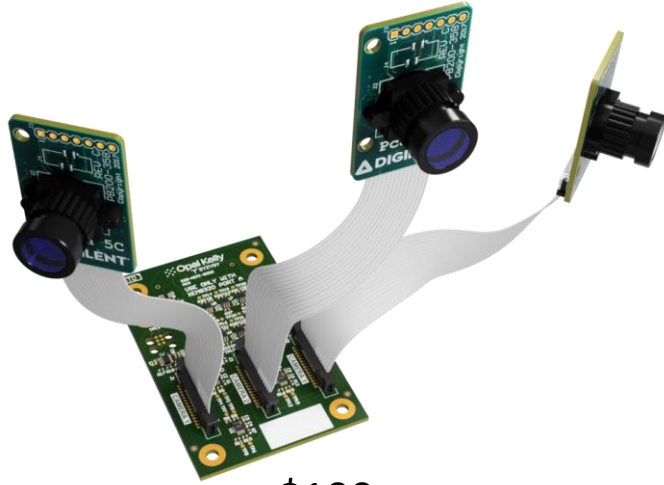




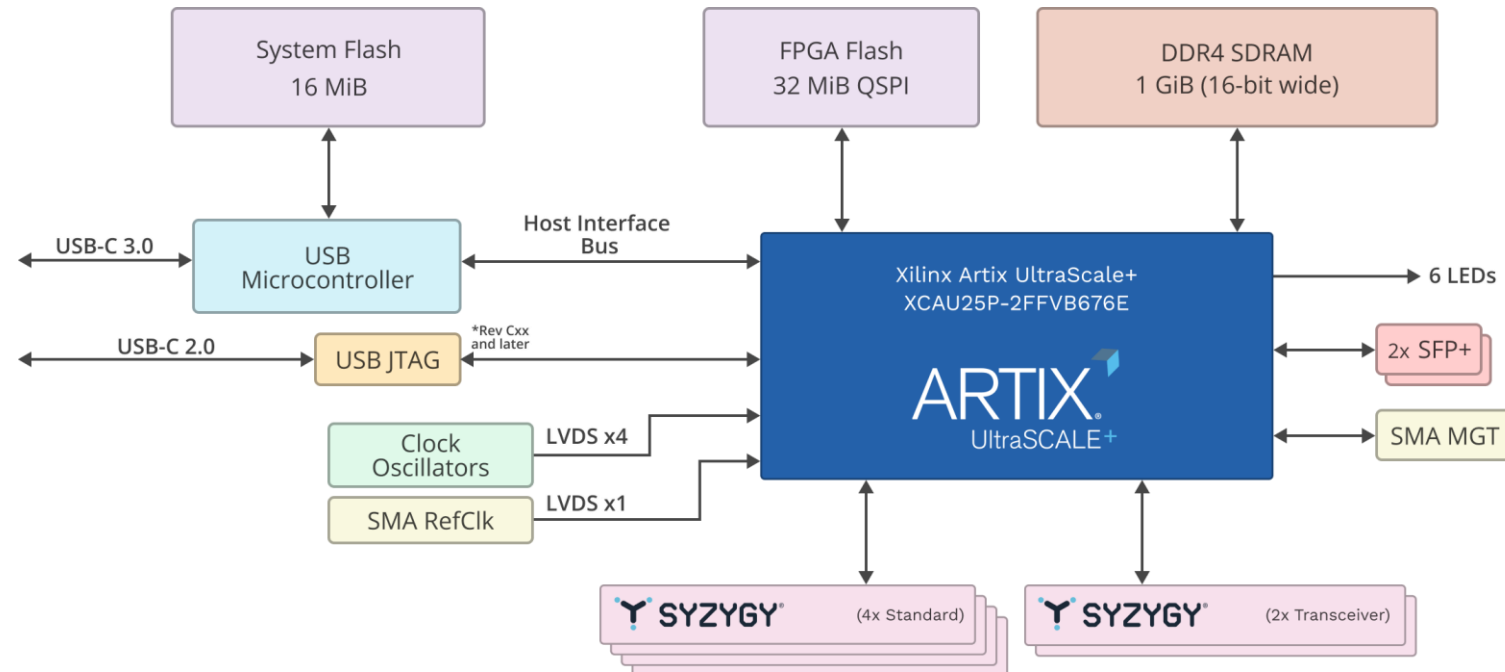
Real time vision IP flow

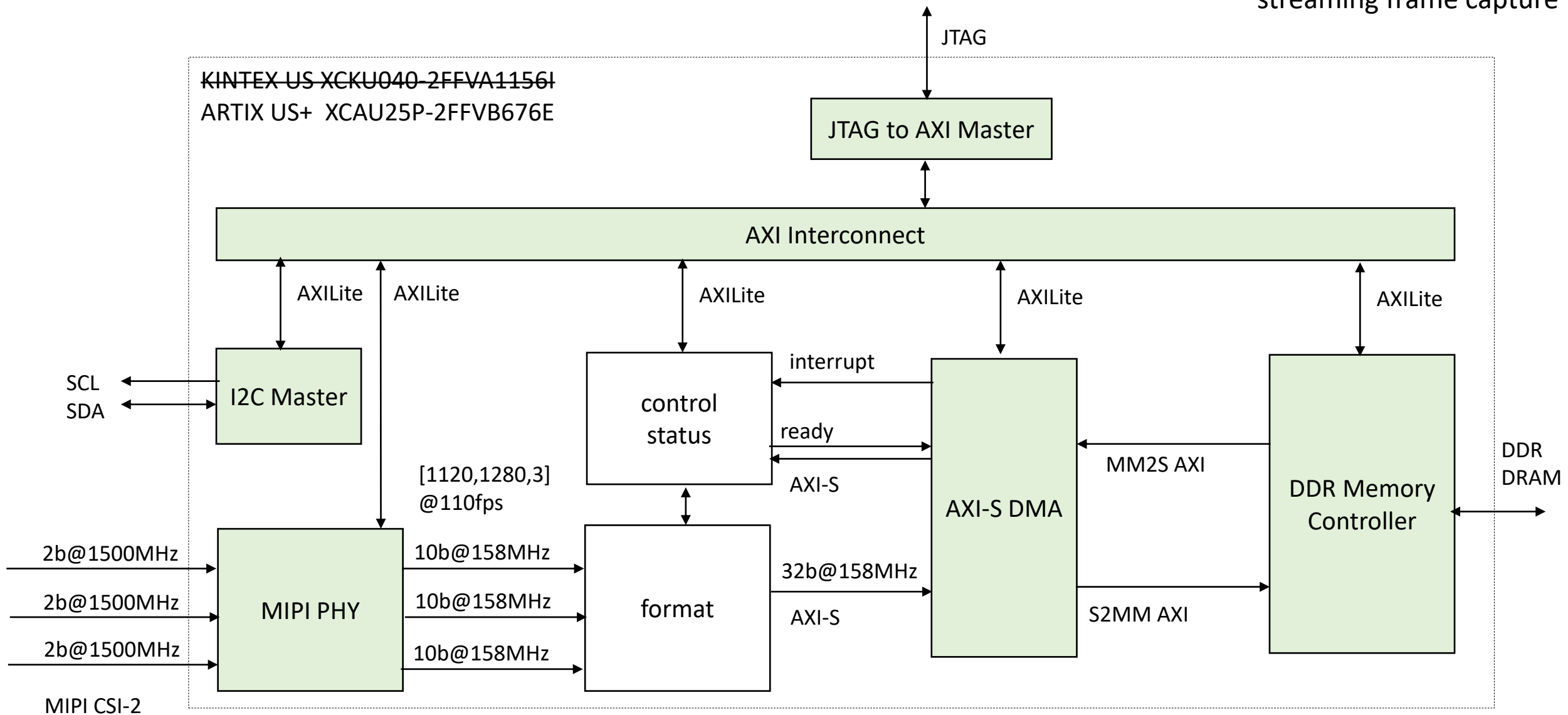


\$1400



\$100





https://www.xilinx.com/products/intellectual-property/jtag_to_axi_master.html

<https://docs.xilinx.com/r/en-US/pg232-mipi-csi2-rx/MIPI-CSI-2-Receiver-Subsystem-Product-Guide>

<https://docs.xilinx.com/v/u/en-US/pg150-ultrascale-memory-ip>

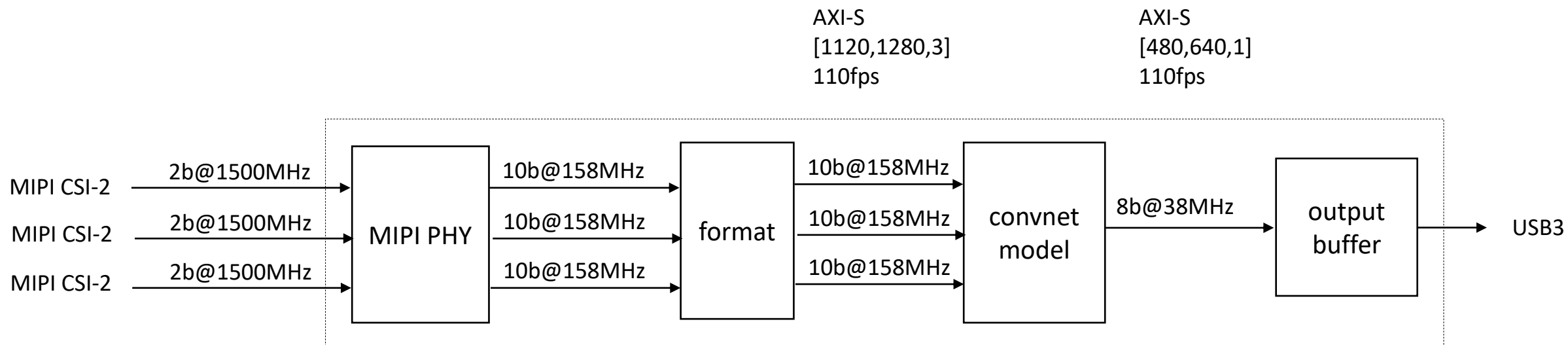
https://docs.xilinx.com/r/en-US/pg021_axi_dma/AXI-DMA-Register-Address-Map?section=XREF_38966_AXI_DMA_Simple_DMA

https://docs.xilinx.com/v/u/en-US/ds768_axi_interconnect

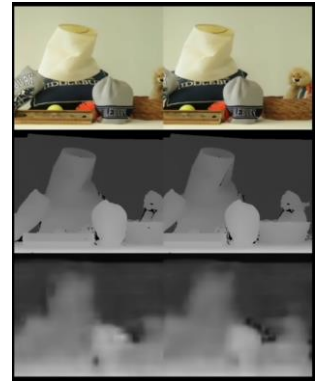
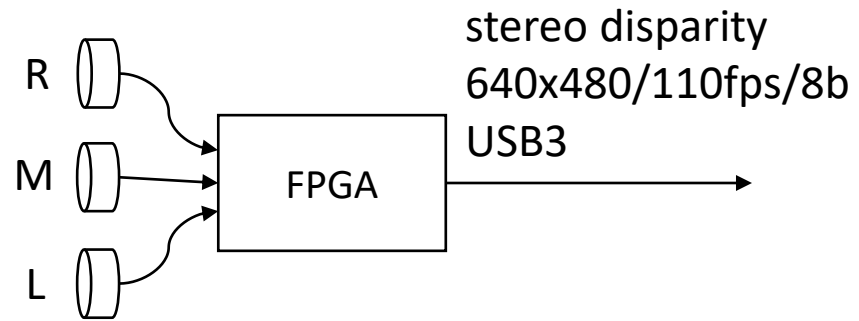
EP address	Type	Bits	Description
0x00	wire_in	31:0	AWADDR
0x01		31:0	WDATA
0x02		31:0	ARADDR
0x03		7:0	AWLEN
0x03		10:8	AWSIZE
0x03		12:11	AWBURST
0x03		13	AWLOCK
0x03		17:14	AWCACHE
0x03		20:18	AWPROT
0x03		24:21	AWQOS
0x04		3:0	WSTRB
0x04		4	WLAST
0x05		7:0	ARLEN
0x05		10:8	ARSIZE
0x05		12:11	ARBURST
0x05		13	ARLOCK
0x05		17:14	ARCACHE
0x05		20:18	ARPROT
0x05		24:21	ARQOS

EP address	Type	Bits	Description
0x40	trigger_in	0	AWVALID
0x40		1	WVALID
0x40		2	BREADY
0x40		3	ARVALID
0x40		4	RREADY
0x40		5	global_resetrn

EP address	Type	Bits	Description
0x20	wire_out	0	AWVALID
0x20		1	WVALID
0x20		2	BVALID
0x20		3	ARVALID
0x20		4	RVALID
0x21		31:0	RDATA
0x22		1:0	BRESP
0x23		1:0	RRESP
0x23		2	RLAST



Mira220, IMG.2/SPEED.1/ADC.2
1280x1120/110fps/10b
MIPI CSI-2

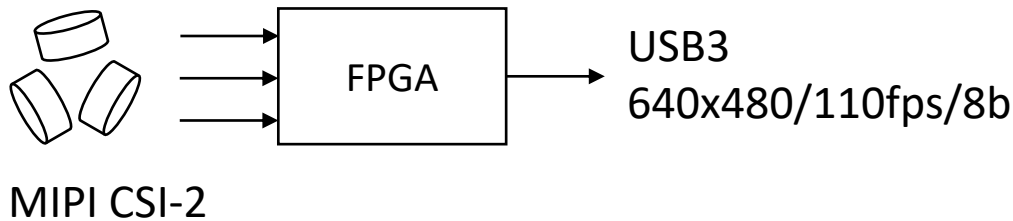
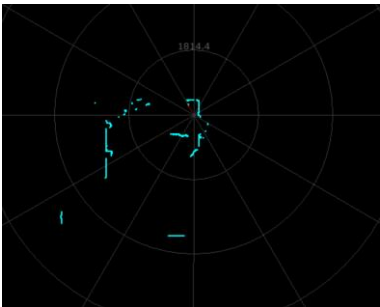


predict RPLidar 2-D LIDAR map from 3X Mira220 in 360 deg configuration (e.g. Roomba view)

demo concept



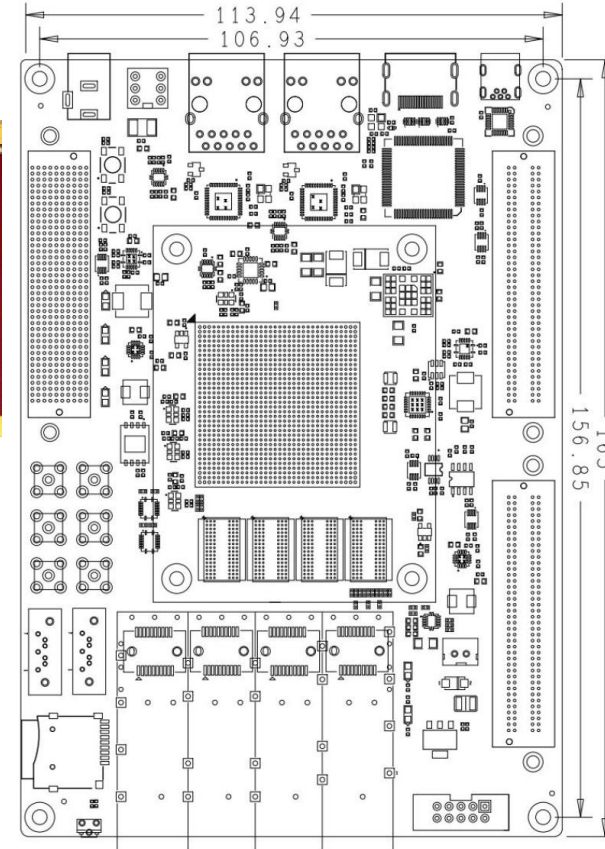
1280x1120/110fps/10b
Mira220, IMG.2/SPEED.1/ADC.2

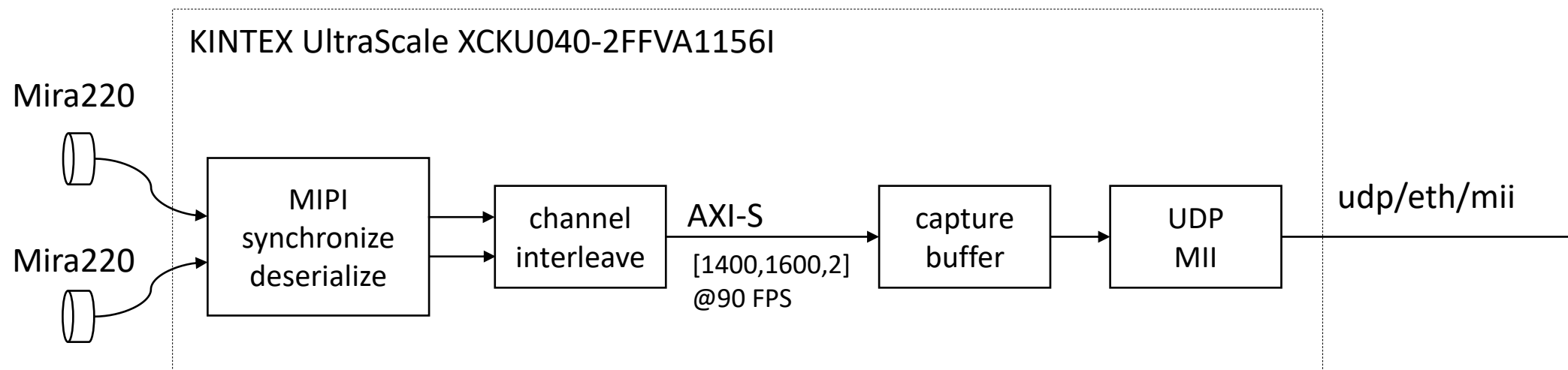


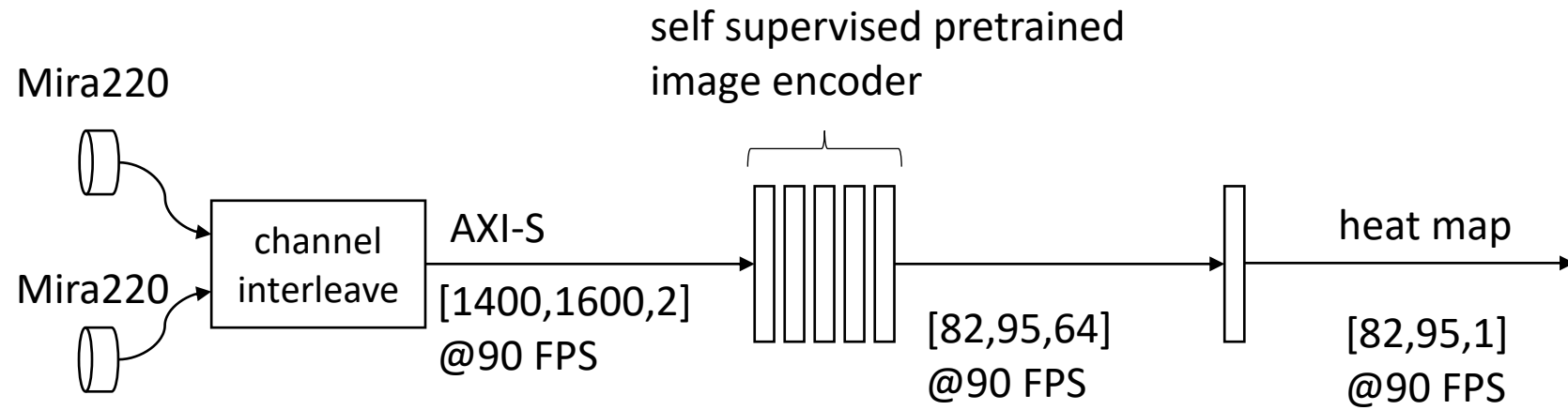
Mira-220

LVDS C+2D@1.5GHz

LVDS C+2D@1.5GHz

FMC-MIPI





pretrain using self-supervised pretext tasks on unlabeled data

1. LR eye swap, predict LR vs RL
2. LR time shift, predict $L[t]R[t]$ vs $L[t-1]R[t]$ vs $L[t]R[t-1]$

Sensing platform

Silicon vision and sensing platform for next generation robotics

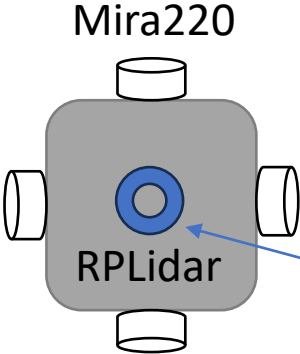
- target next generation robotics based on multimodal LLM technology
- the image encoder socket is the first stage in a multimodal LLM [e.g. CLIP](#)
- foundation/backbone vision models encode raw pixels into a compressed feature map
- domain specific FPGA enables low latency "pixels to features" at sensor speed
- extends to 360-degree vision, hyperspectral, video, tactile sensors, audio, RF

Strategy

- Open software
- Optimized FPGA silicon initially targeting the emerging **image encoder socket**
- Scale using chiplets
- Partner with image sensor, robotics ecosystems

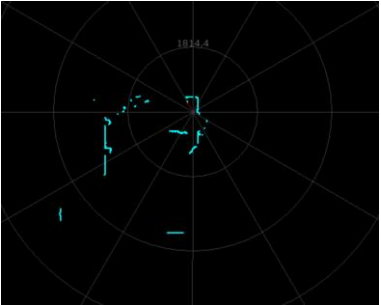
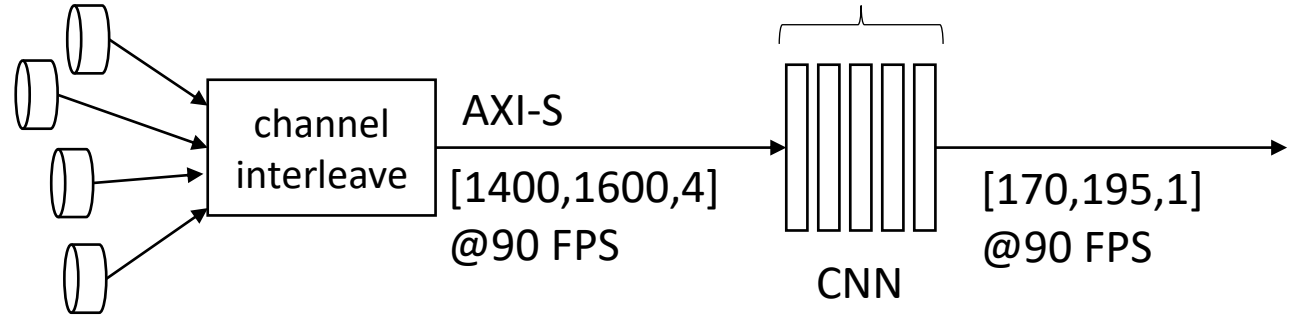
Execution

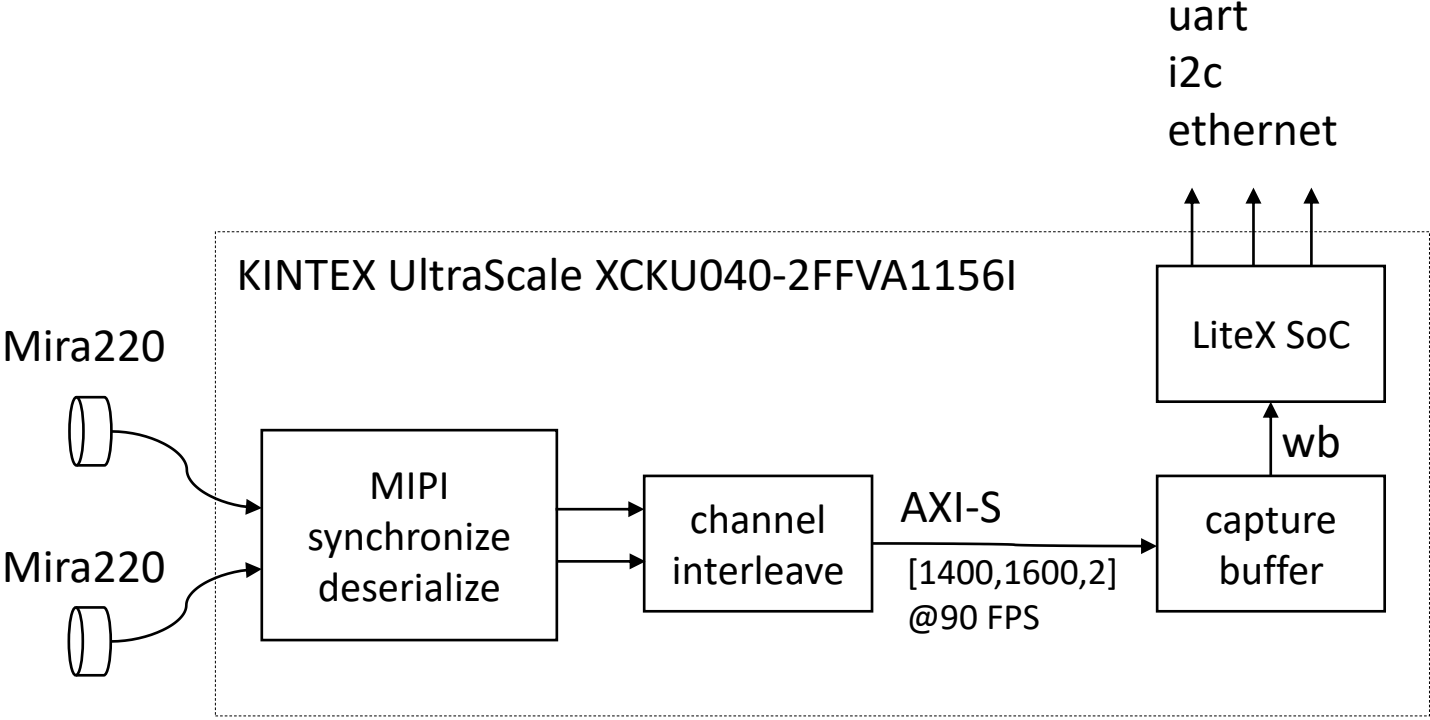
- TF to Verilog demo, real time CNN inference at high resolution and frame rate
- demonstrate vision model using self supervised pretraining on egocentric image data

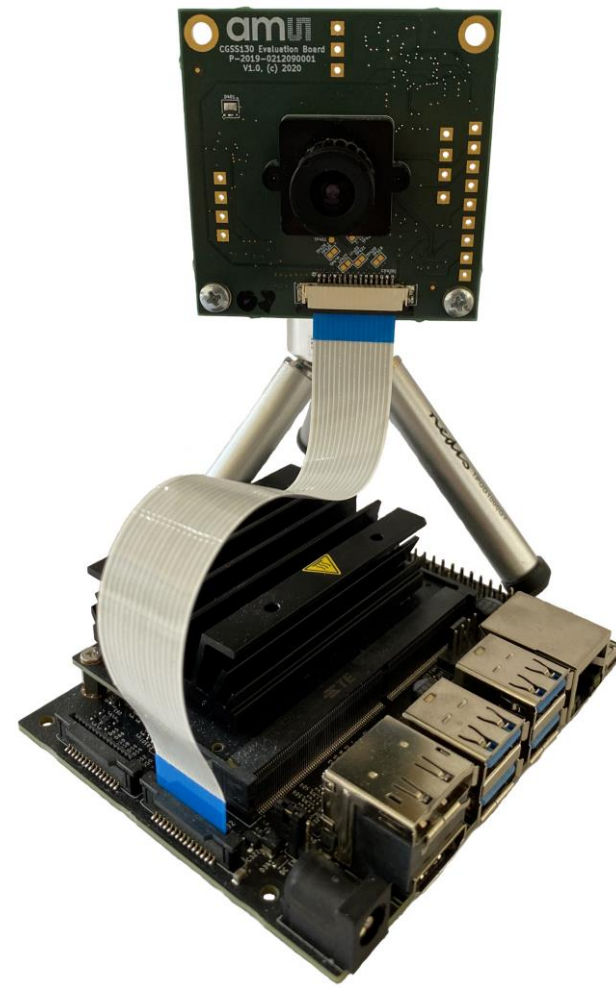


supervised training
predict overhead Lidar view
from 4xMira220

Mira220







<https://www.terasic.com.tw/cgi-bin/page/archive.pl?Language=English&CategoryNo=142&No=1295#contents>

<https://www.xilinx.com/products/boards-and-kits/1-so55i9.html>

FPGA Compute Platform

Vision

- FPGA as a flexible compute platform
 - embedded AI, edge and cloud applications, GPU alternative

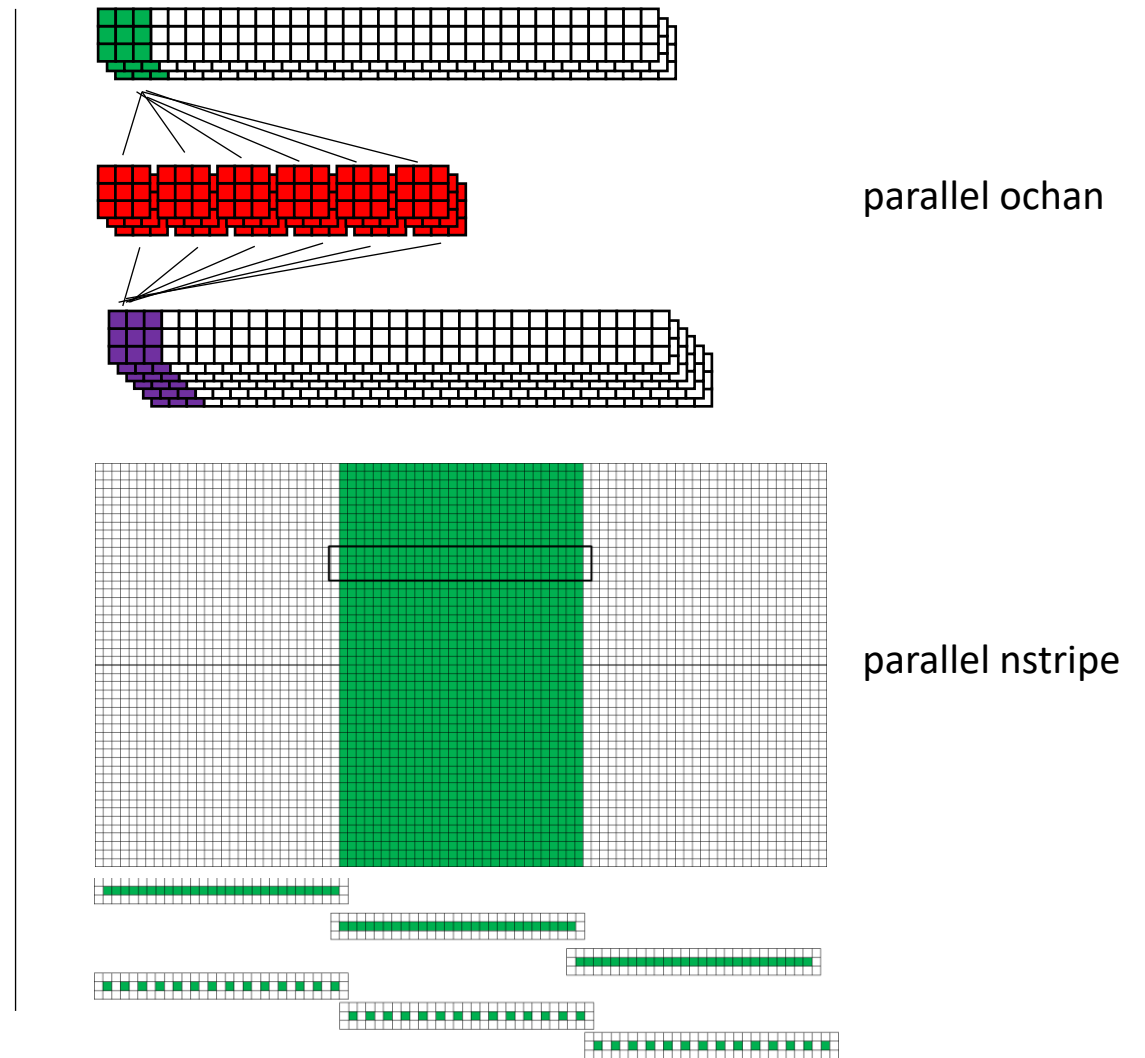
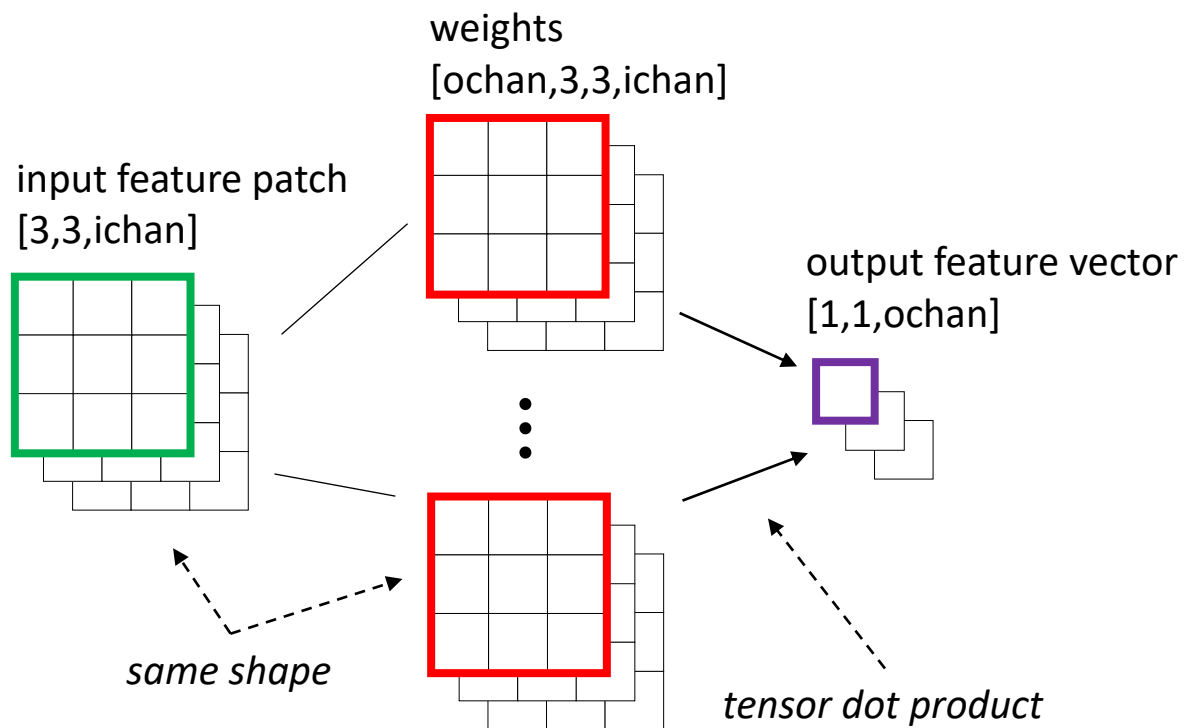
Strategy

- GPT-friendly open IP library
- GPT-automated development flow
 - Raptor, HW/SW co-design (CUDA, RTOS)
- GPT-assisted dataflow algorithm implementation
 - graph based computation for deep models (CNN, Transformer)
 - TF to Verilog
 - traditional DSP pipelines (vision, RF)
 - MATLAB to Verilog
 - multiple architectures (training accelerators, embedded streaming inference)
- Compute-optimized FPGA silicon product line

Execution

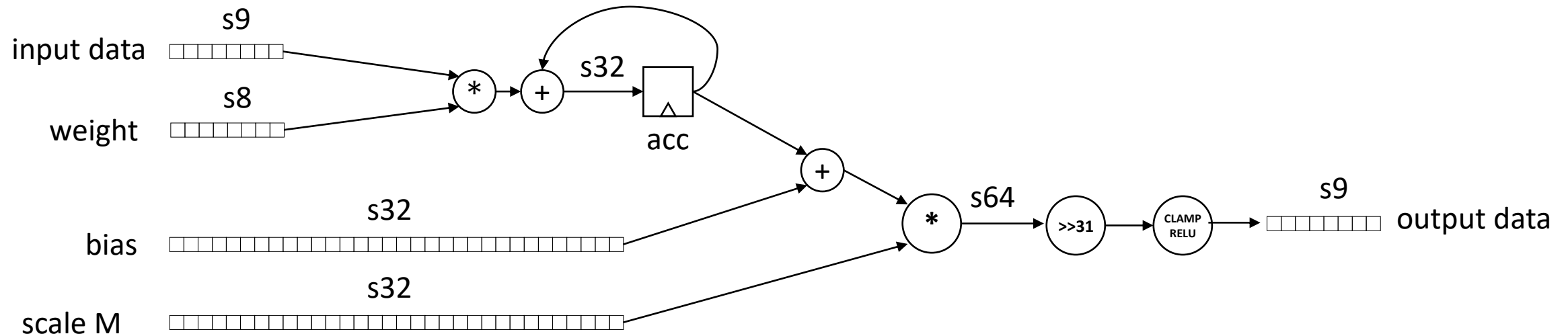
- "hello world" TF to Verilog demo (embedded real time edge inference)

Parallel computation of CNN dot products

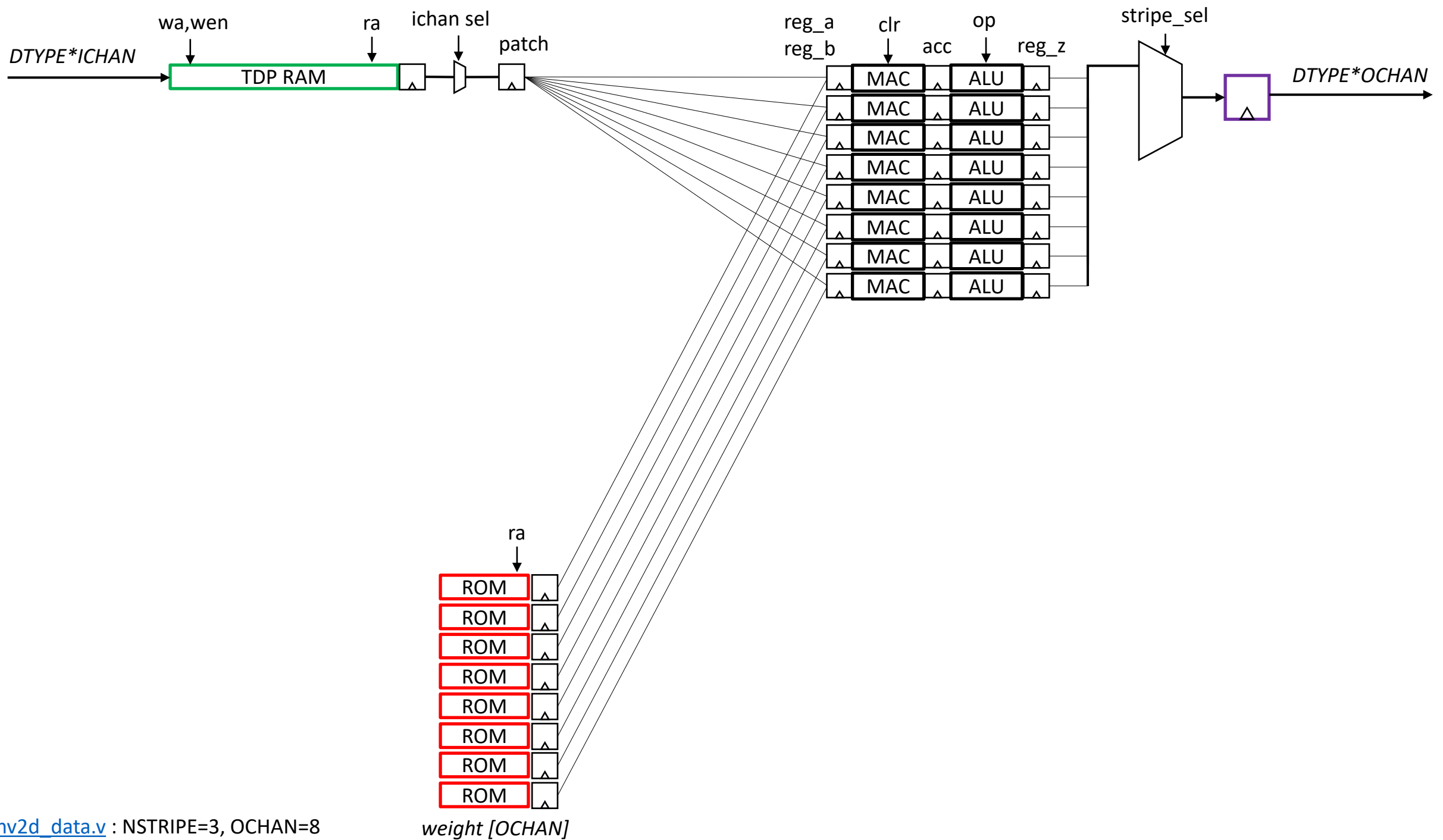


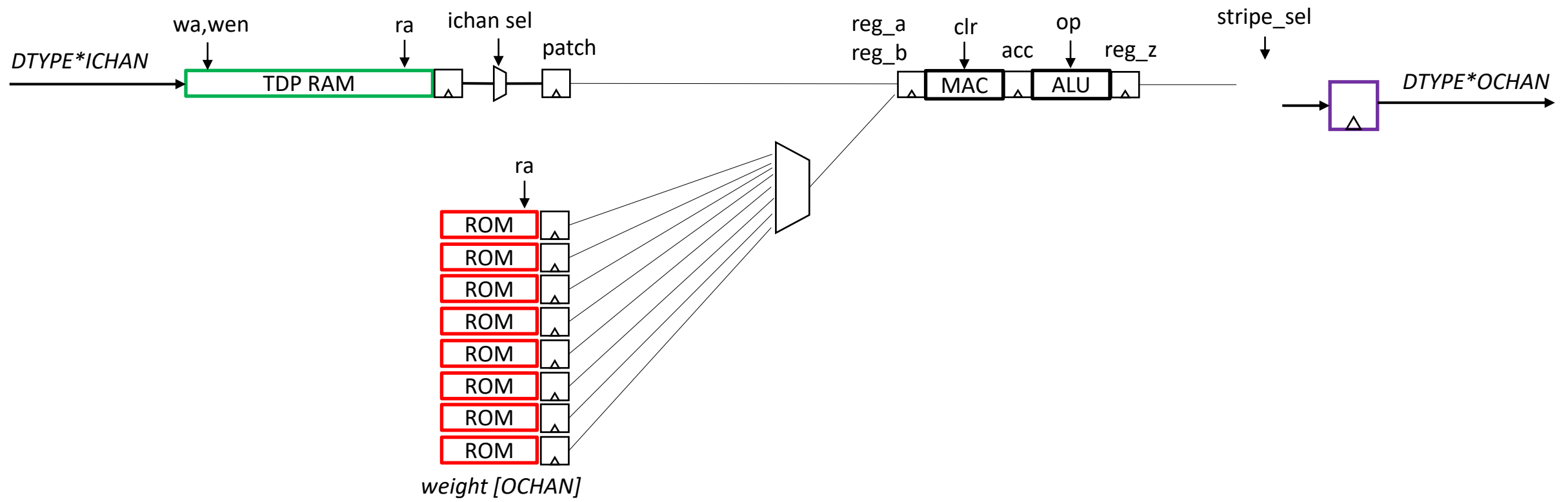
Backup

TFLite int8 quantization



- 1: compute dot product ($s9 * s9 + s32$)
- 2: add bias ($s32 + s32$)
3. multiply by scale ($s32 * s32$)
4. shift right $\gg 31$
5. clamp to s9 bits and RELU





fill row of
padded stripes
from previous
layer

t_{row}

compute NSTRIPES*OCHAN parallel
dot products, quant scale, RELU
stride through input row data

t_{dot}

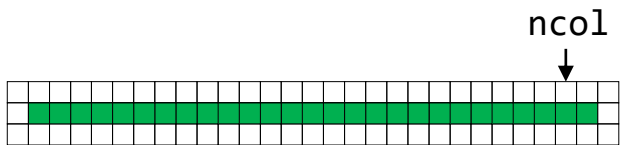
$(\text{owidth}/\text{nstripe}) \cdot t_{\text{dot}} < t_{\text{row}}$

serialize nstripe features over
output TDATA*OCHAN

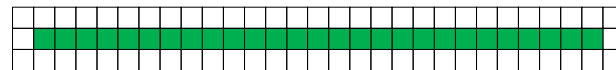
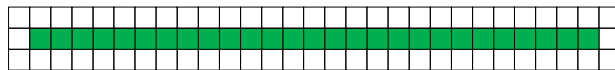
$\text{nstripe} \cdot t_{\text{clk}} < t_{\text{dot}}$

$\text{ndata} = \# \text{ of input features since TLAST (for current layer)}$
 $\text{stripe} = \text{ndata} \% \text{istripe}$

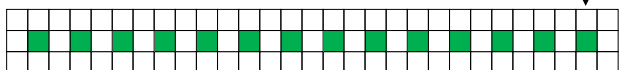
owidth=27*3
stride=1



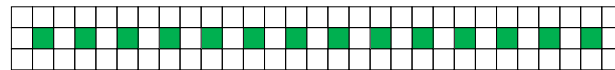
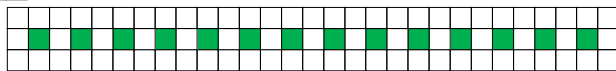
ncol



owidth=14*3
stride=2

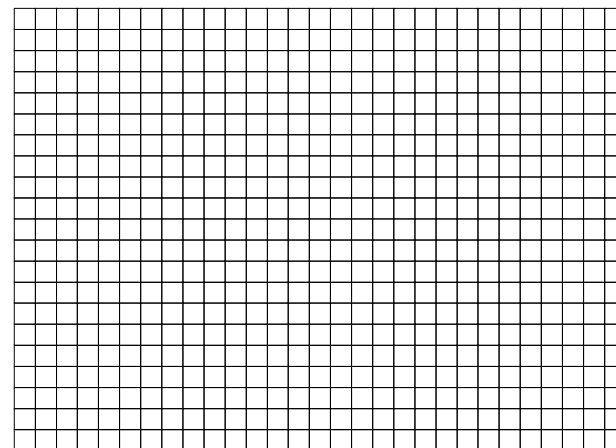


ncol



col

row



t3.py (4K video @30fps)

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 2158, 3838, 8)	224
conv2d_1 (Conv2D)	(None, 1078, 1918, 8)	584
conv2d_2 (Conv2D)	(None, 1076, 1916, 16)	1168
conv2d_3 (Conv2D)	(None, 537, 957, 16)	2320
conv2d_4 (Conv2D)	(None, 535, 955, 32)	4640
conv2d_5 (Conv2D)	(None, 267, 477, 32)	9248
conv2d_6 (Conv2D)	(None, 265, 475, 32)	9248
conv2d_7 (Conv2D)	(None, 132, 237, 32)	9248
conv2d_8 (Conv2D)	(None, 130, 235, 32)	9248
conv2d_9 (Conv2D)	(None, 64, 117, 32)	9248
conv2d_10 (Conv2D)	(None, 62, 115, 32)	9248
conv2d_11 (Conv2D)	(None, 30, 57, 32)	9248
conv2d_12 (Conv2D)	(None, 28, 55, 32)	9248
conv2d_13 (Conv2D)	(None, 13, 27, 32)	9248
conv2d_14 (Conv2D)	(None, 11, 25, 16)	4624
conv2d_15 (Conv2D)	(None, 5, 12, 16)	2320

conv2d_16 (Conv2D)	(None, 3, 10, 8)	1160
conv2d_17 (Conv2D)	(None, 1, 4, 8)	584
flatten (Flatten)	(None, 32)	0

=====
Total params: 100,856
Trainable params: 100,856
Non-trainable params: 0

Ref Name	Used	Functional Category
LUT6	18400	LUT
FDRE	12853	Flop & Latch
MUXF7	5218	MuxFx
RAMD64E	2112	Distributed Memory
RAMB36E1	1851	Block Memory
CARRY4	1568	CarryLogic
LUT4	1517	LUT
DSP48E1	1032	Block Arithmetic
LUT5	807	LUT
LUT2	249	LUT
FDSE	146	Flop & Latch
LUT1	131	LUT
LUT3	117	LUT
OBUF	66	IO
IBUF	26	IO
MUXF8	15	MuxFx
OBUFFT	1	IO
BUFG	1	Clock

andromeda.py log file

```
rliston@uscompute1:~/andromeda$ python3 andromeda.py --model t3 --fps 30
```

```
numpy 1.23.5
```

```
Namespace(model='t3', clk=500000000.0, fps=30.0, tdata=8)
```

layer	0	nstripe	14	stride	1	sdepth	3291	wdepth	31	rate	5.375e+10	nmac	108	shapes	[	1	2160	3840	3]	[	1	2158	3838	8]	[8	3	3	3]	[8]
layer	1	nstripe	9	stride	2	sdepth	17057	wdepth	76	rate	3.578e+10	nmac	72	shapes	[	1	2158	3838	8]	[	1	1078	1918	8]	[8	3	3	8]	[8]
layer	2	nstripe	9	stride	1	sdepth	6819	wdepth	76	rate	7.146e+10	nmac	143	shapes	[	1	1078	1918	8]	[	1	1076	1916	16]	[16	3	3	8]	[16]
layer	3	nstripe	5	stride	2	sdepth	30656	wdepth	148	rate	3.562e+10	nmac	72	shapes	[	1	1076	1916	16]	[	1	537	957	16]	[16	3	3	16]	[16]
layer	4	nstripe	5	stride	1	sdepth	12249	wdepth	148	rate	7.104e+10	nmac	143	shapes	[	1	537	957	16]	[	1	535	955	32]	[32	3	3	16]	[32]
layer	5	nstripe	3	stride	2	sdepth	50933	wdepth	292	rate	3.532e+10	nmac	71	shapes	[	1	535	955	32]	[	1	267	477	32]	[32	3	3	32]	[32]
layer	6	nstripe	3	stride	1	sdepth	20352	wdepth	292	rate	3.521e+10	nmac	71	shapes	[	1	267	477	32]	[	1	265	475	32]	[32	3	3	32]	[32]
layer	7	nstripe	1	stride	2	sdepth	76000	wdepth	292	rate	8.700e+09	nmac	18	shapes	[	1	265	475	32]	[	1	132	237	32]	[32	3	3	32]	[32]
layer	8	nstripe	1	stride	1	sdepth	30336	wdepth	292	rate	8.649e+09	nmac	18	shapes	[	1	132	237	32]	[	1	130	235	32]	[32	3	3	32]	[32]
layer	9	nstripe	1	stride	2	sdepth	37600	wdepth	292	rate	2.112e+09	nmac	5	shapes	[	1	130	235	32]	[	1	64	117	32]	[32	3	3	32]	[32]
layer	10	nstripe	1	stride	1	sdepth	14976	wdepth	292	rate	2.070e+09	nmac	5	shapes	[	1	64	117	32]	[	1	62	115	32]	[32	3	3	32]	[32]
layer	11	nstripe	1	stride	2	sdepth	18400	wdepth	292	rate	4.928e+08	nmac	1	shapes	[	1	62	115	32]	[	1	30	57	32]	[32	3	3	32]	[32]
layer	12	nstripe	1	stride	1	sdepth	7296	wdepth	292	rate	4.728e+08	nmac	1	shapes	[	1	30	57	32]	[	1	28	55	32]	[32	3	3	32]	[32]
layer	13	nstripe	1	stride	2	sdepth	8800	wdepth	292	rate	1.064e+08	nmac	1	shapes	[	1	28	55	32]	[	1	13	27	32]	[32	3	3	32]	[32]
layer	14	nstripe	1	stride	1	sdepth	3456	wdepth	292	rate	4.852e+07	nmac	1	shapes	[	1	13	27	32]	[	1	11	25	16]	[16	3	3	32]	[16]
layer	15	nstripe	1	stride	2	sdepth	2000	wdepth	148	rate	4.752e+06	nmac	1	shapes	[	1	11	25	16]	[	1	5	12	16]	[16	3	3	16]	[16]
layer	16	nstripe	1	stride	1	sdepth	768	wdepth	148	rate	2.074e+06	nmac	1	shapes	[	1	5	12	16]	[	1	3	10	8]	[8	3	3	16]	[8]
layer	17	nstripe	1	stride	2	sdepth	400	wdepth	76	rate	1.296e+05	nmac	1	shapes	[	1	3	10	8]	[	1	1	4	8]	[8	3	3	8]	[8]

```
total stripe RAM bits    147841520
```

```
total weight RAM bits    816832
```

```
total required MAC units    733
```

```
total used MAC units    1032
```

```
rliston@uscompute1:~/andromeda$
```

3. Memory

Site Type	Used	Fixed	Prohibited	Available	Util%
Block RAM Tile	1851	0	0	1880	98.46
RAMB36/FIFO*	1851	0	0	1880	98.46
RAMB36E1 only	1851				
RAMB18	0	0	0	3760	0.00

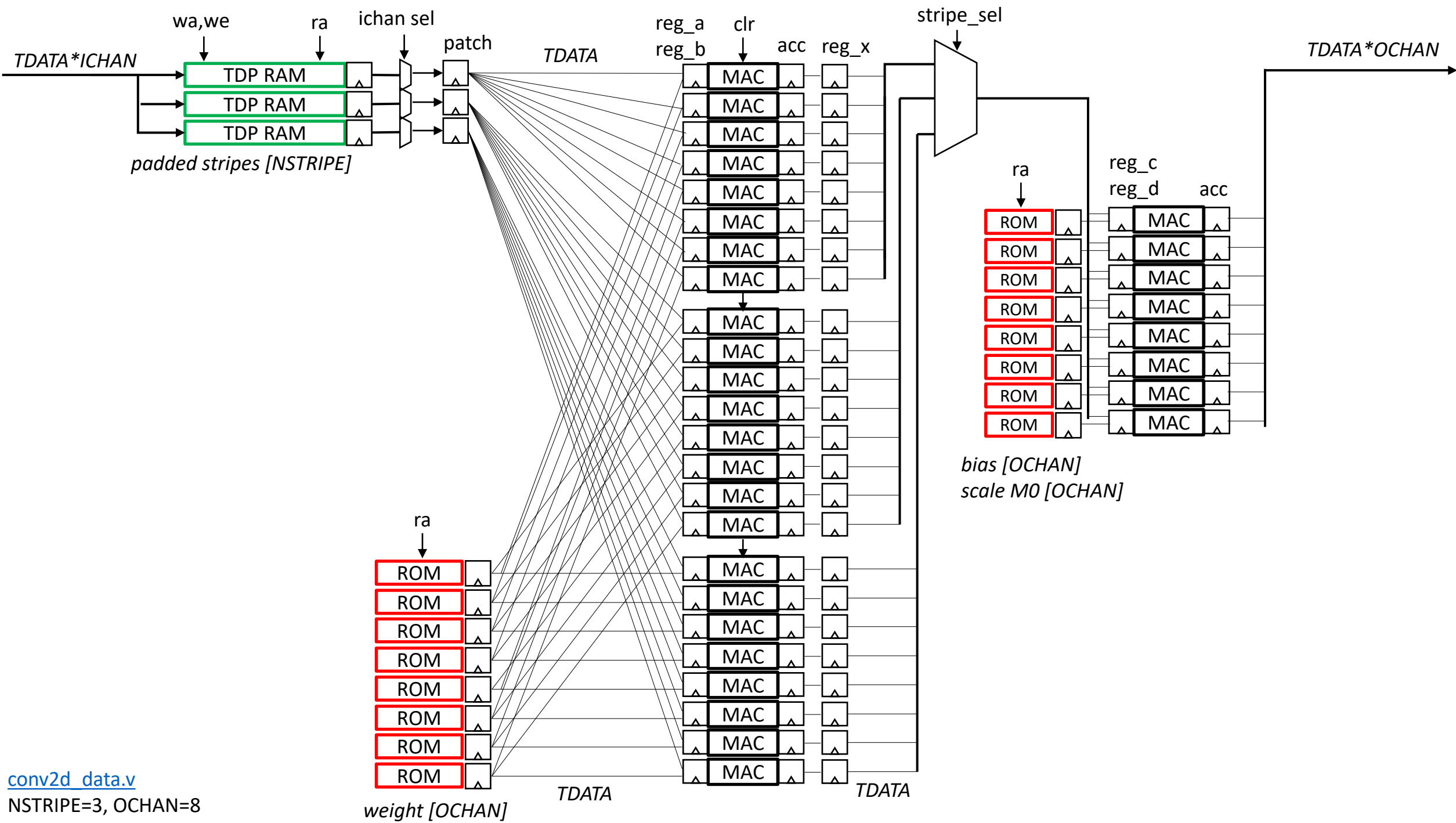
* Note: Each Block RAM Tile only has one FIFO logic available and therefore can accommodate only one FIFO36E1 or one FIFO18E1. However, if a FIFO18E1 occupies a Block RAM Tile, that tile can still accommodate a RAMB18E1

4. DSP

Site Type	Used	Fixed	Prohibited	Available	Util%
DSPs	1032	0	0	3360	30.71
DSP48E1 only	1032				

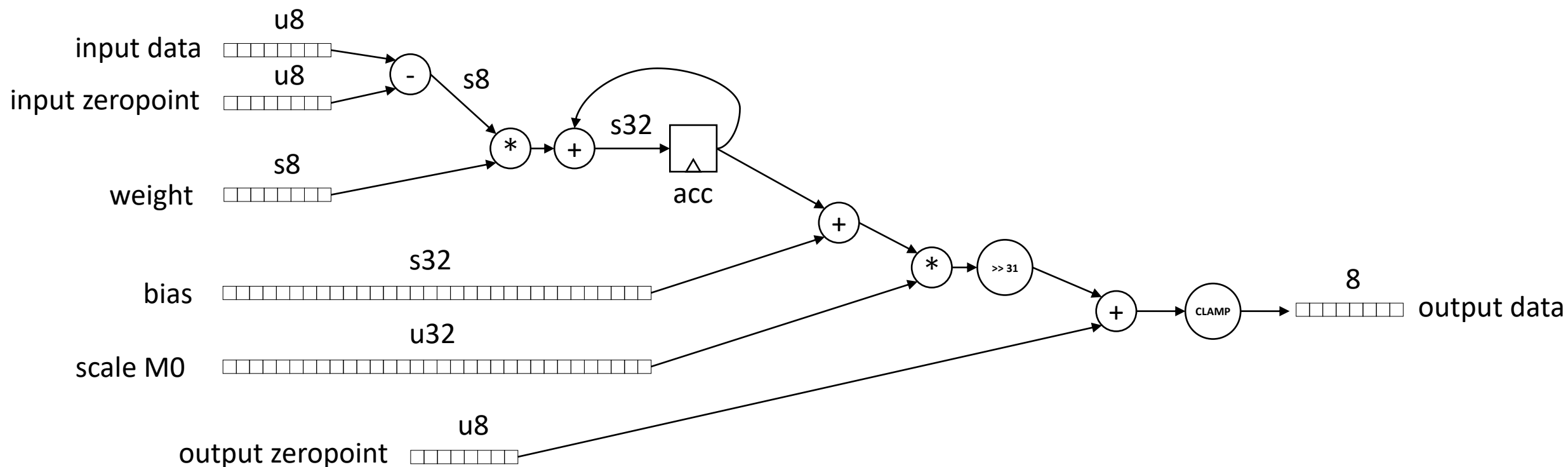
Target applications

- high speed, low latency inference at the edge
 - ex: 3840x2160 @100 fps image classification for industrial applications
 - process raw uncompressed pixels directly from sensor (e.g. HDMI, MIPI)
 - hyperspectral image sensors, RF IQ baseband, 300+ frames/s
- inference only, supports small but useful subset of TFLite (conv2d only, padding='valid', stride={1,2}, activation=RELU)
- no external memory required
- real time data stream -> real time prediction stream (or feature map)
- tool to compile TFLite model into synthesizable Verilog
- generated Verilog code is self contained, includes streaming inference and weights
- compiler generates parallel code for OCHAN*NSTRIPE MAC operations
- compiler generates ROM file with the weights



[conv2d_data.v](#)

NSTRIPE=3, OCHAN=8

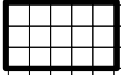


- 1: compute dot product (8*8+32)
- 2: add bias (32+32)
3. multiply by M0 scale (32*32)
4. shift right 31 bits with round
5. clamp to 8 bits + RELU

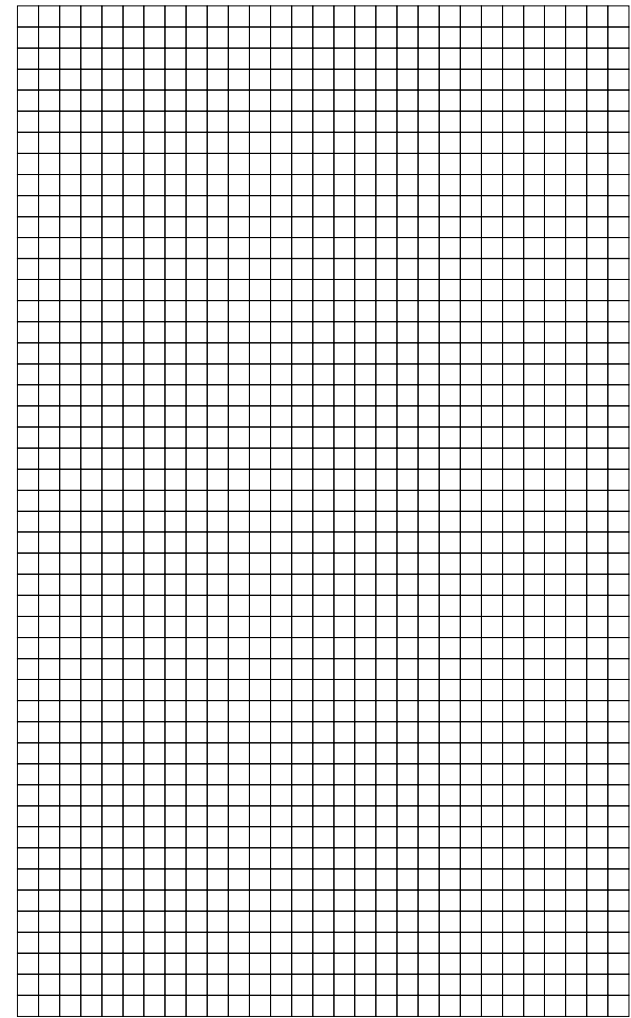
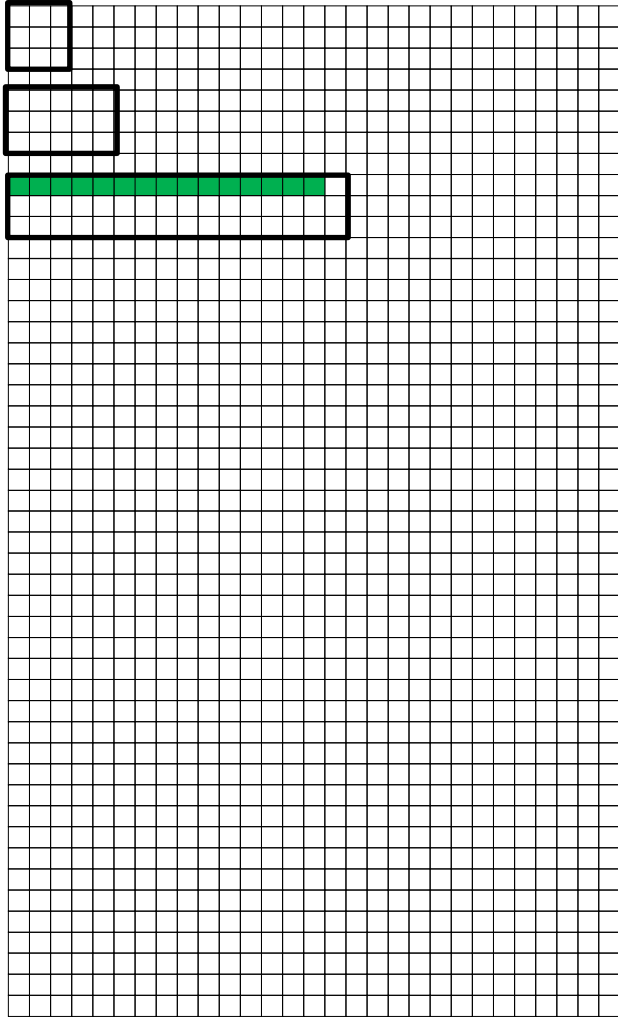
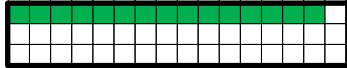
nstripes=1



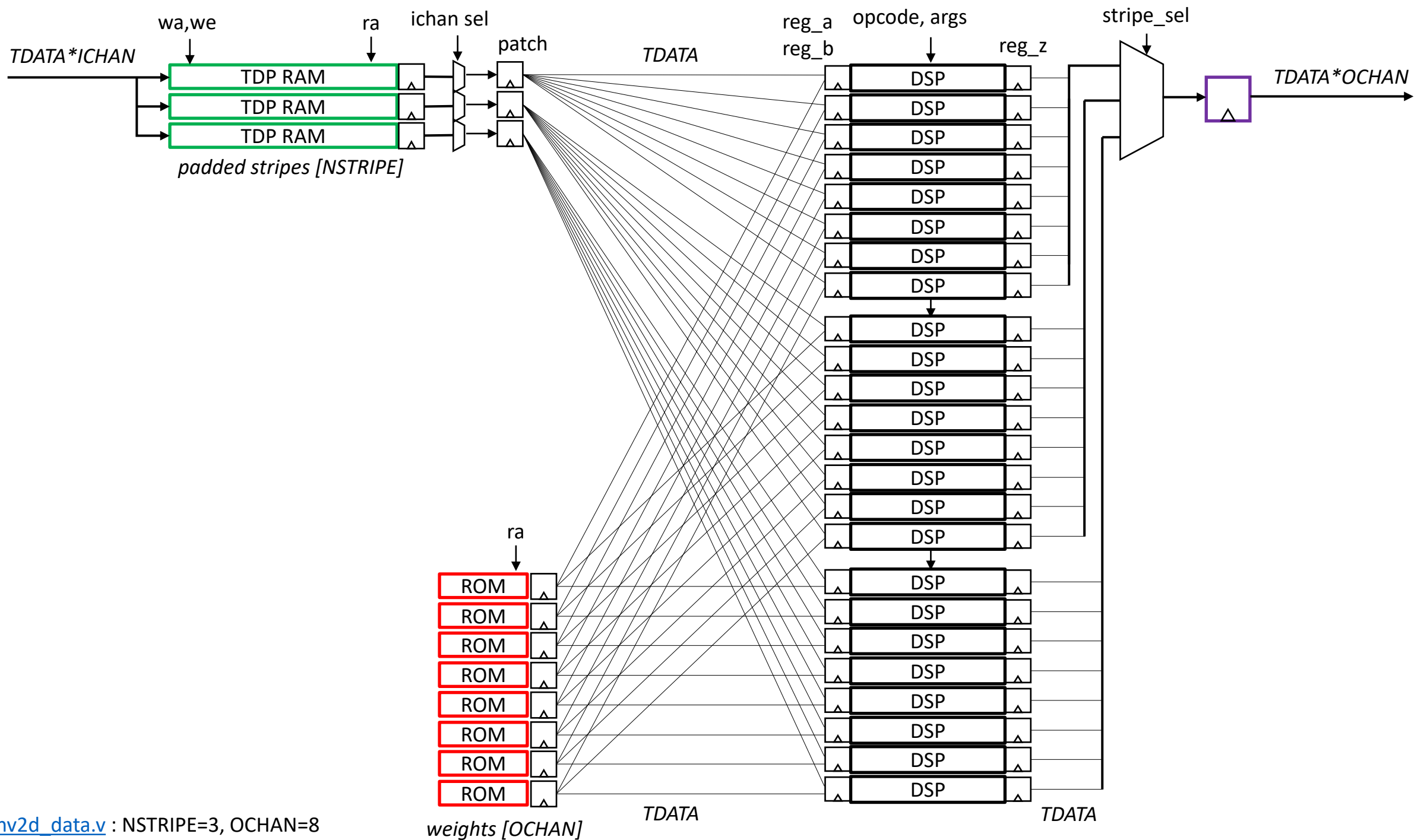
nstripes=3

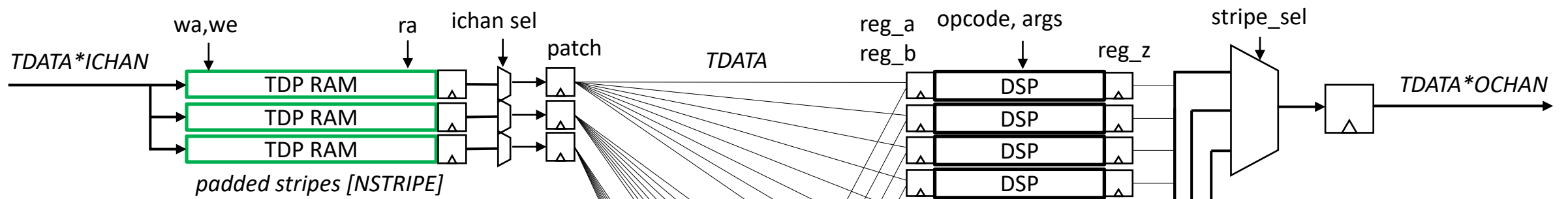


nstripes=15



fps	100														
width	3,840														
height	2,160														
channels	3														
clk	500,000,000														
layer	iwidth	iheight	ichan	stride	kheight	kwidth	owidth	oheight	ochan	patches/s	MAC/patch	MAC/s	required MAC	nstripe	nstripe*ochan
1	3,840	2160	3	1	3	3	3,838	2,158	16	829,440,000	432	358,318,080,000	717	45	720
2	3,838	2,158	16	2	3	3	1,918	1,078	16	206,760,400	2,304	476,375,961,600	953	60	960
3	1,918	1,078	16	1	3	3	1,916	1,076	32	206,161,600	4,608	949,992,652,800	1,900	60	1,920
4	1,916	1,076	32	2	3	3	957	537	32	51,390,900	9,216	473,618,534,400	948	30	960
5	957	537	32	1	3	3	955	535	64	51,092,500	18,432	941,736,960,000	1,884	30	1,920
6	955	535	64	2	3	3	477	267	64	12,735,900	36,864	469,496,217,600	939	15	960
7	477	267	64	1	3	3	475	265	64	12,587,500	36,864	464,025,600,000	929	15	960
8	475	265	64	2	3	3	237	132	64	3,128,400	36,864	115,325,337,600	231	4	256
9	237	132	64	1	3	3	235	130	64	3,055,000	36,864	112,619,520,000	226	4	256
10	235	130	64	2	3	3	117	64	64	748,800	36,864	27,603,763,200	56	1	64
11	117	64	64	1	3	3	115	62	64	713,000	36,864	26,284,032,000	53	1	64
12	115	62	64	2	3	3	57	30	64	171,000	36,864	6,303,744,000	13	1	64
13	57	30	64	1	3	3	55	28	64	154,000	36,864	5,677,056,000	12	1	64
14	55	28	64	2	3	3	27	13	64	35,100	36,864	1,293,926,400	3	1	64
15	27	13	64	1	3	3	25	11	32	27,500	18,432	506,880,000	2	1	32
16	25	11	32	2	3	3	12	5	32	6,000	9,216	55,296,000	1	1	32
17	12	5	32	1	3	3	10	3	16	3,000	4,608	13,824,000	1	1	16
18	10	3	16	2	3	3	4	1	16	400	2,304	921,600	1	1	16
												4,429,178,265,600	8,866		9,264

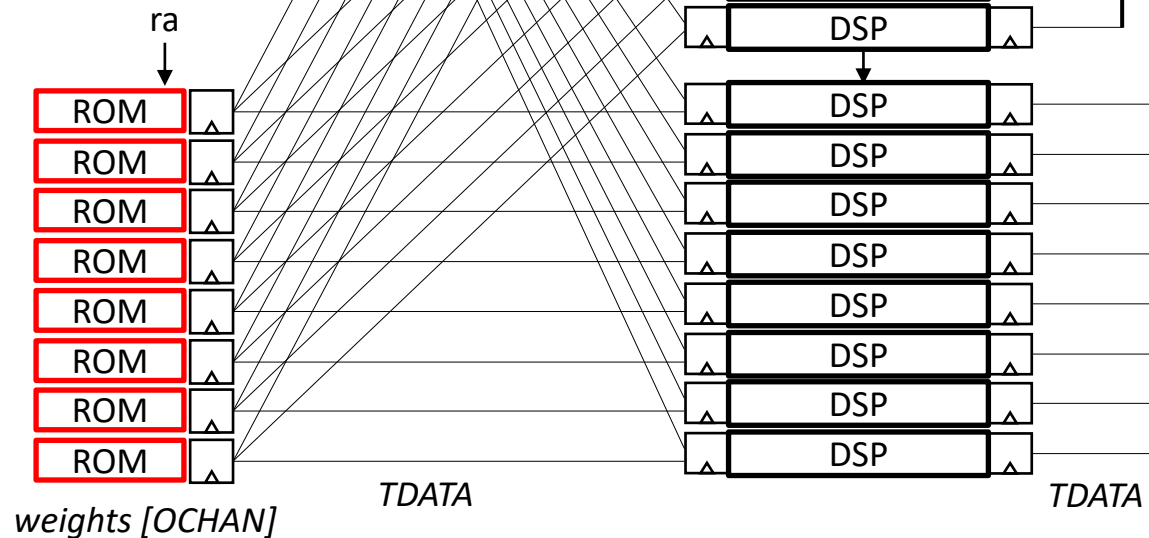




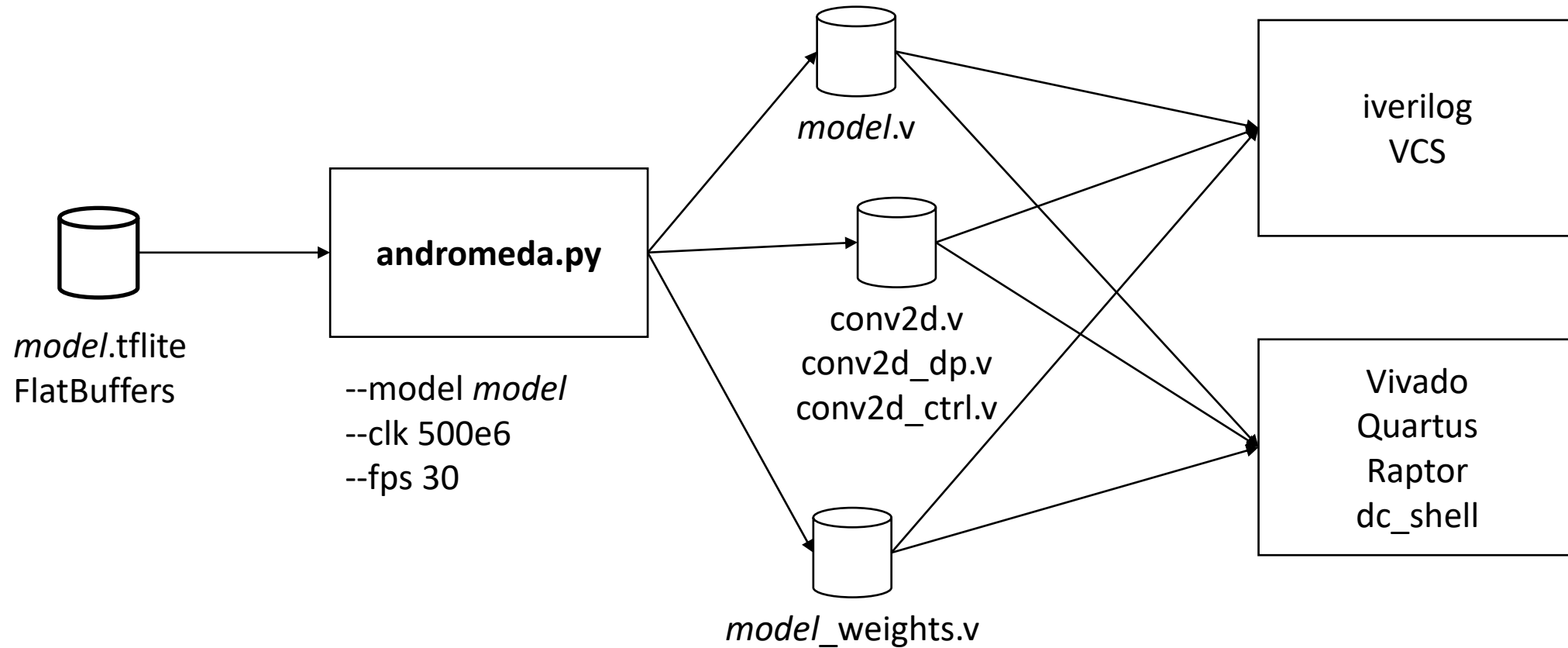
FDRE	3735	Flop & Latch
LUT6	1634	LUT
LUT4	695	LUT
CARRY4	586	CarryLogic
DSP48E1	510	Block Arithmetic
RAMB18E1	181	Block Memory
LUT2	177	LUT
RAMB36E1	105	Block Memory
OBUF	82	IO
LUT1	64	LUT
LUT3	54	LUT
FDSE	39	Flop & Latch
LUT5	24	LUT
IBUF	10	IO
FDCE	6	Flop & Latch
OBUFT	1	IO
BUFG	1	Clock

$TDATA = \{int8, bfloat16\}$
 $ICHAN = \{1..256\}$
 $OCHAN = \{1..256\}$
 $NSTRIPE = \{1..64\}$

[conv2d_dp.v](#) : NSTRIPE=3, OCHAN=8

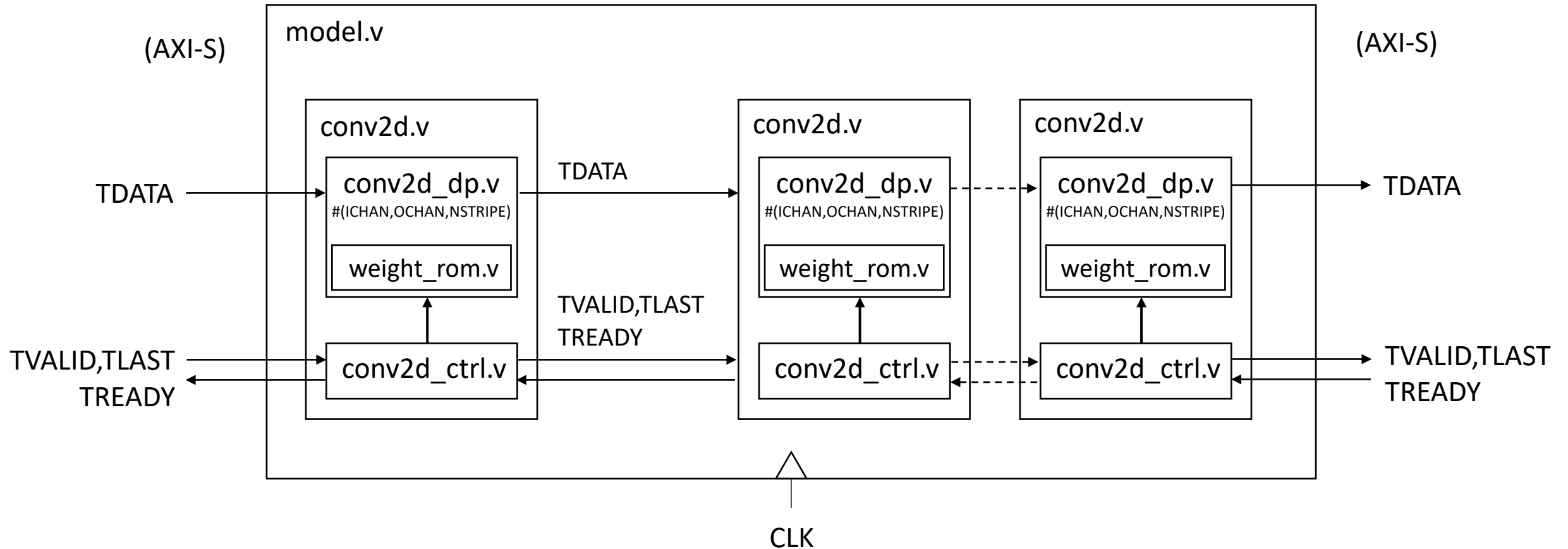


andromeda.py streaming CNN compiler

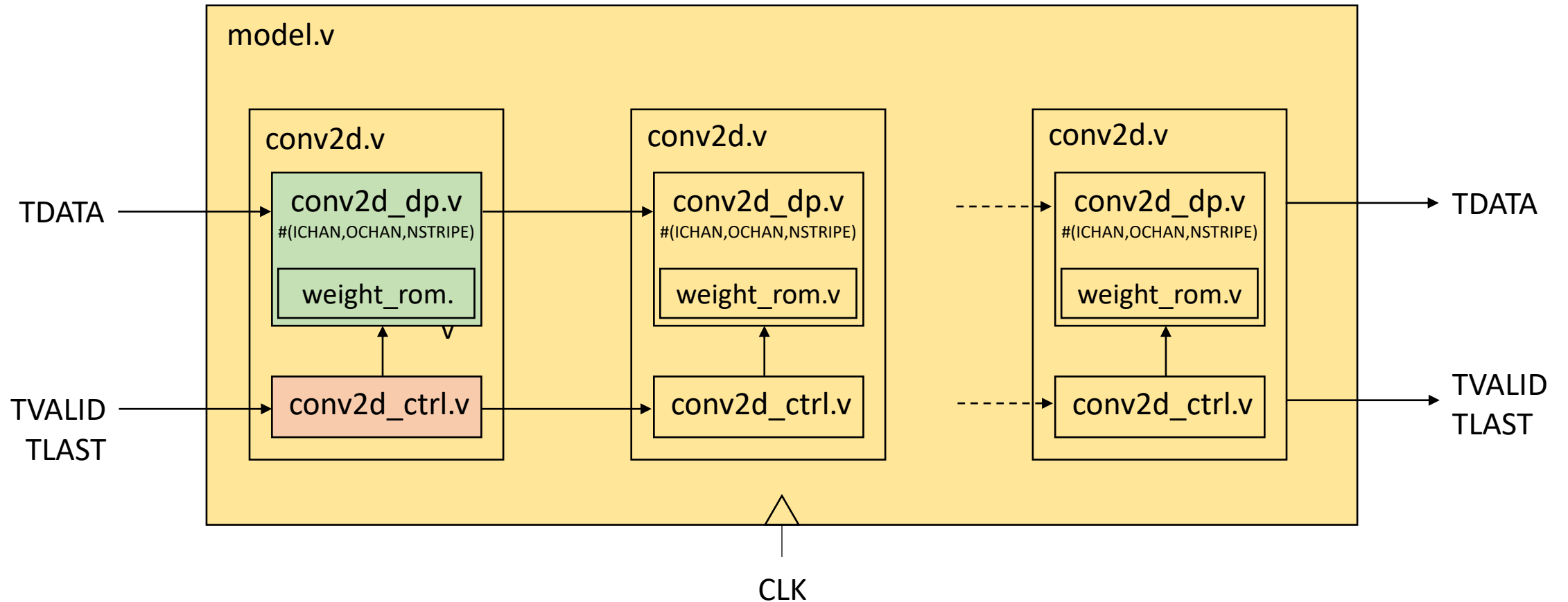


automated tool to compile TFLite model into synthesizable Verilog
compiler generates parallel code for $OCHAN * NSTRIDE$ MAC operations per layer

Streaming CNN Verilog Hierarchy



Streaming CNN Verilog Hierarchy



<https://arxiv.org/pdf/1712.05877.pdf>

https://www.tensorflow.org/lite/performance/quantization_spec#int8_quantized_operator_specifications

requirements for TFLite quant:

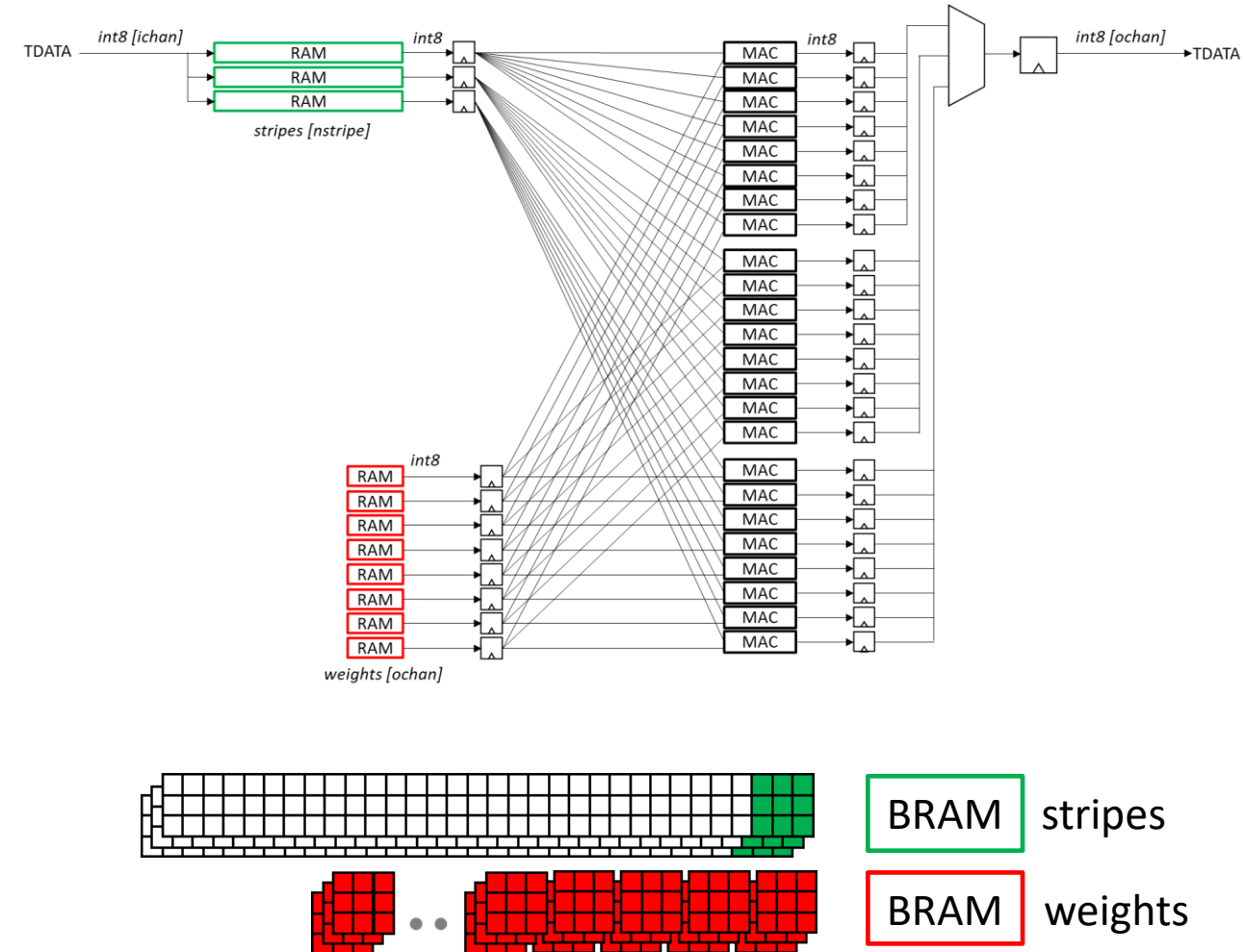
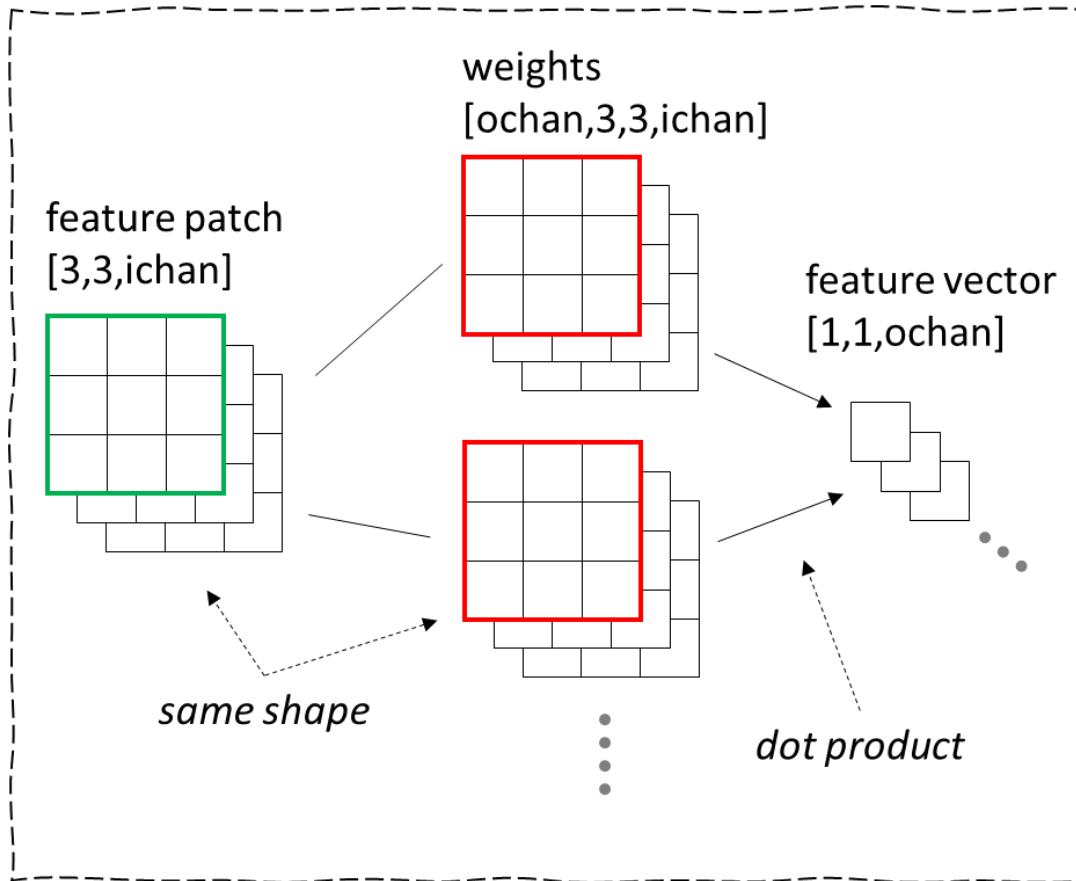
1. saturating $8 \times 8 + 32$ bit MAC
2. rounding-to-nearest right-shift

https://en.wikipedia.org/wiki/Wafer-scale_integration

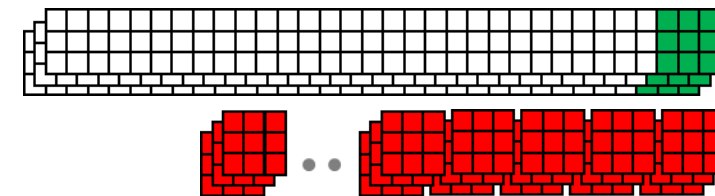
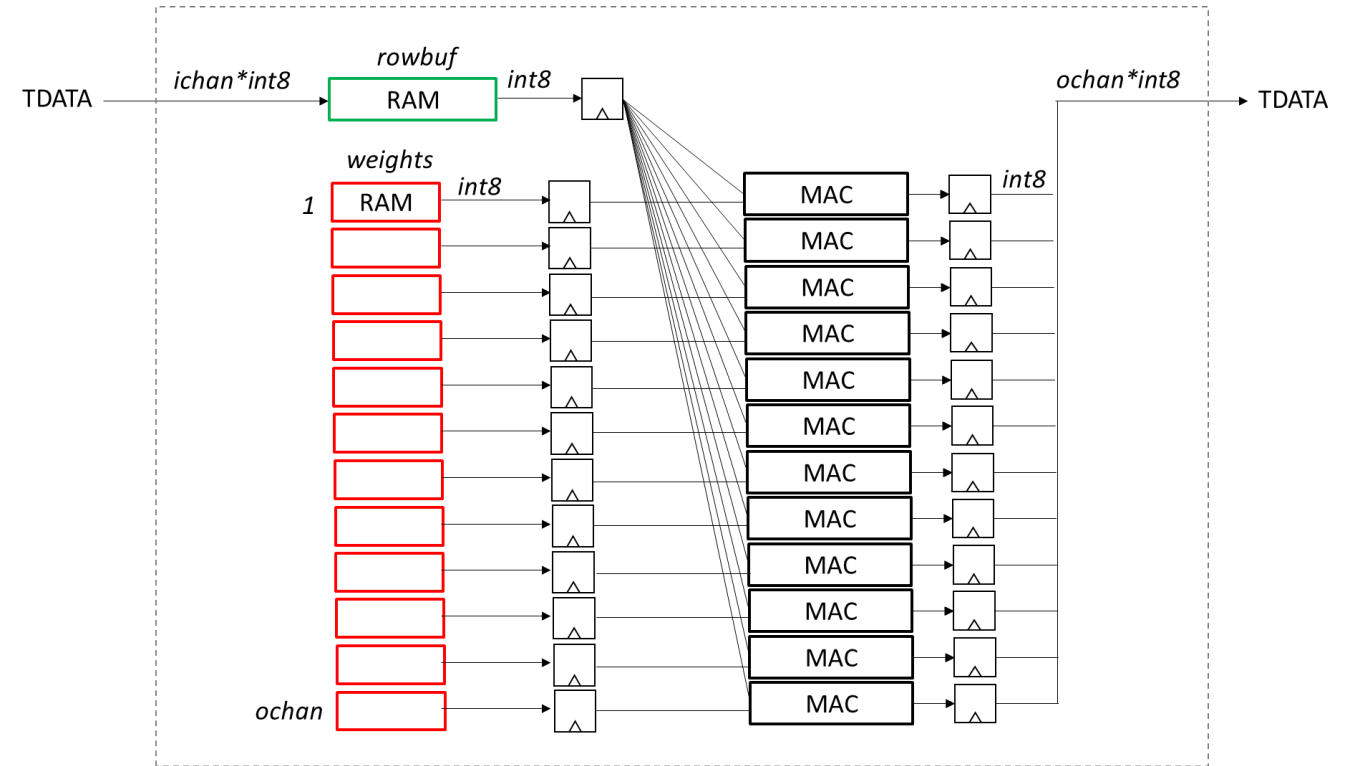
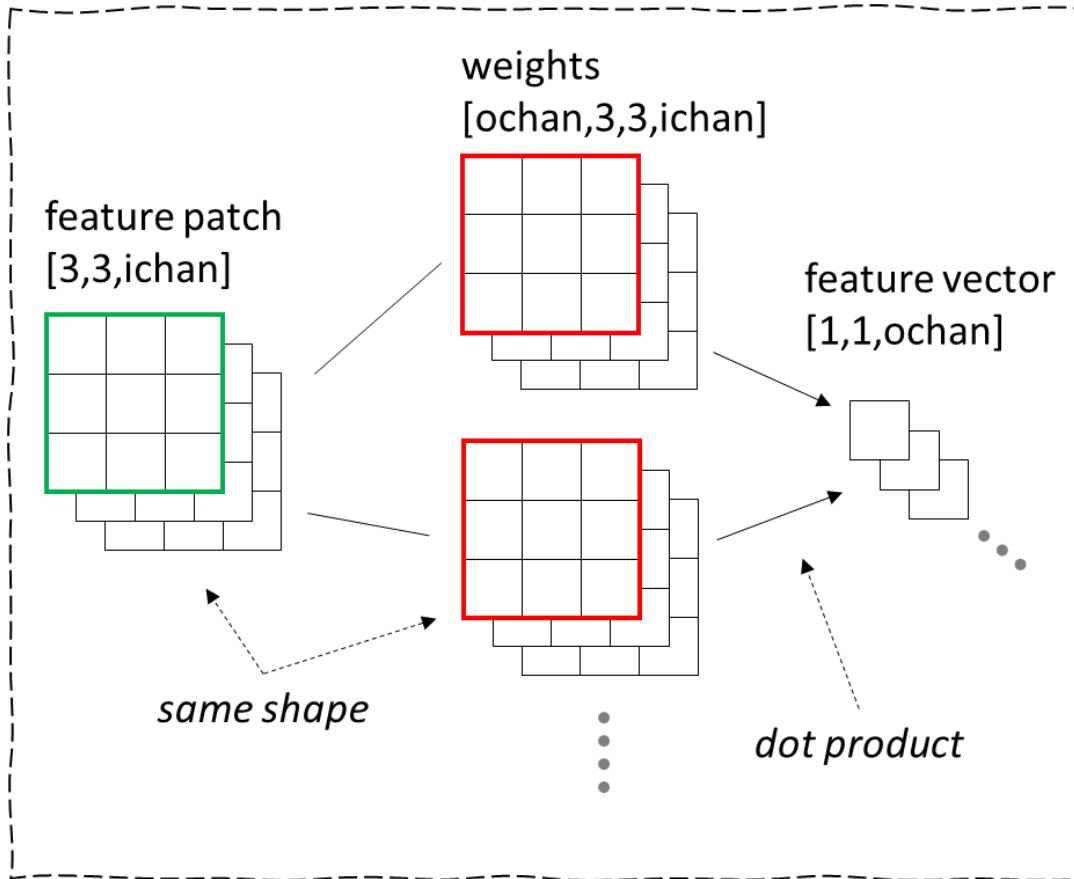
<https://en.wikipedia.org/wiki/Cerebras>

<https://arxiv.org/pdf/2112.07571.pdf>

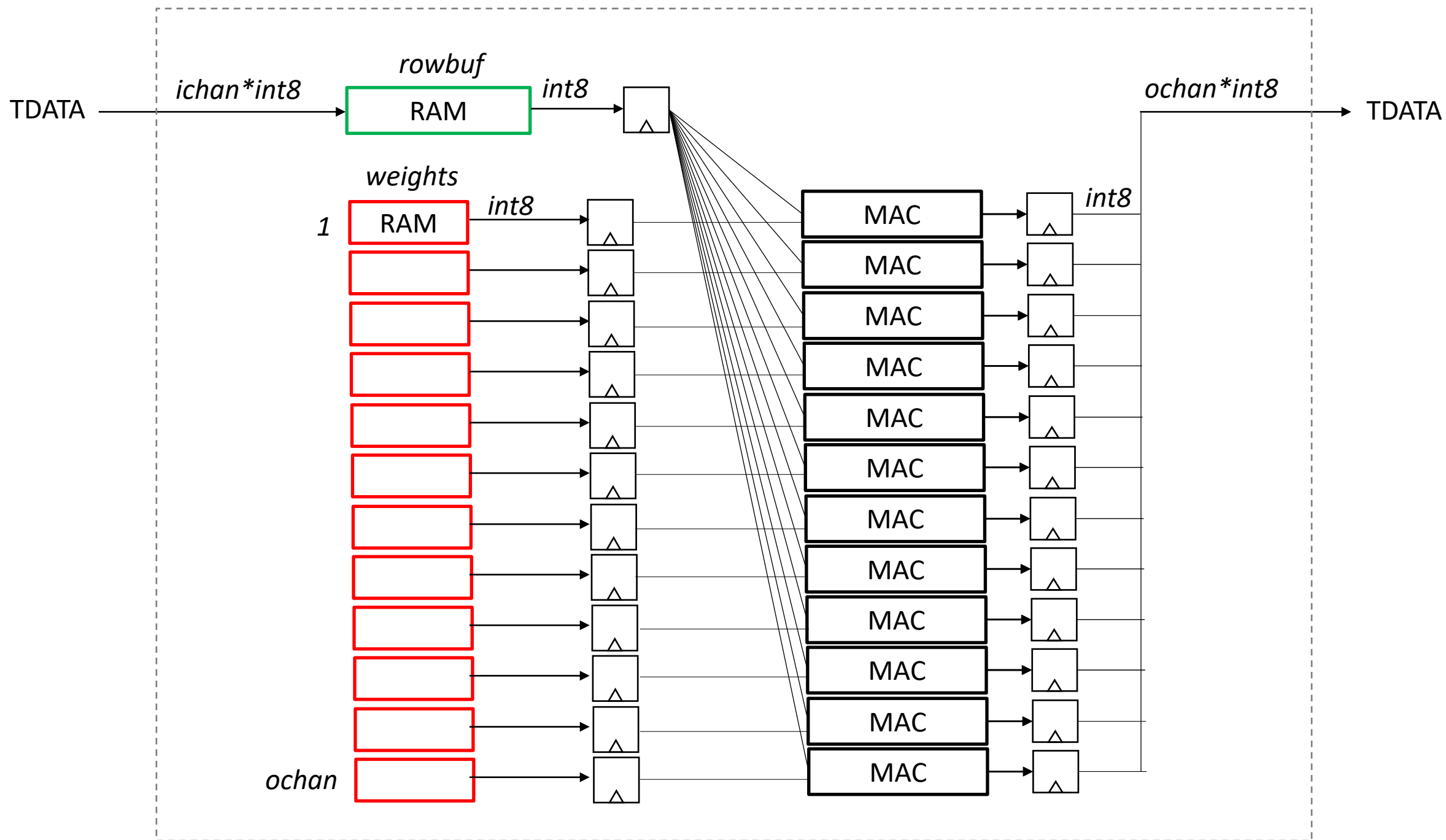
CNN dot product using FPGA

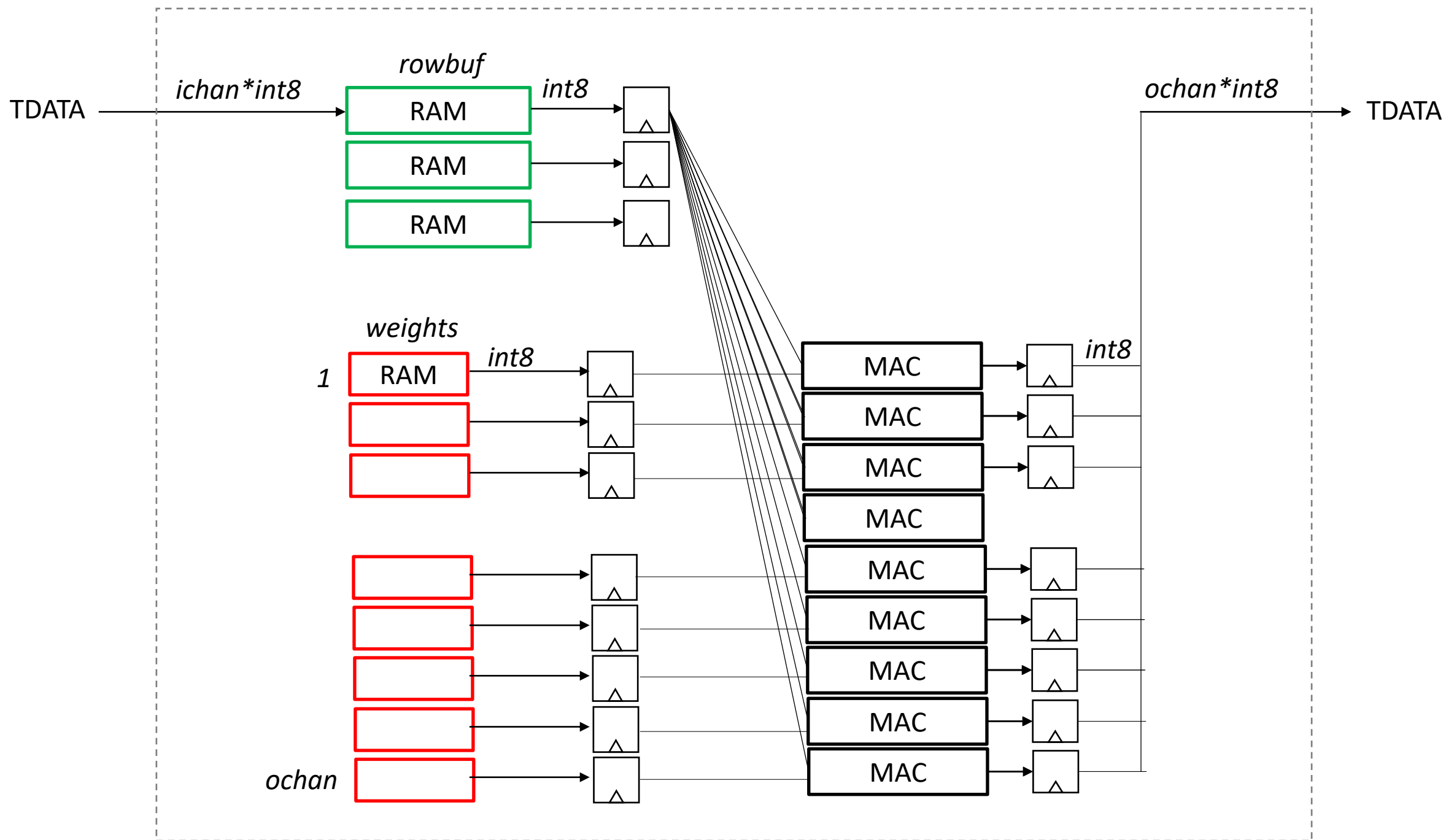


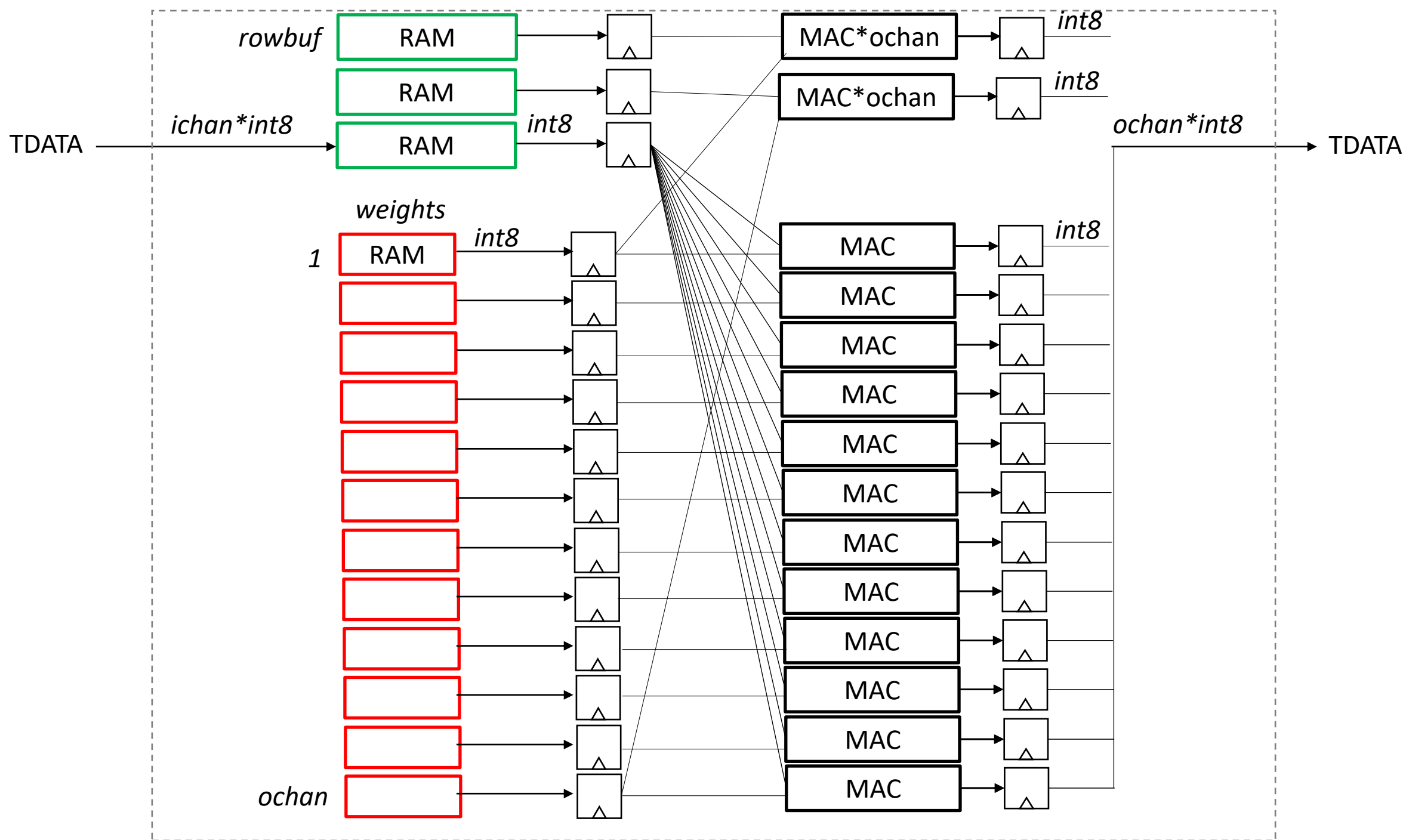
CNN dot product using FPGA

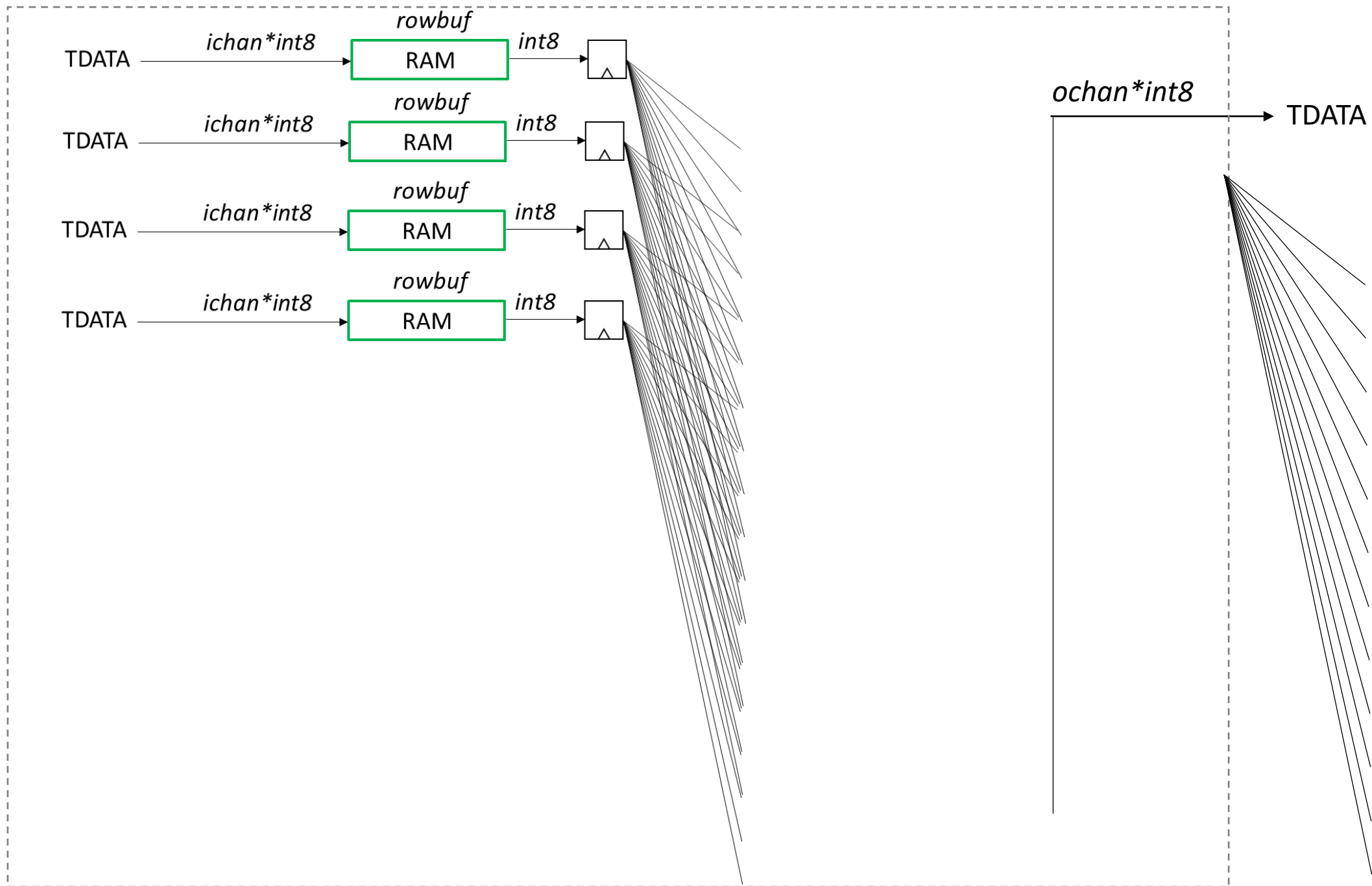


BRAM row buffer
BRAM weights

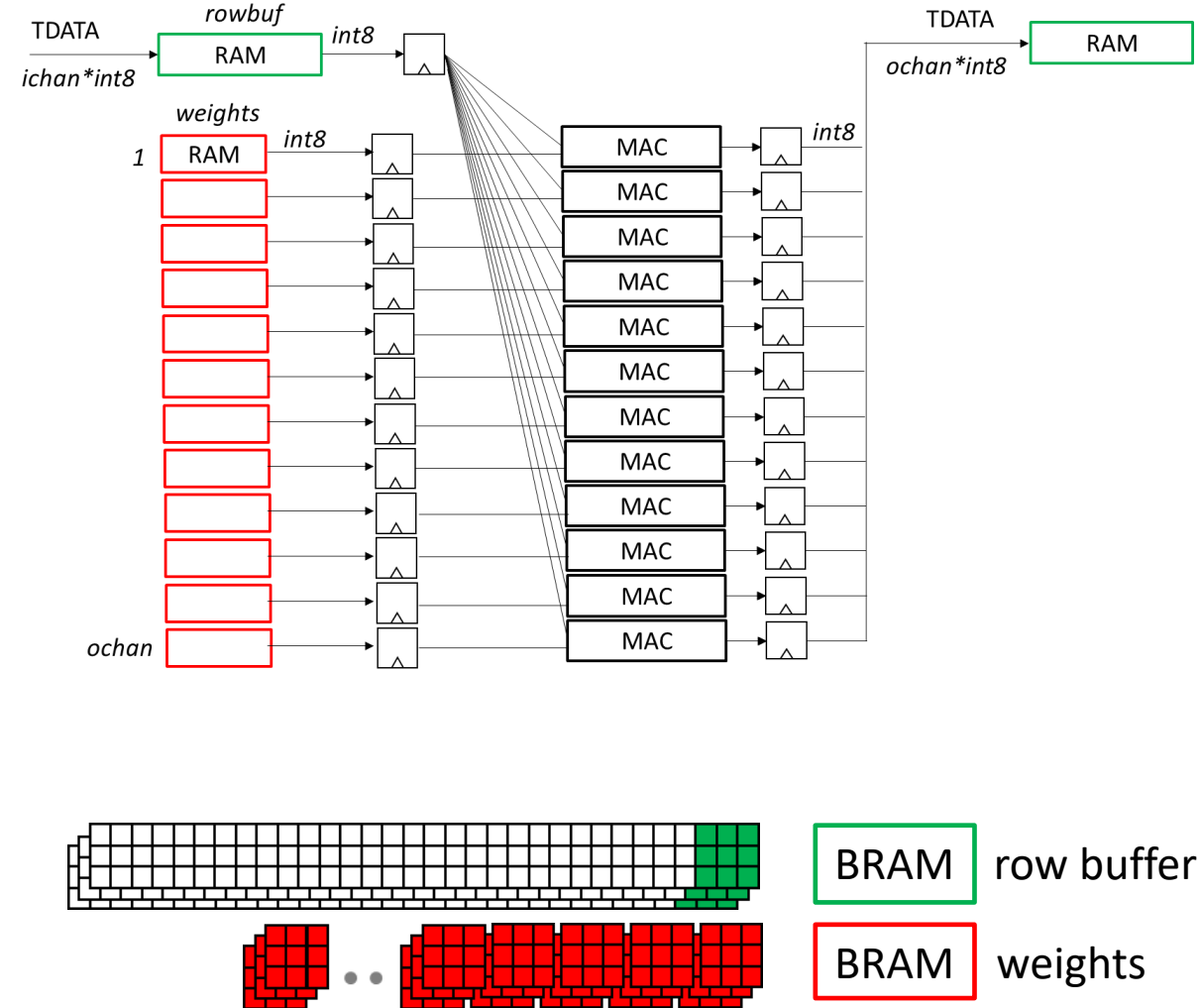
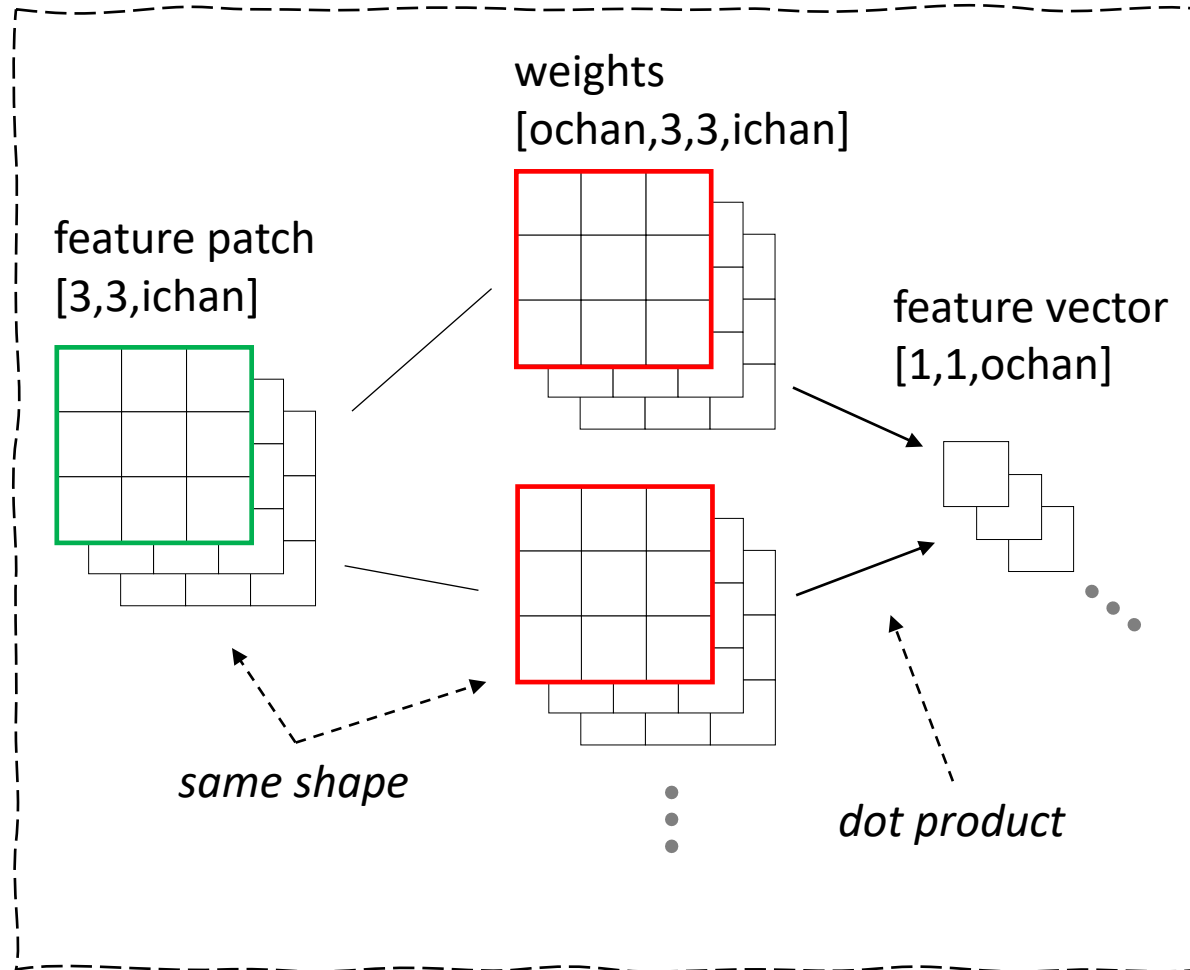


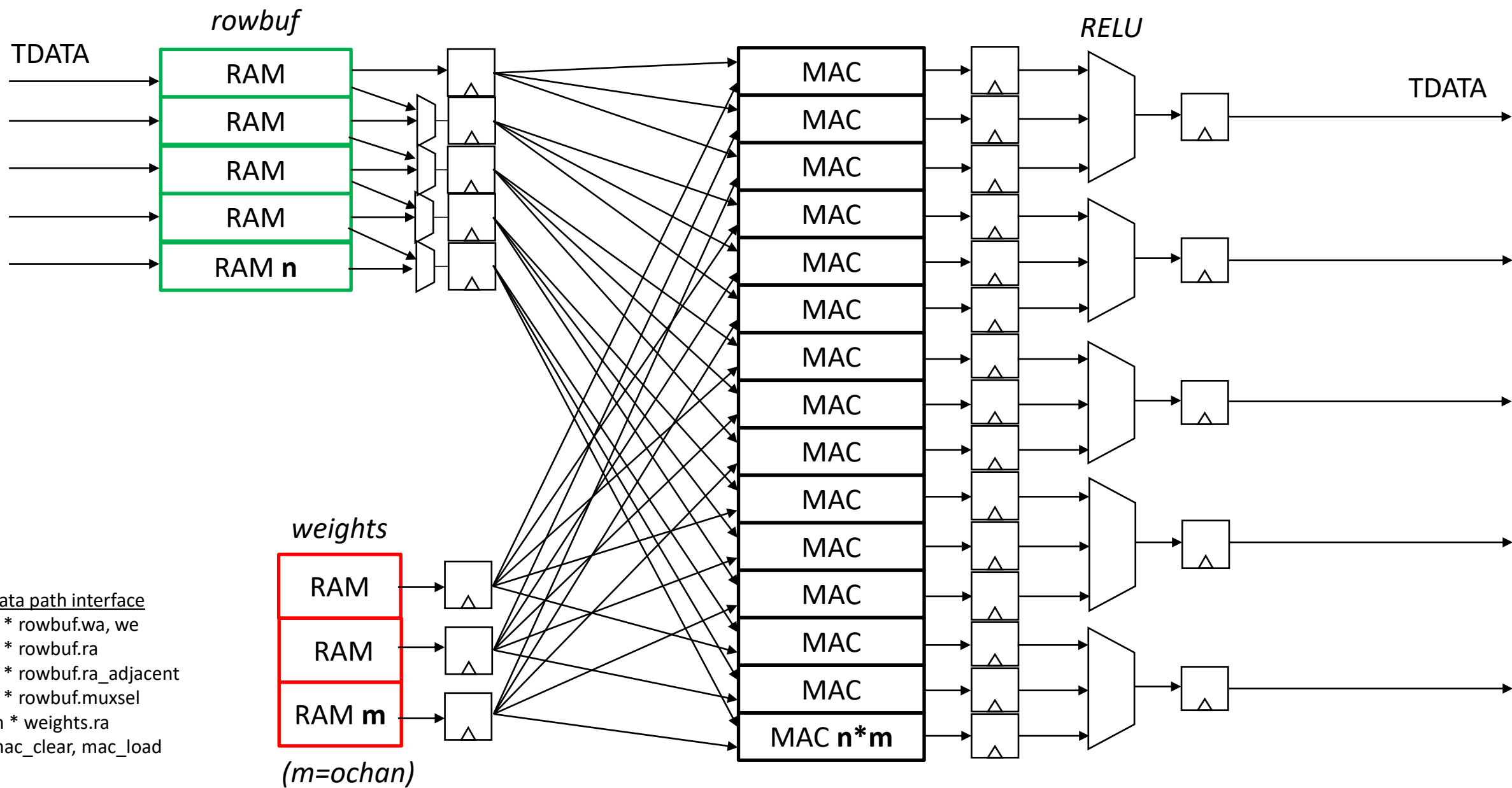


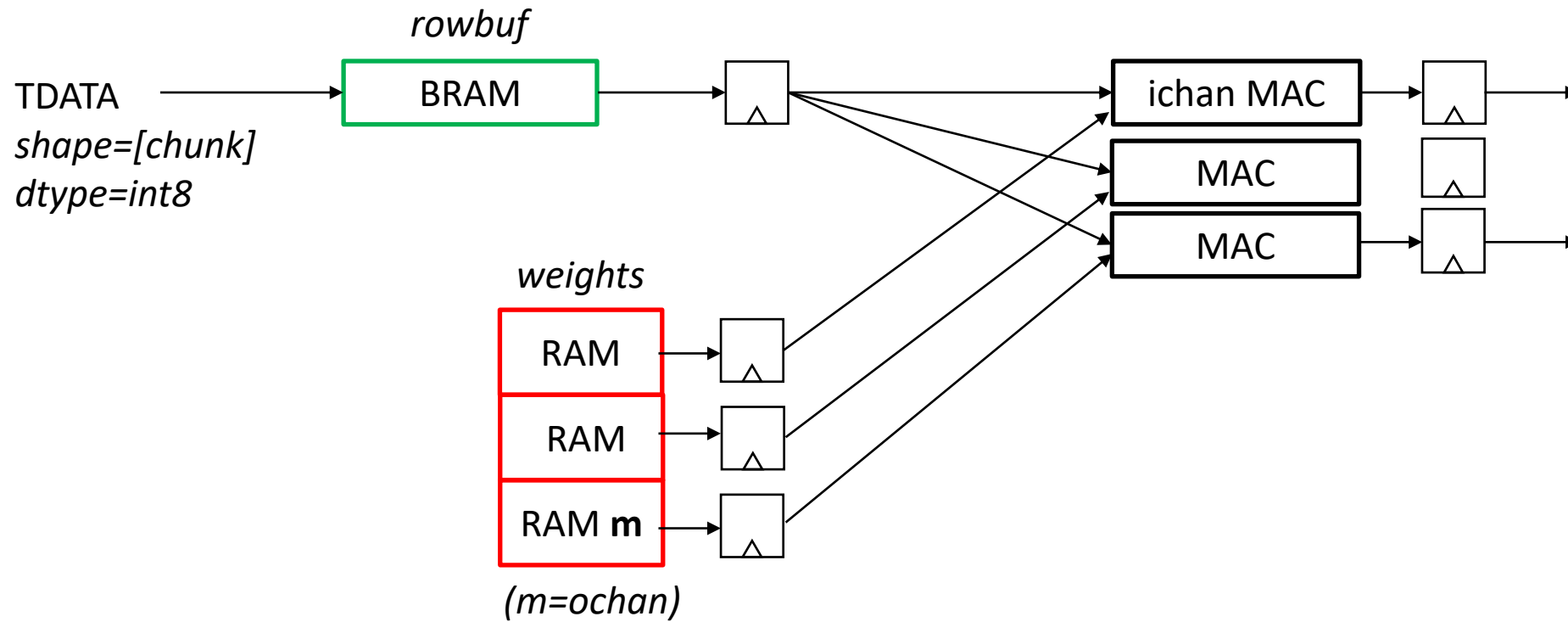




CNN dot product using FPGA





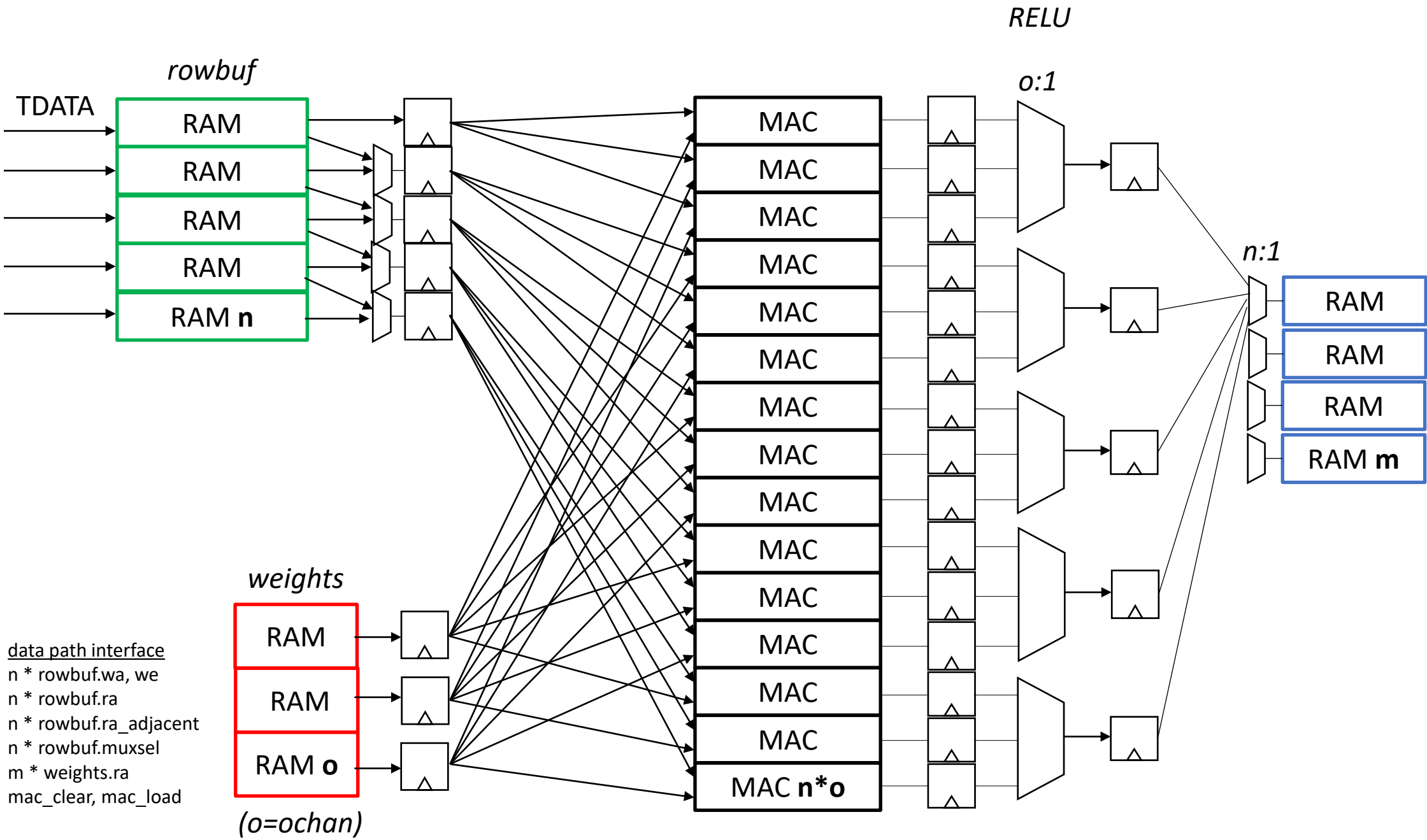


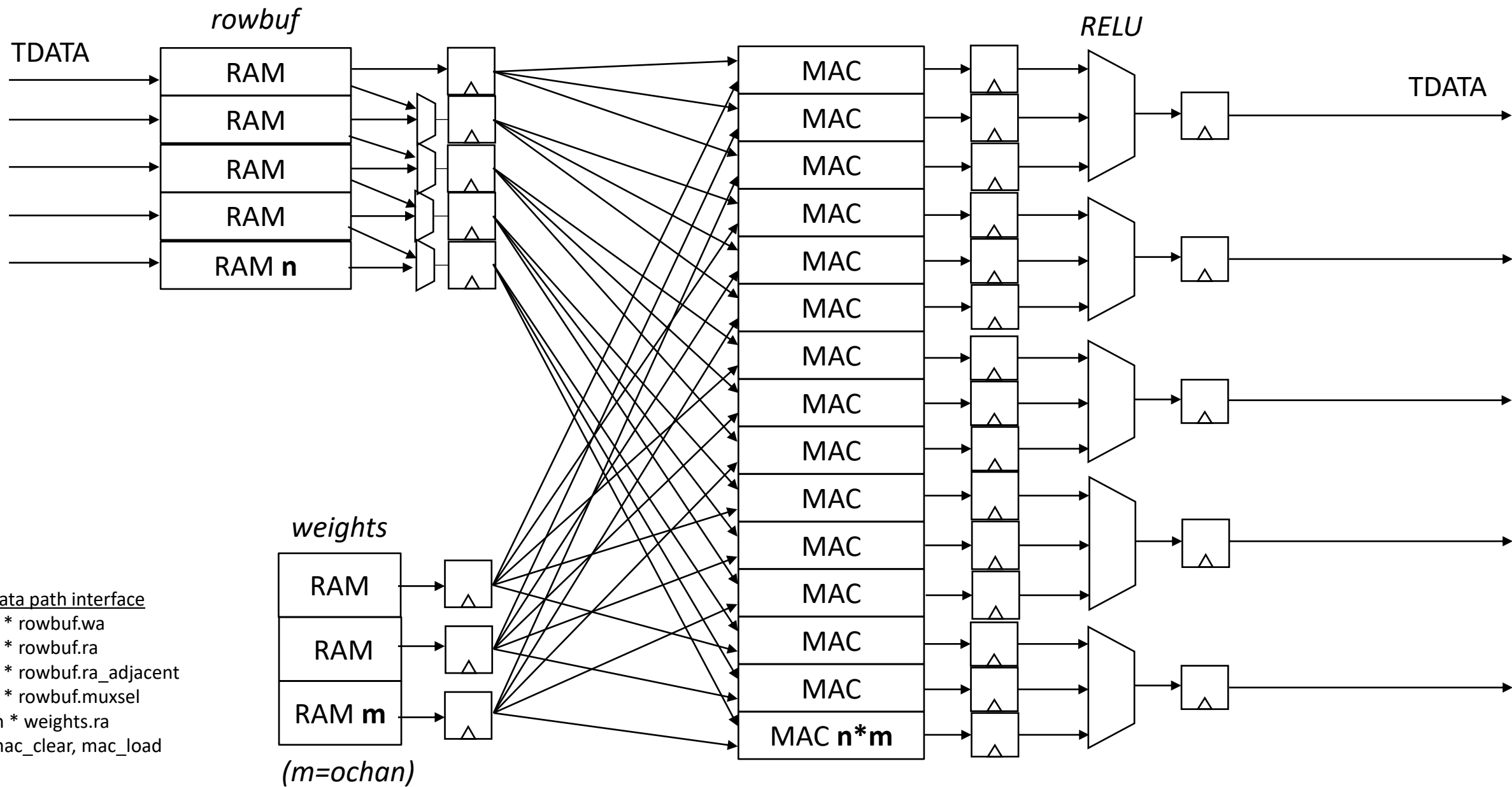
1. serialize flattened tensor
2. send tensor over TDATA in $S \times \text{dtype}$ size chunks
3. write and read BRAM in $S \times \text{dtype}$ size chunks

data path interface
 $n * \text{rowbuf.wa}$, we
 $n * \text{rowbuf.ra}$
 $n * \text{rowbuf.ra_adjacent}$
 $n * \text{rowbuf.muxsel}$
 $m * \text{weights.ra}$
 mac_clear , mac_load

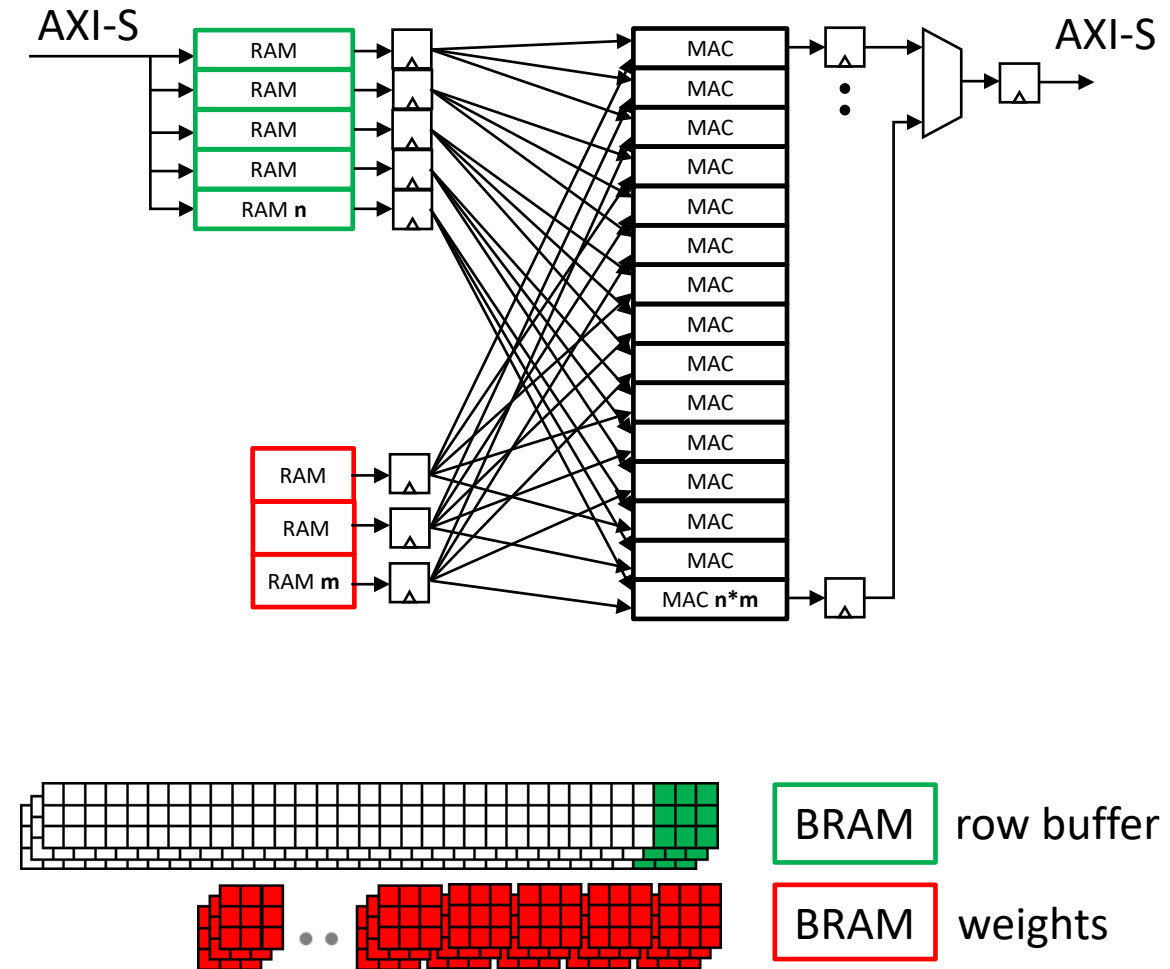
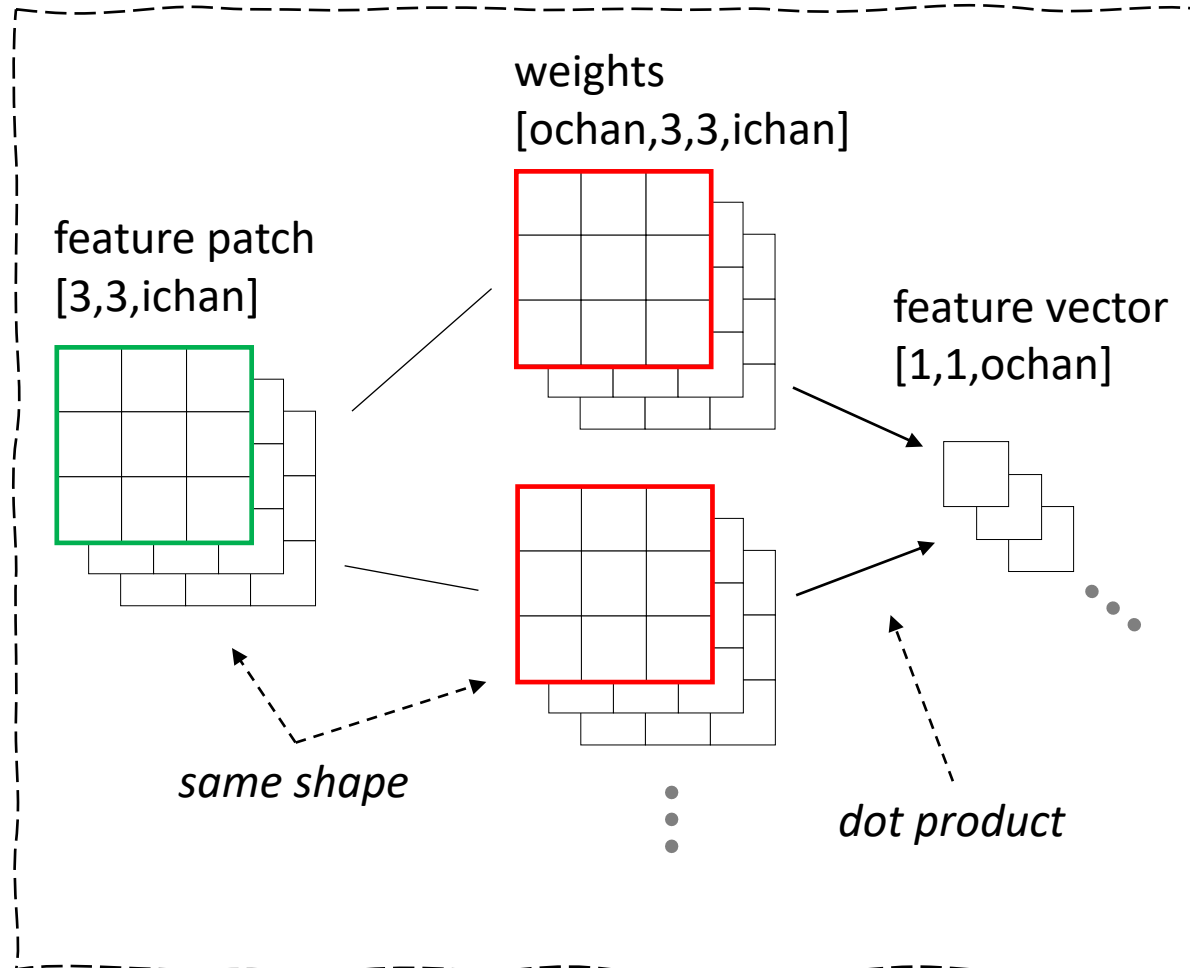
fps	289																	
width	728																	
height	544																	
channels	3																	
clk	500,000,000																	
layer	iwidth	iheight	ichan	stride	kheight	kwidth	owidth	oheight	ochan	patches/s	MAC/patch	MAC/s	required MAC	row buffer words	weight words	n	n*m	rowbuf depth
1	728	544	3	1	3	3	726	542	32	114,453,248	864	98,887,606,272	198	6,552	864	7	224	936
2	726	542	32	2	3	3	362	270	32	28,246,860	9,216	260,323,061,760	521	69,696	9,216	17	544	4,100
3	362	270	32	1	3	3	360	268	32	27,882,720	9,216	256,967,147,520	514	34,752	9,216	17	544	2,044
4	360	268	32	2	3	3	179	133	32	6,880,223	9,216	63,408,135,168	127	34,560	9,216	4	128	8,640
5	179	133	32	1	3	3	177	131	32	6,701,043	9,216	61,756,812,288	124	17,184	9,216	4	128	4,296
6	177	131	32	2	3	3	88	65	32	1,653,080	9,216	15,234,785,280	31	16,992	9,216	1	32	16,992
7	88	65	32	1	3	3	86	63	32	1,565,802	9,216	14,430,431,232	29	8,448	9,216	1	32	8,448
8	86	63	32	2	3	3	42	31	32	376,278	9,216	3,467,778,048	7	8,256	9,216	1	32	8,256
9	42	31	32	1	3	3	40	29	32	335,240	9,216	3,089,571,840	7	4,032	9,216	1	32	4,032
10	40	29	32	2	3	3	19	14	32	76,874	9,216	708,470,784	2	3,840	9,216	1	32	3,840
11	19	14	32	1	3	3	17	12	32	58,956	9,216	543,338,496	2	1,824	9,216	1	32	1,824
12	17	12	32	2	3	3	8	5	32	11,560	9,216	106,536,960	1	1,632	9,216	1	32	1,632
13	8	5	32	1	3	3	6	3	32	5,202	9,216	47,941,632	1	768	9,216	1	32	768
14	6	3	32	2	3	3	2	1	32	578	9,216	5,326,848	1	576	9,216	1	32	576
15	2	1	32	1	3	3	0	-1	32	-	9,216	-	-	192	9,216	-	-	#DIV/0!
												778,976,944,128	1,565	209,304	129,888	58	1,856	

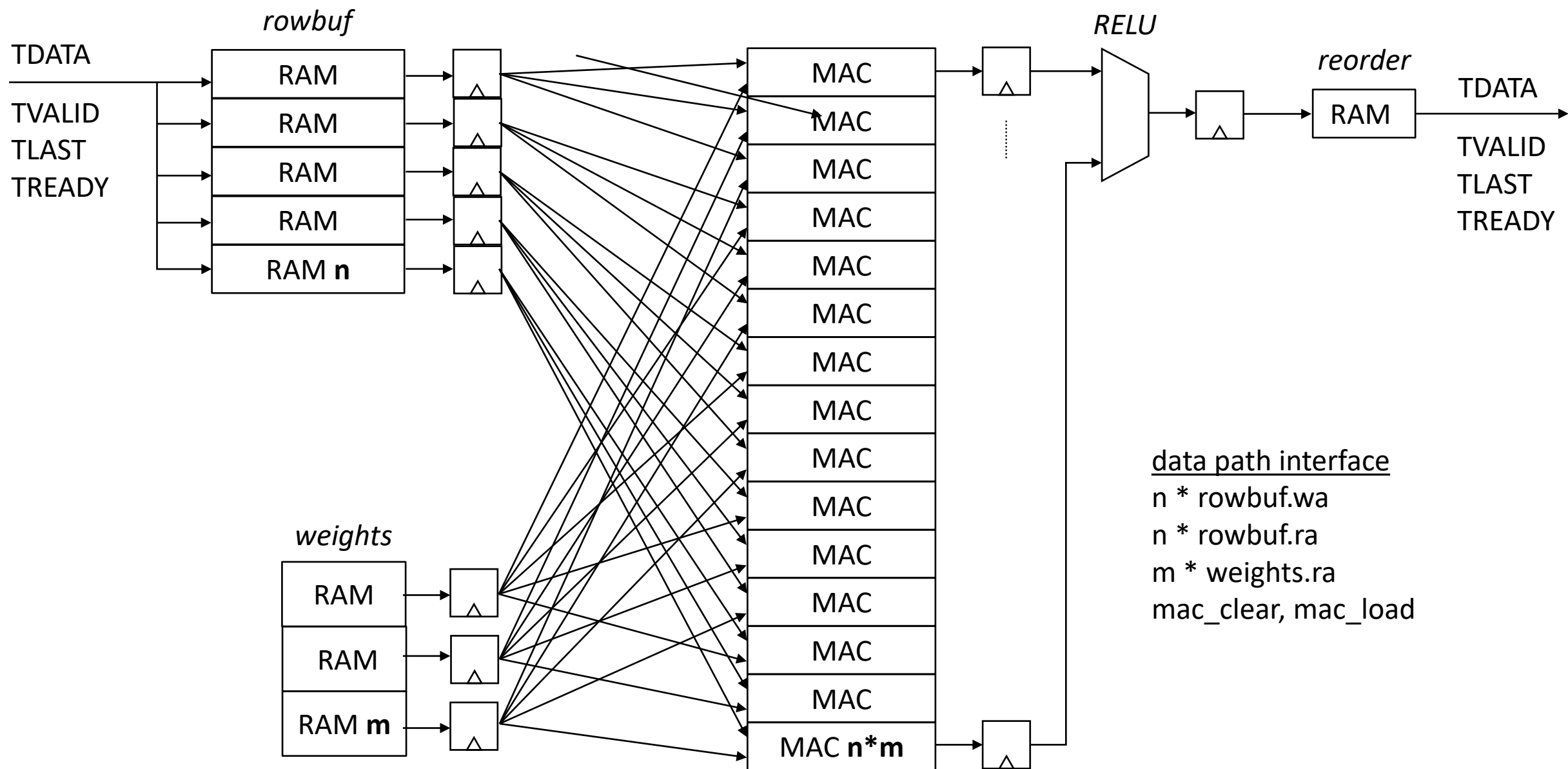
1. write incoming features from AXI-S to rowbuf[]
2. when a complete row of patches has arrived, begin computing dot products per-inputslice per-CHANOUT
3. n*m features will complete simultaneously, write them sequentially into the reorder buffer

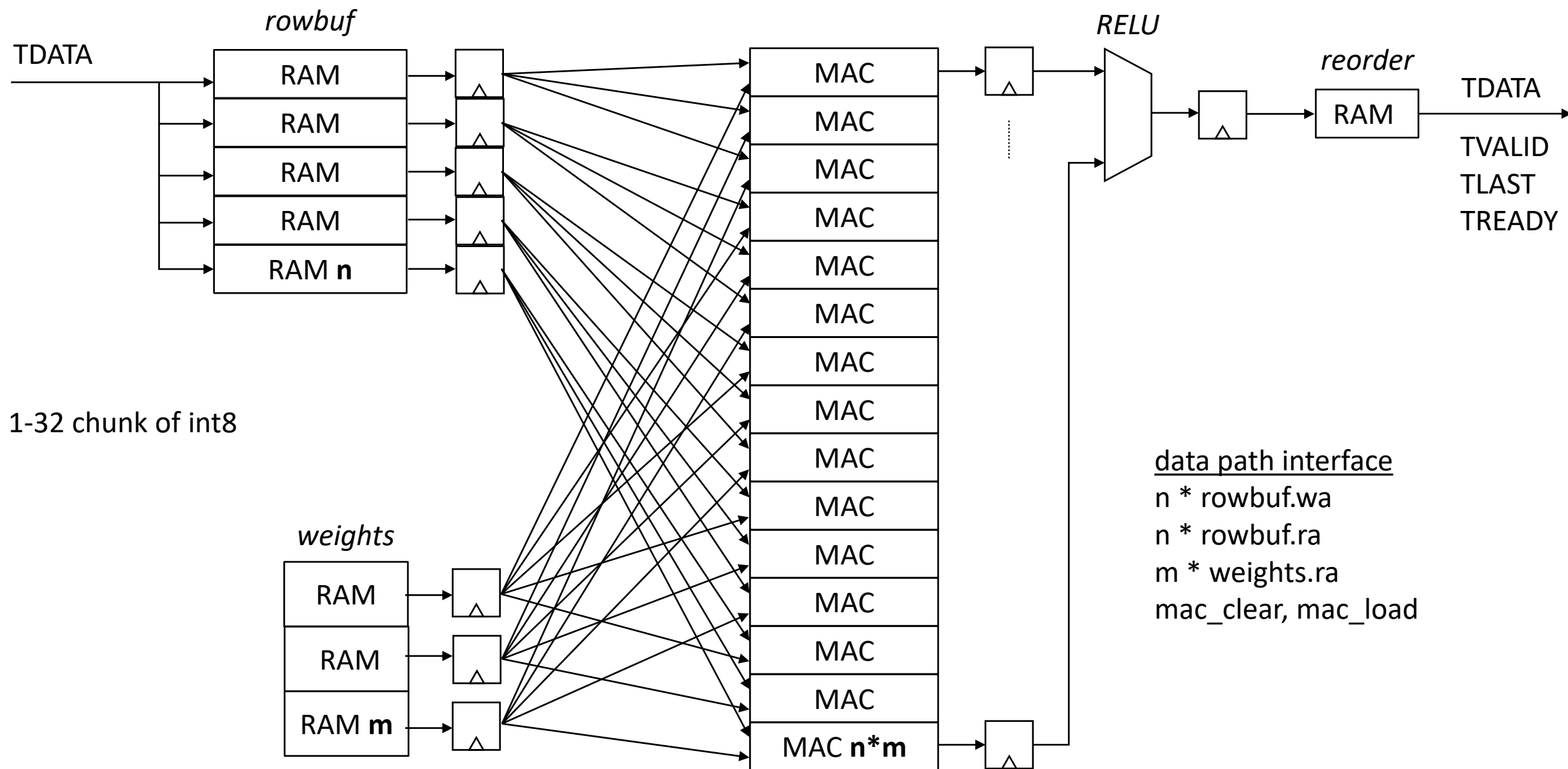




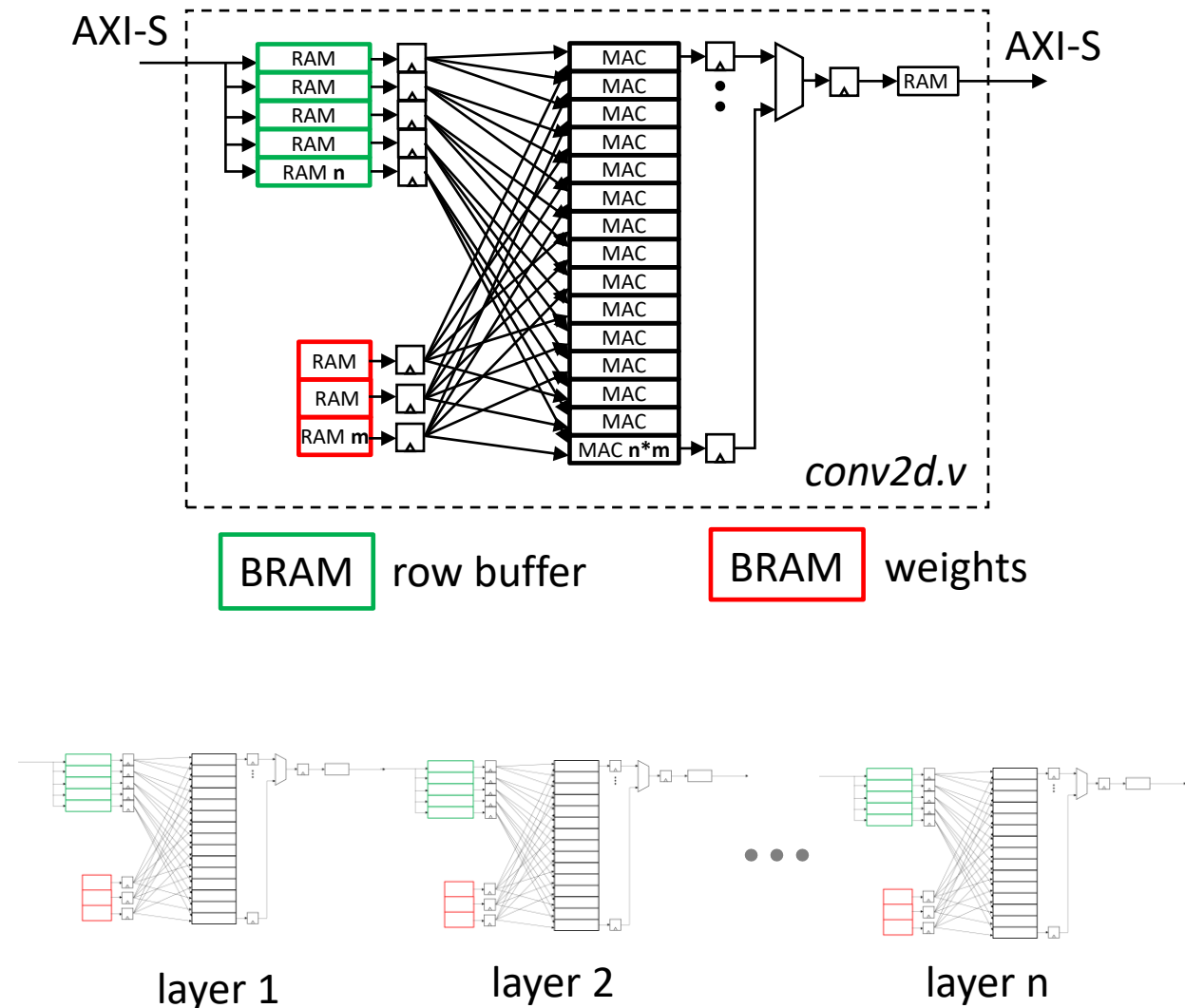
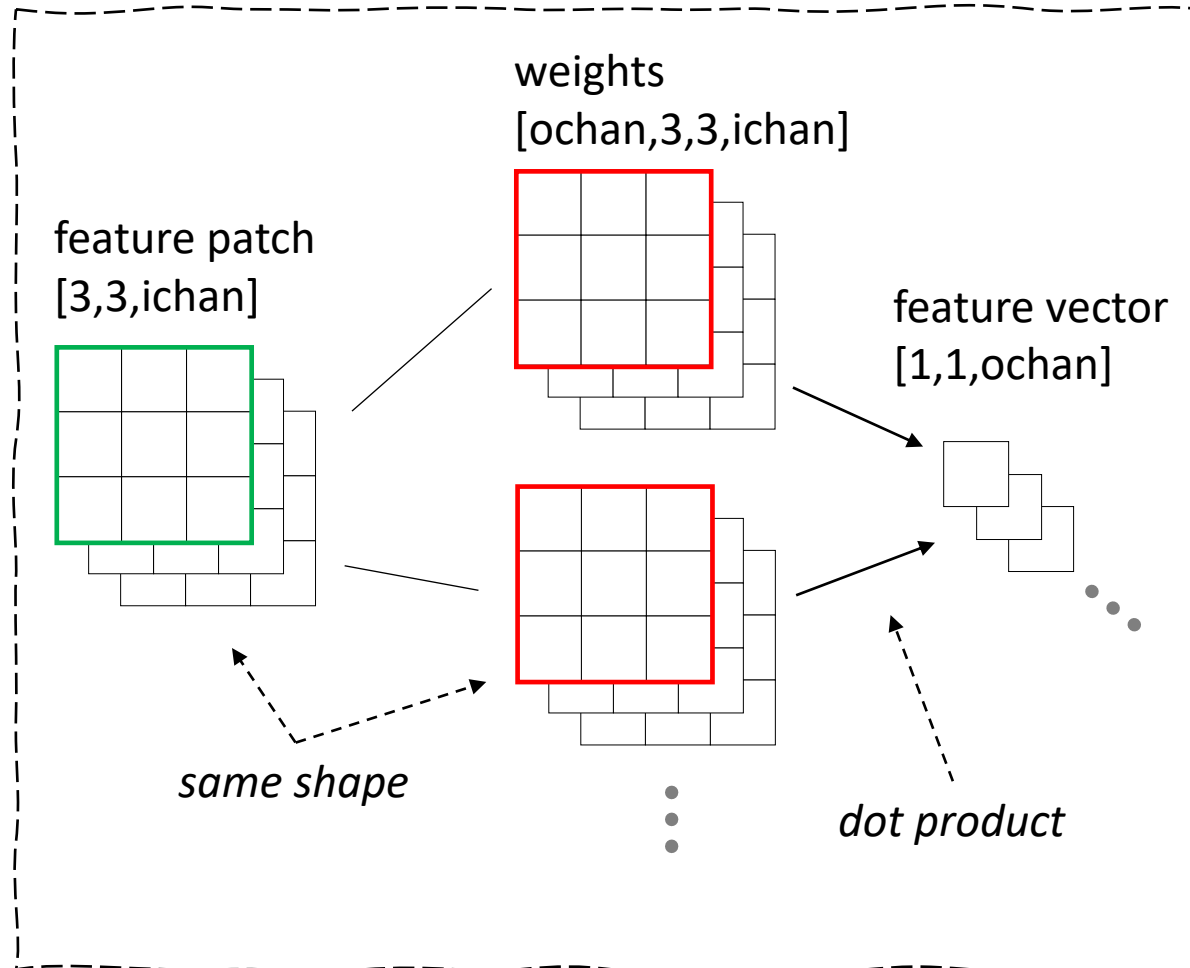
CNN dot product using FPGA

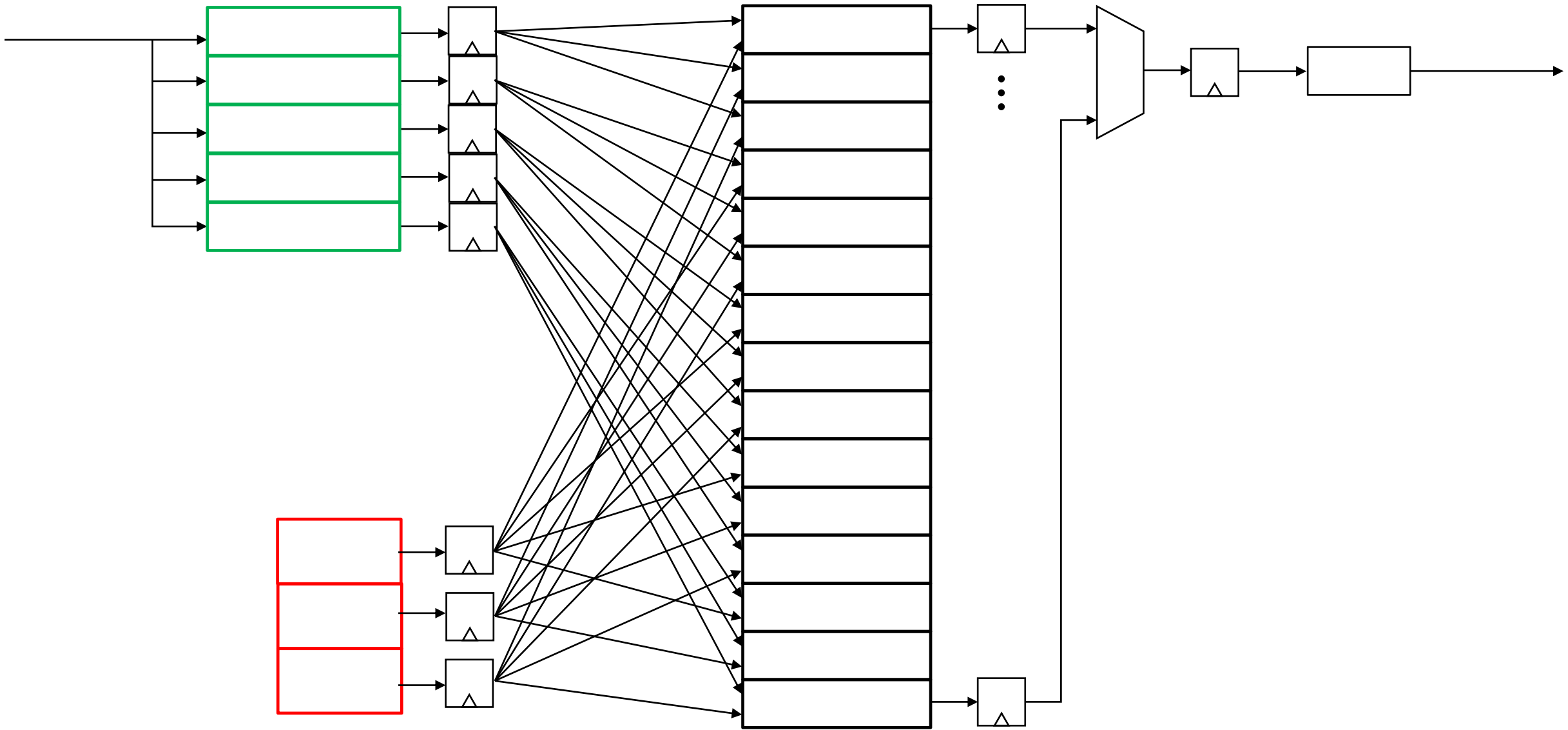




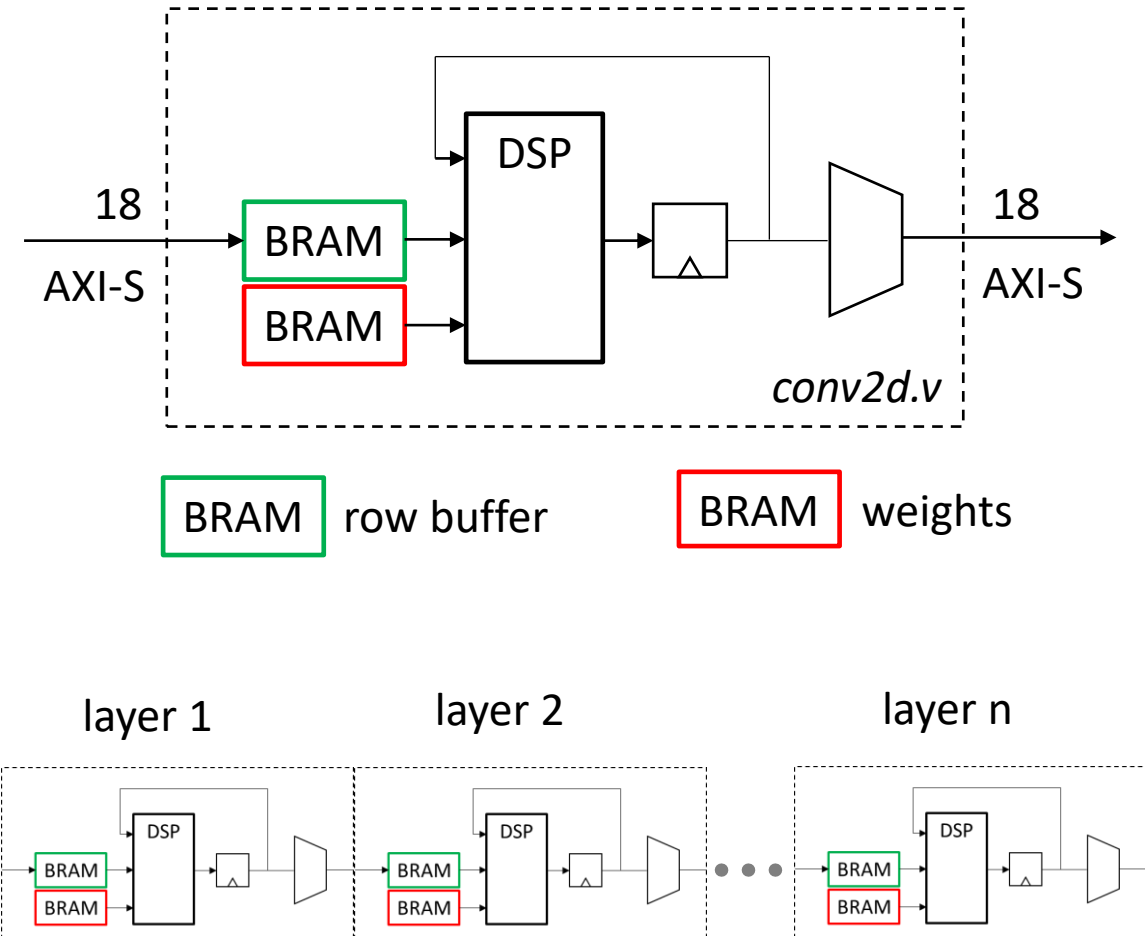
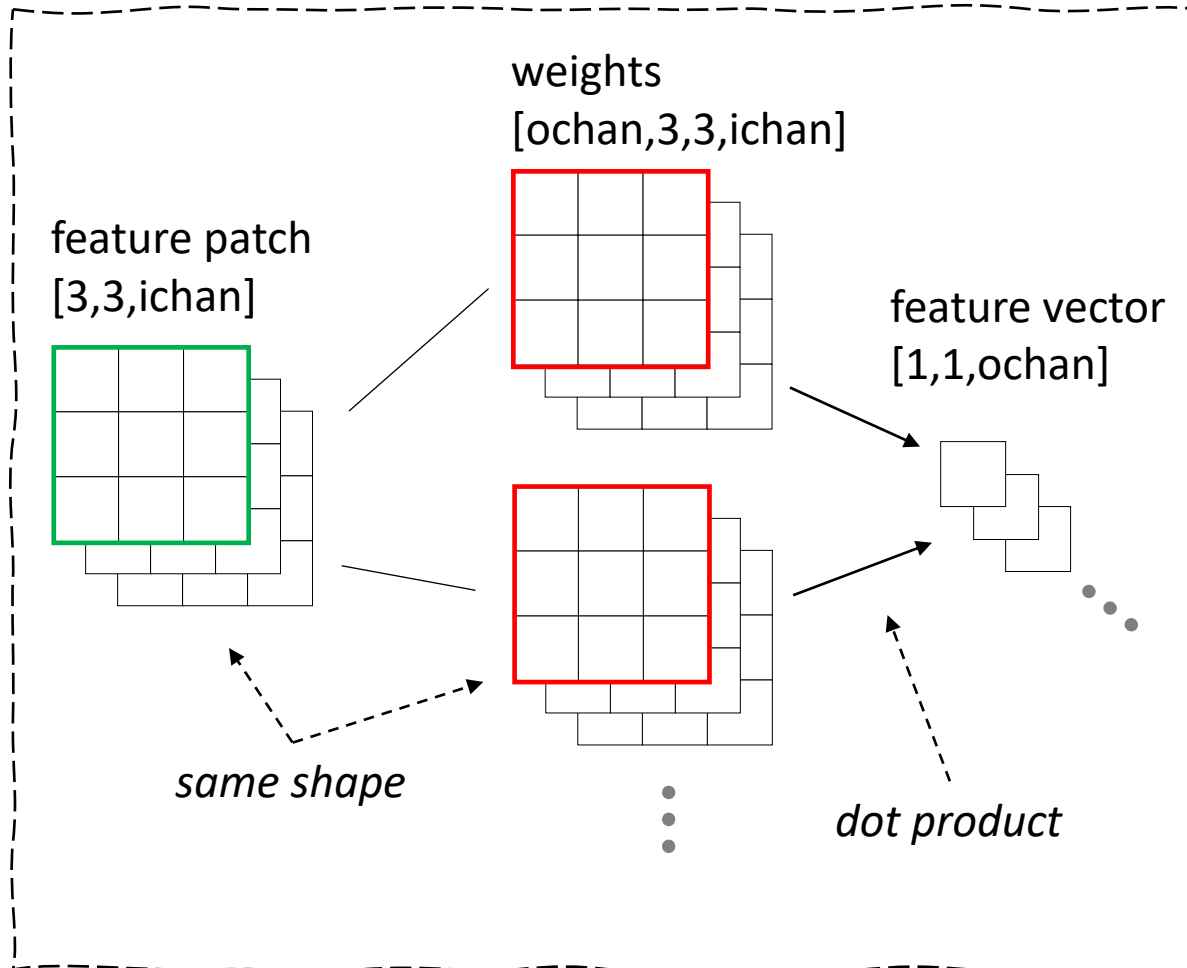


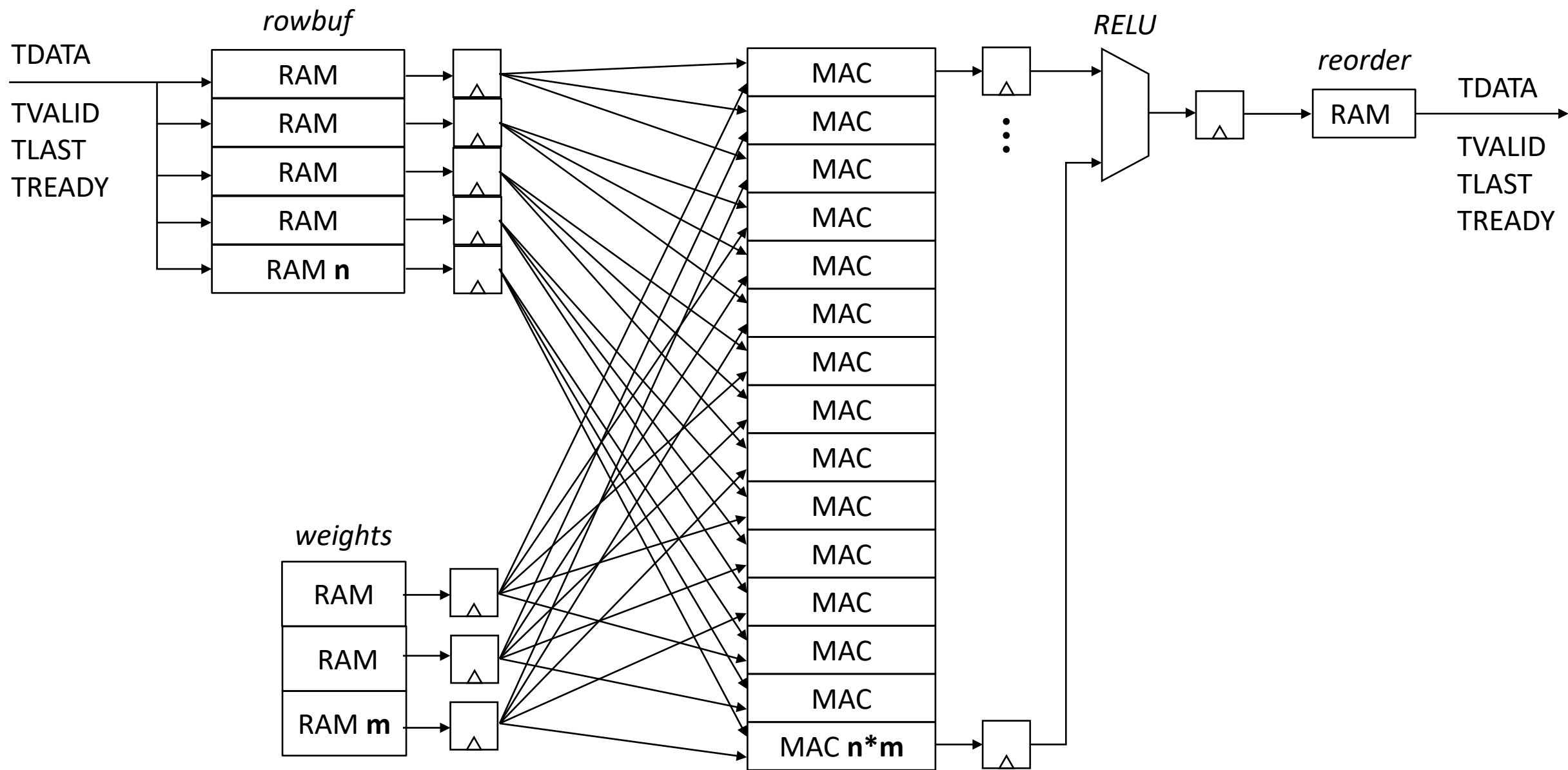
CNN dot product using FPGA

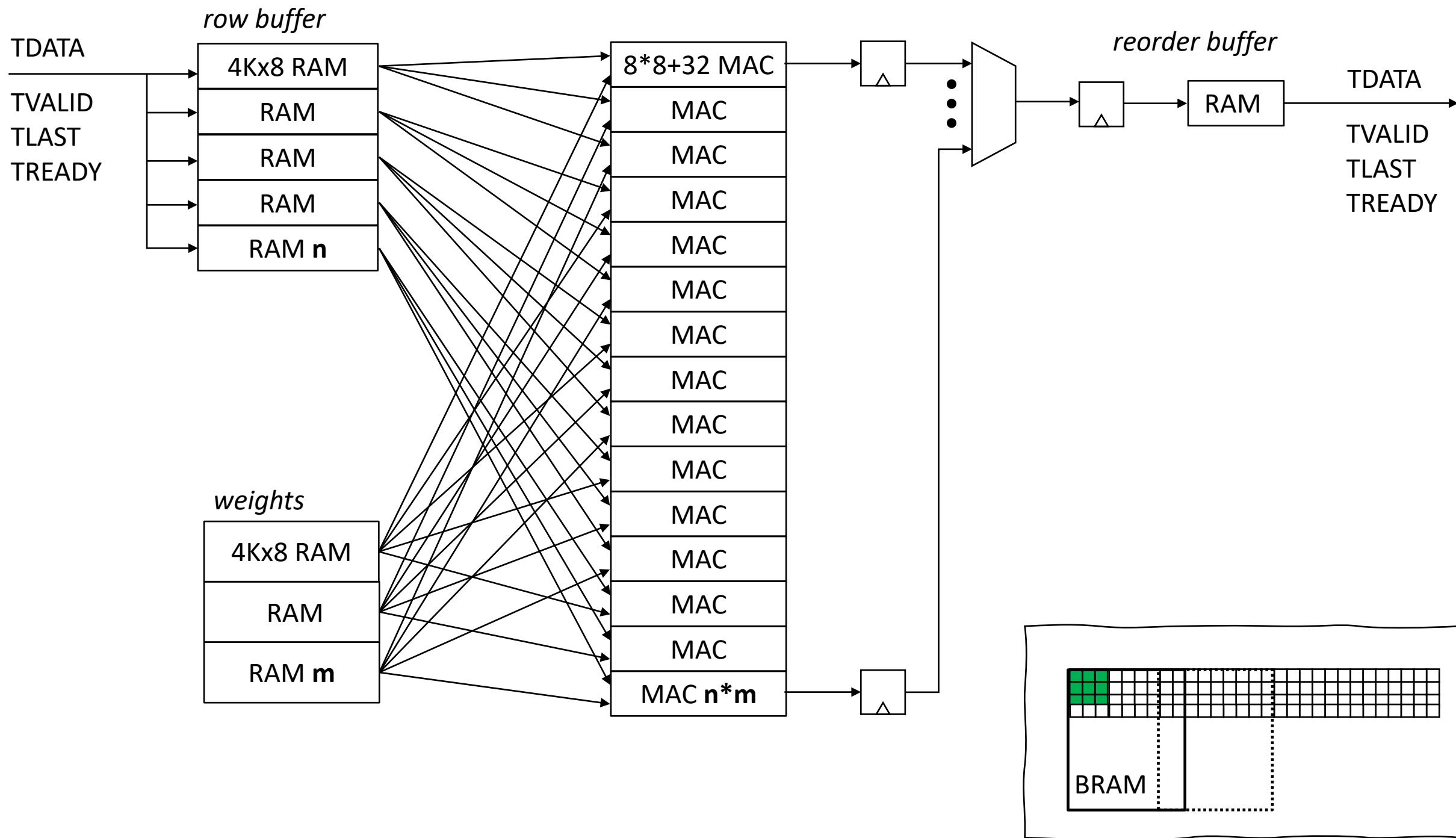


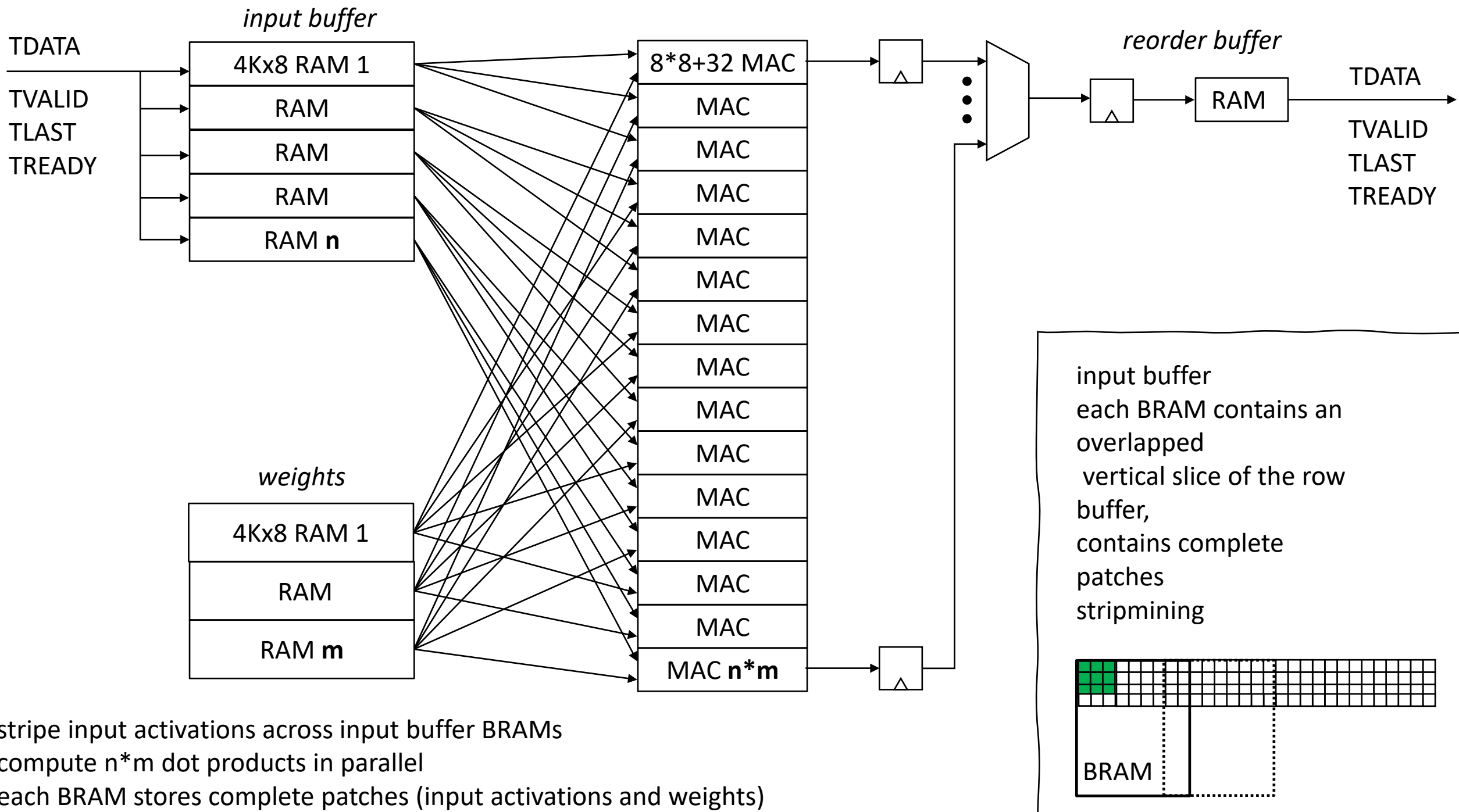


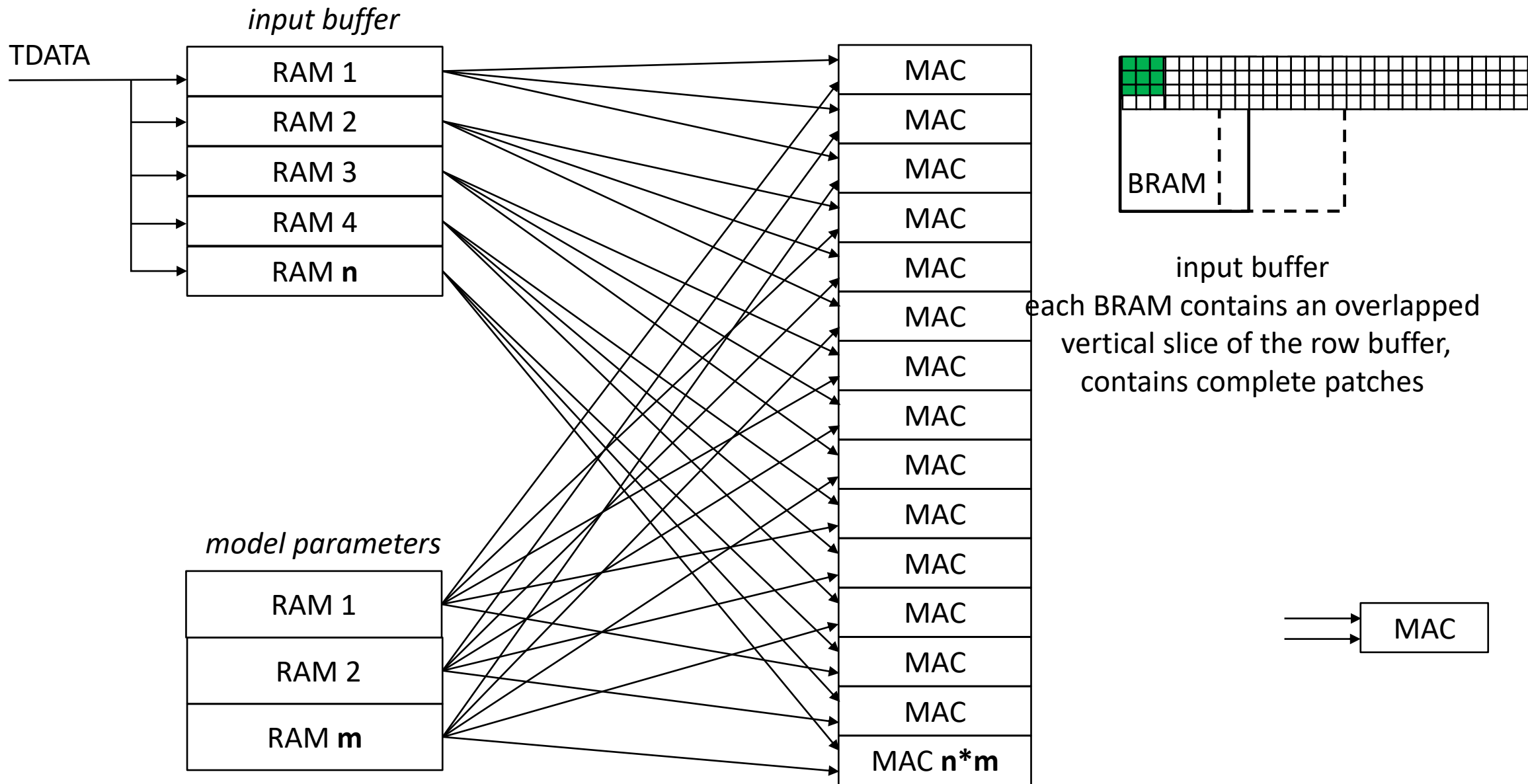
CNN dot product using FPGA



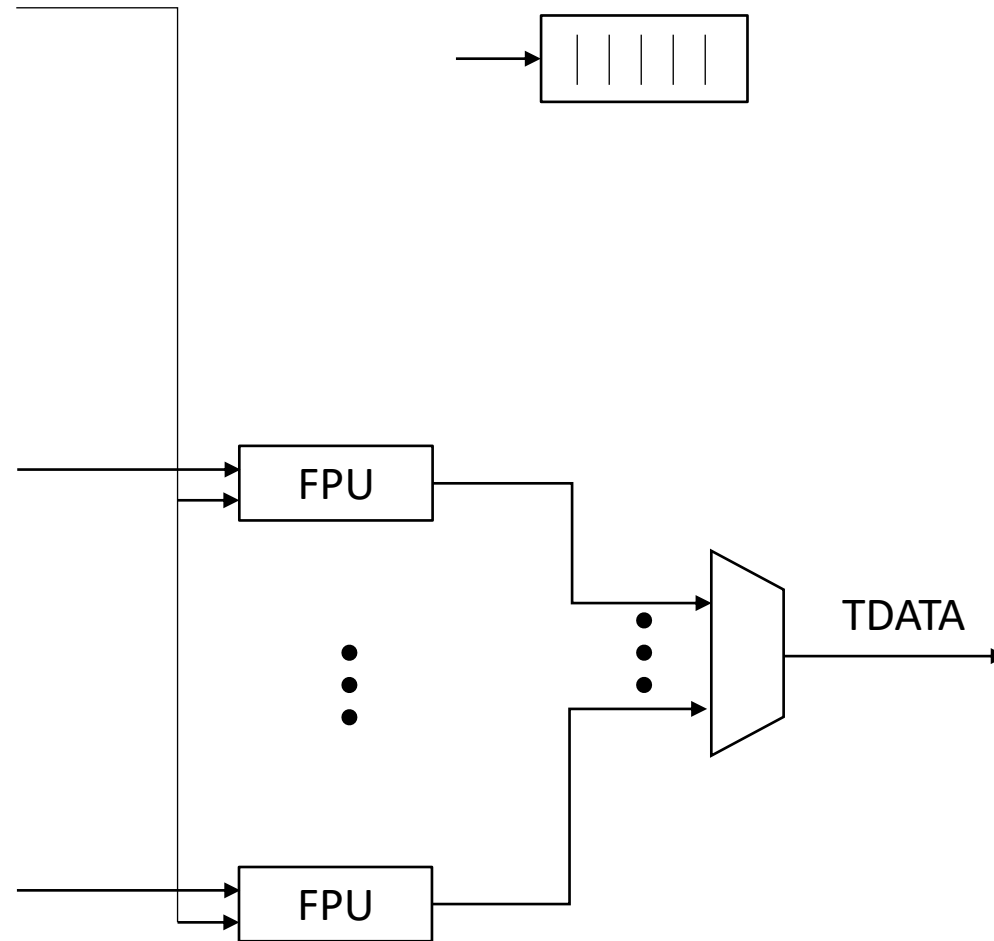
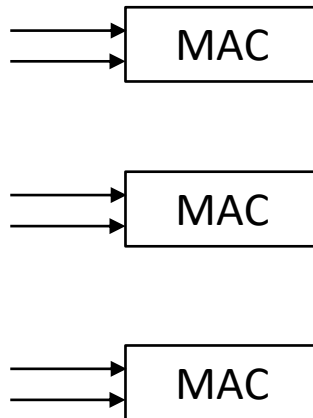
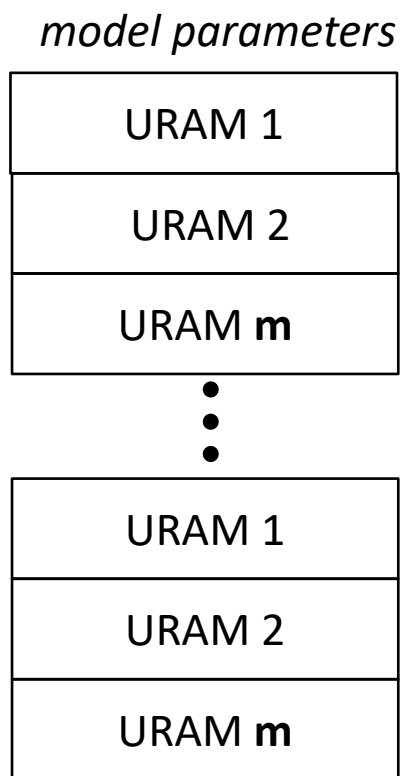
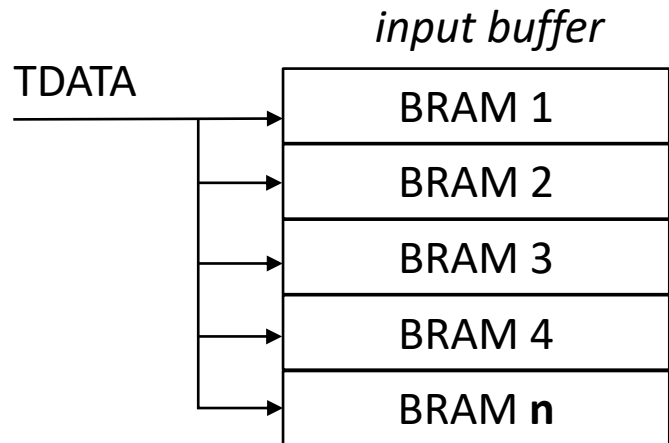


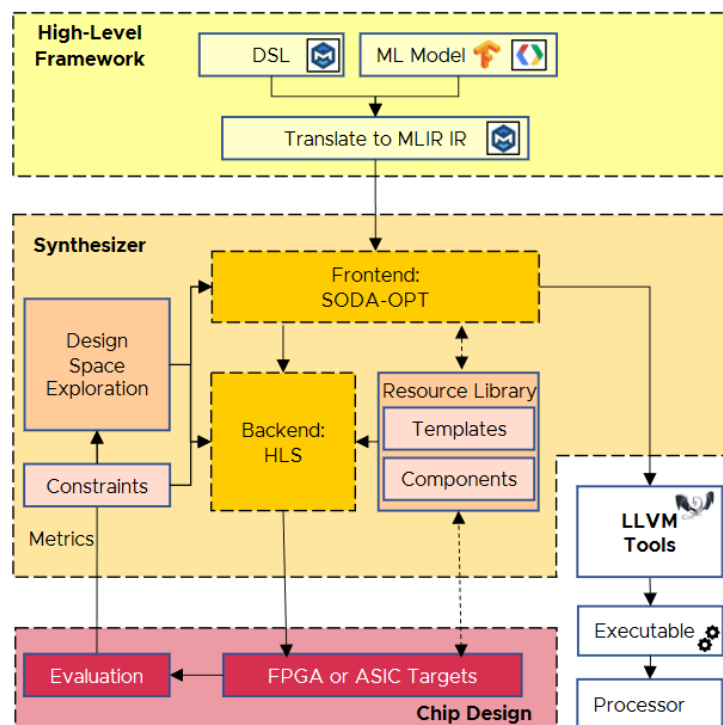






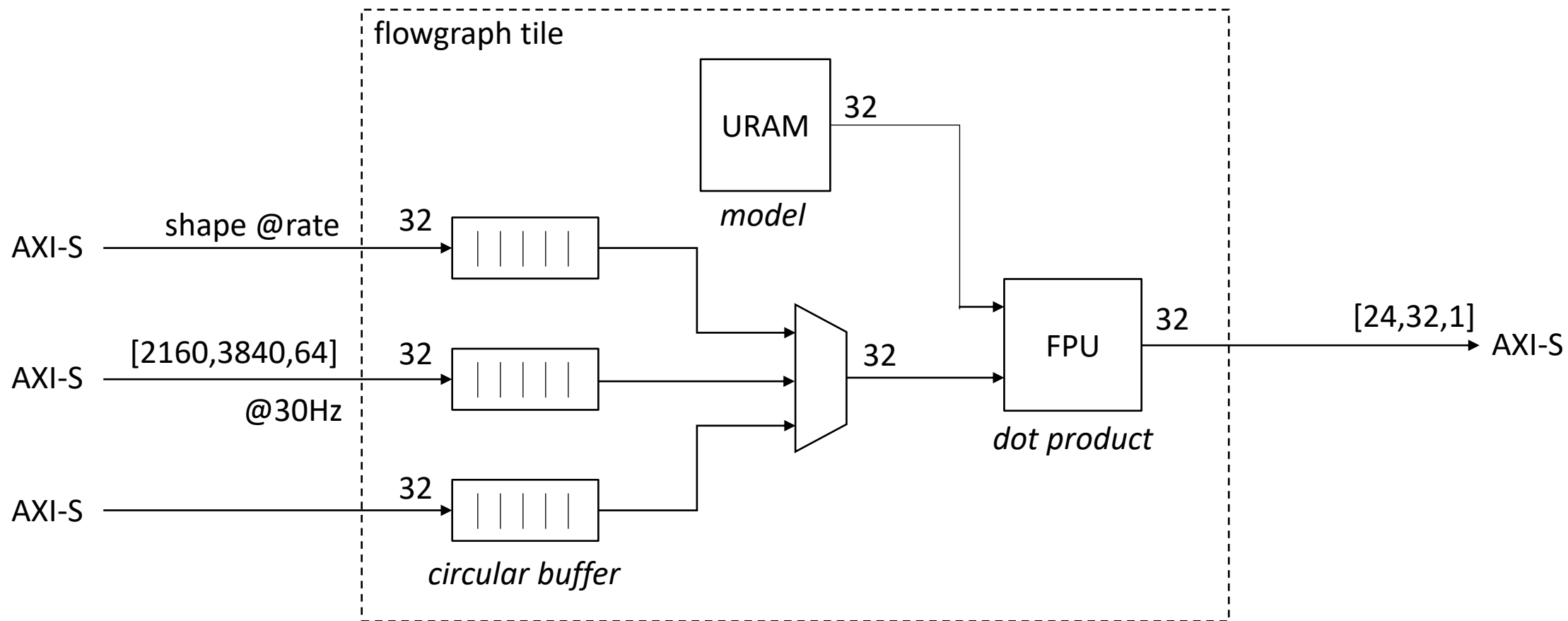
compute $n \times m$ dot products in parallel

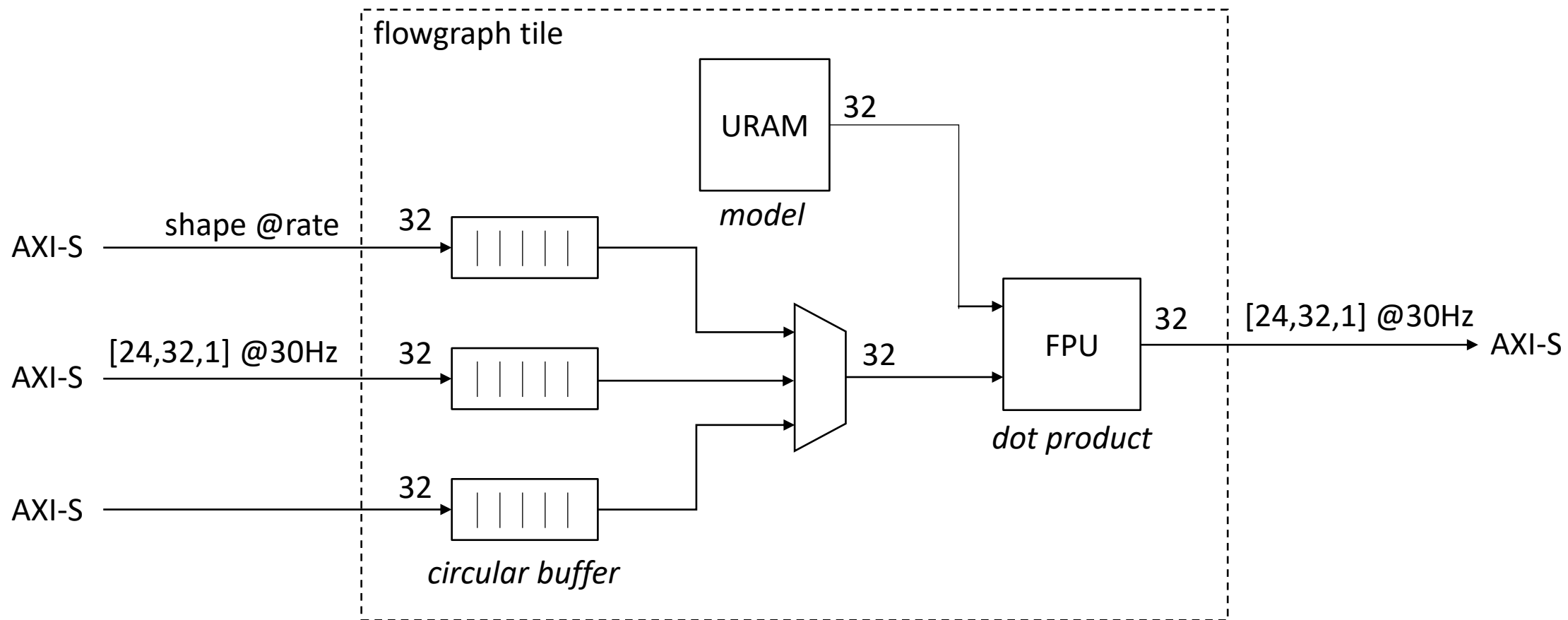


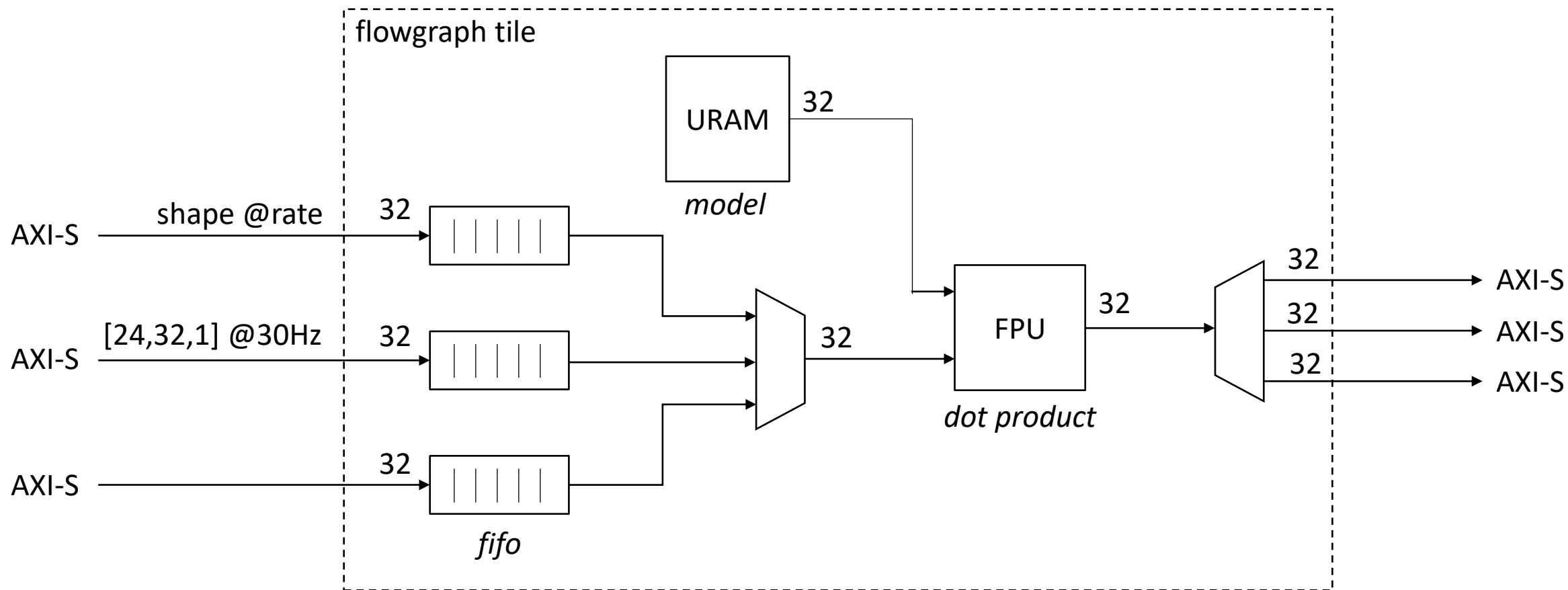


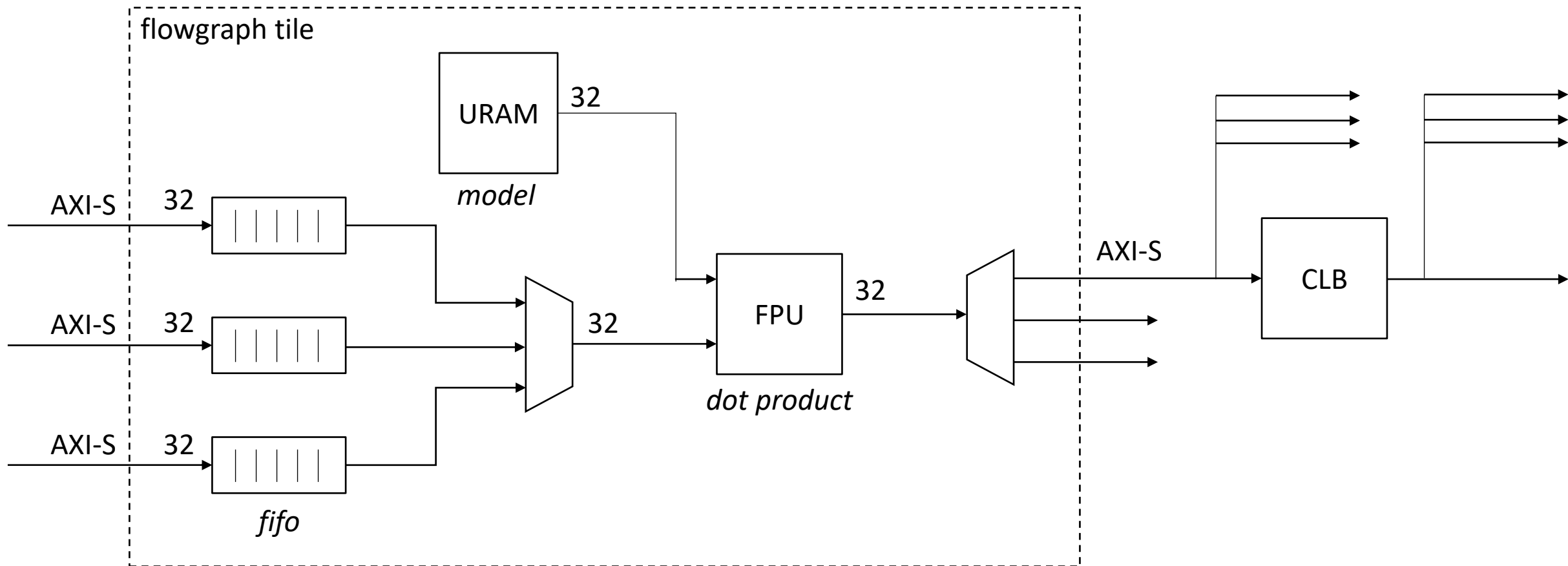
← TFLite to MLIR, starting with Conv2D

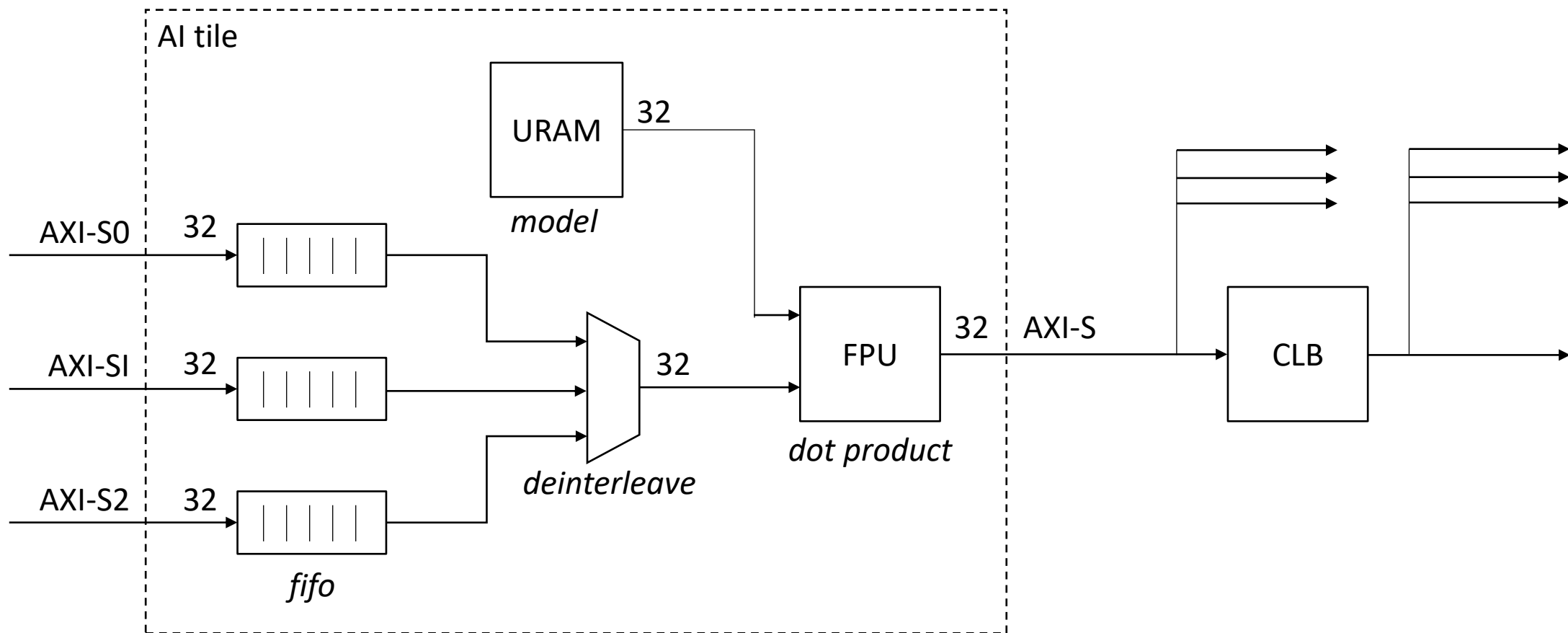
Fig. 1: The SODA Synthesizer.

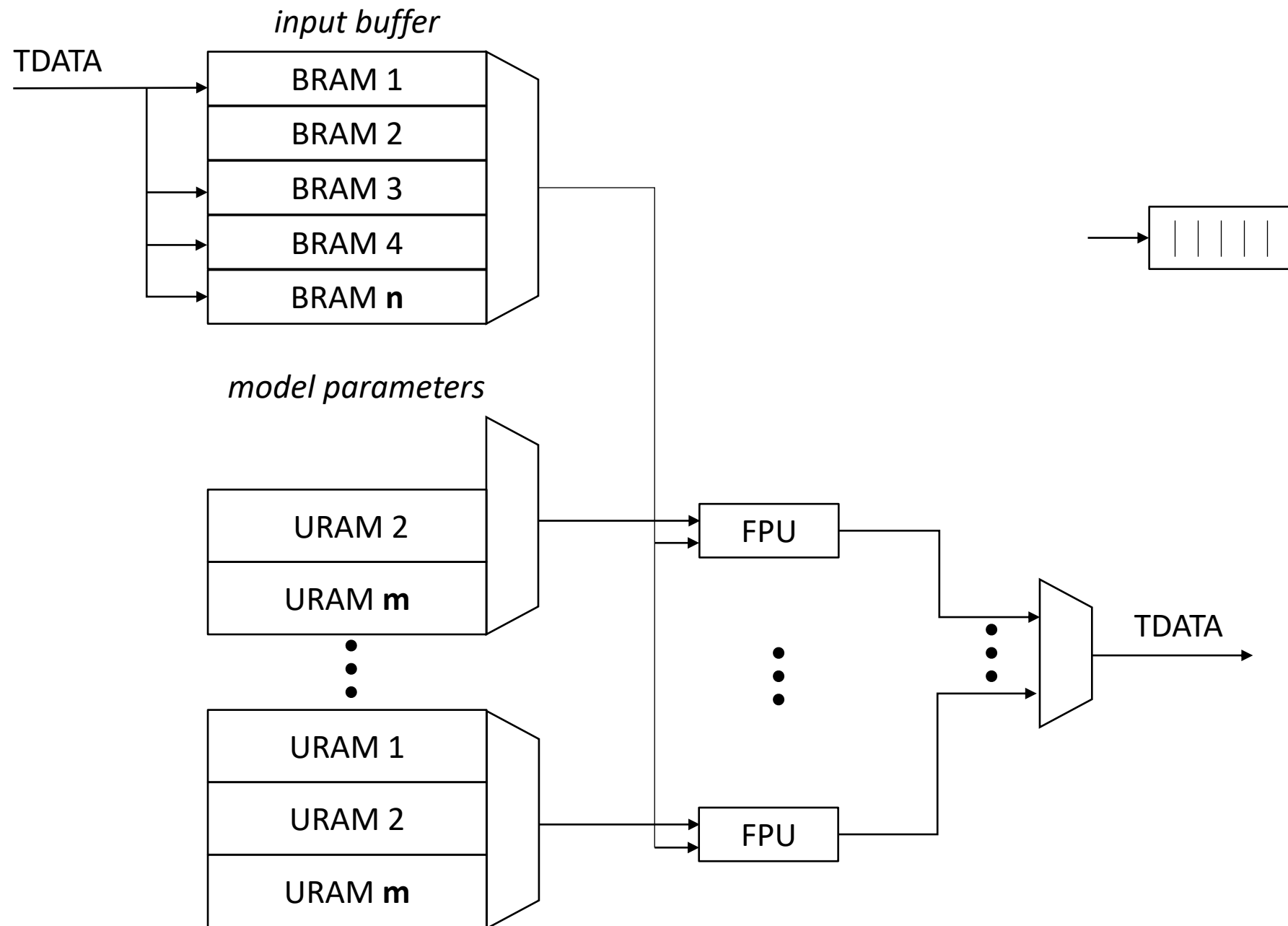




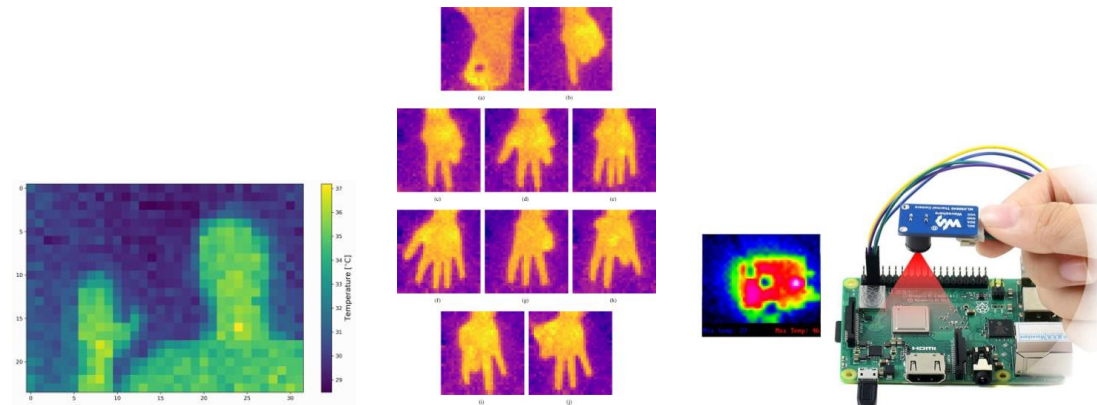
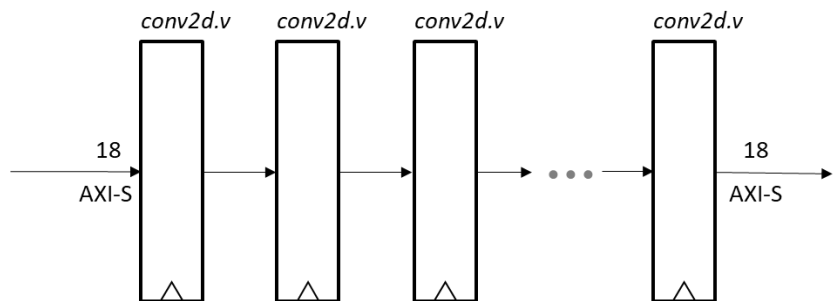








- conv2d.v is a Tensorflow compatible CNN layer implementation
- Each layer in the model is evaluated in parallel using pipelining
- Weights are stored in on-chip BRAM for high performance
- Can be used stand alone or as a front end to a larger model



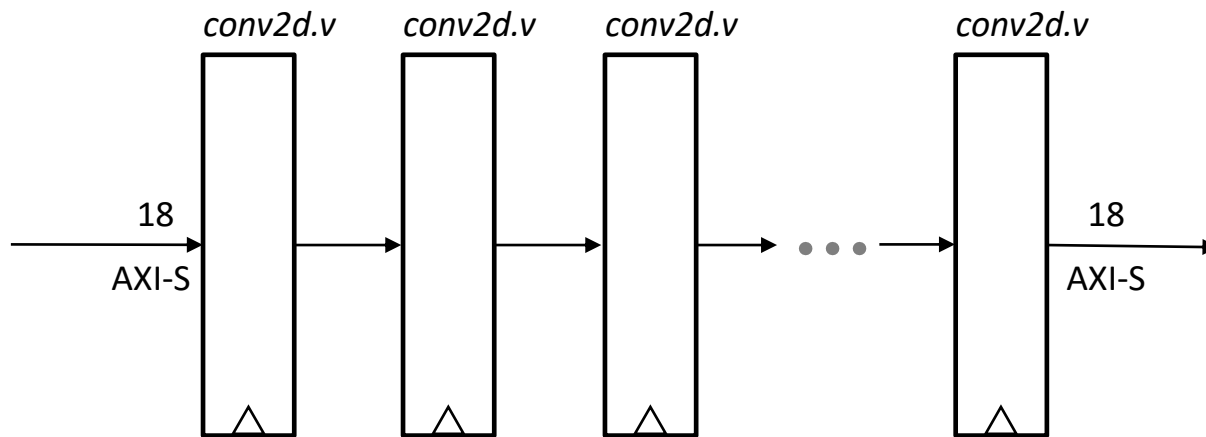
Real time classification demo for Gemini

- Code complete, runs at 200MHz using Artix-7/Vivado
- Next step is functional verification using MNIST

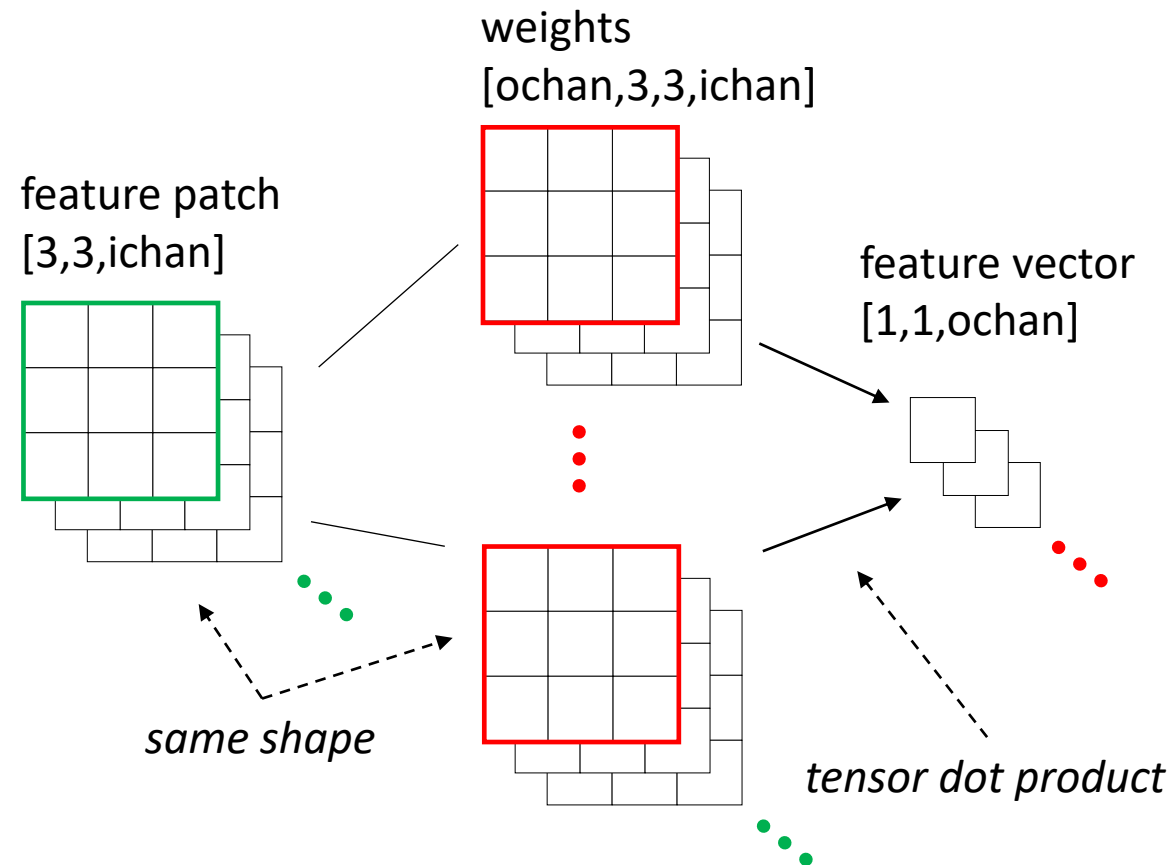
conv2d.v

Objectives

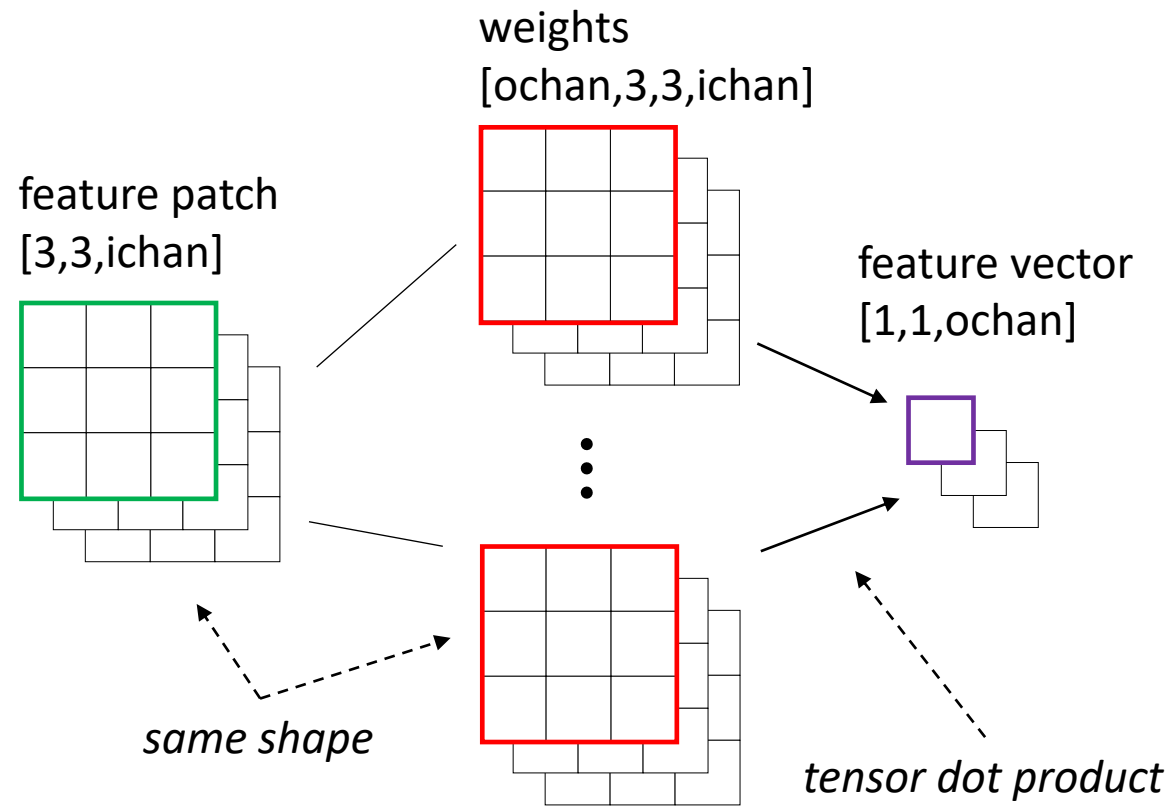
- initially target Gemini fabric
- high throughput, low latency deep model evaluation
- efficiently utilize FPGA BRAM and DSP resources
- package as a parameterized, reusable IP



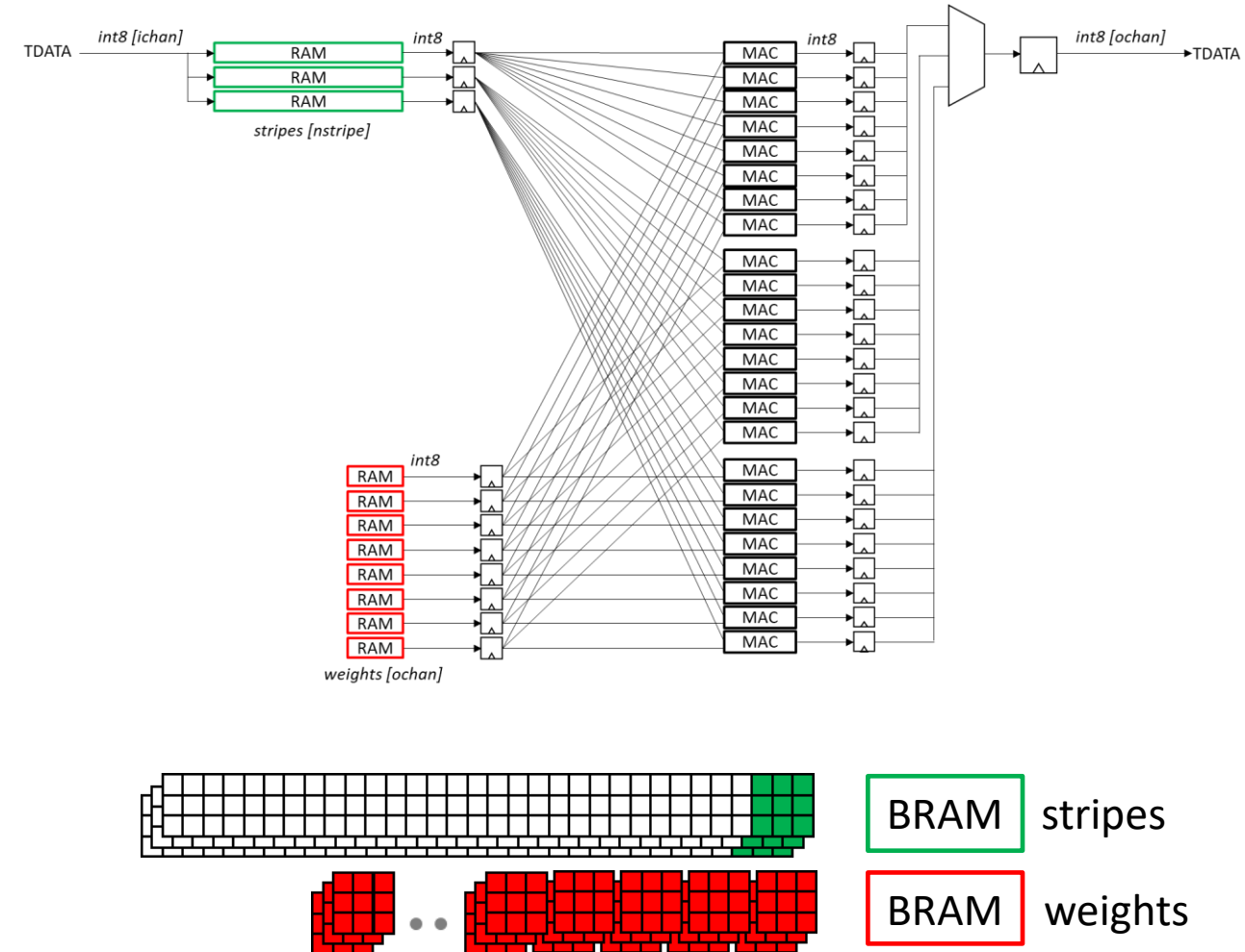
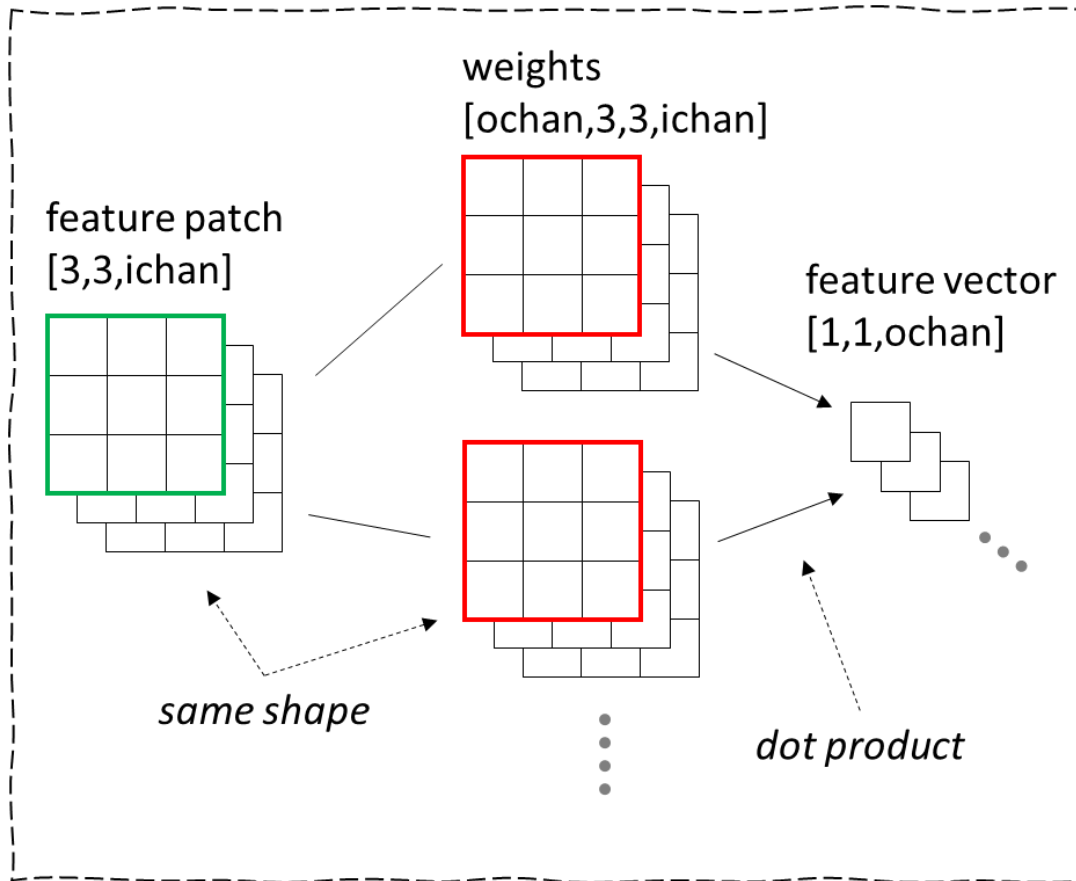
CNN primitive operation



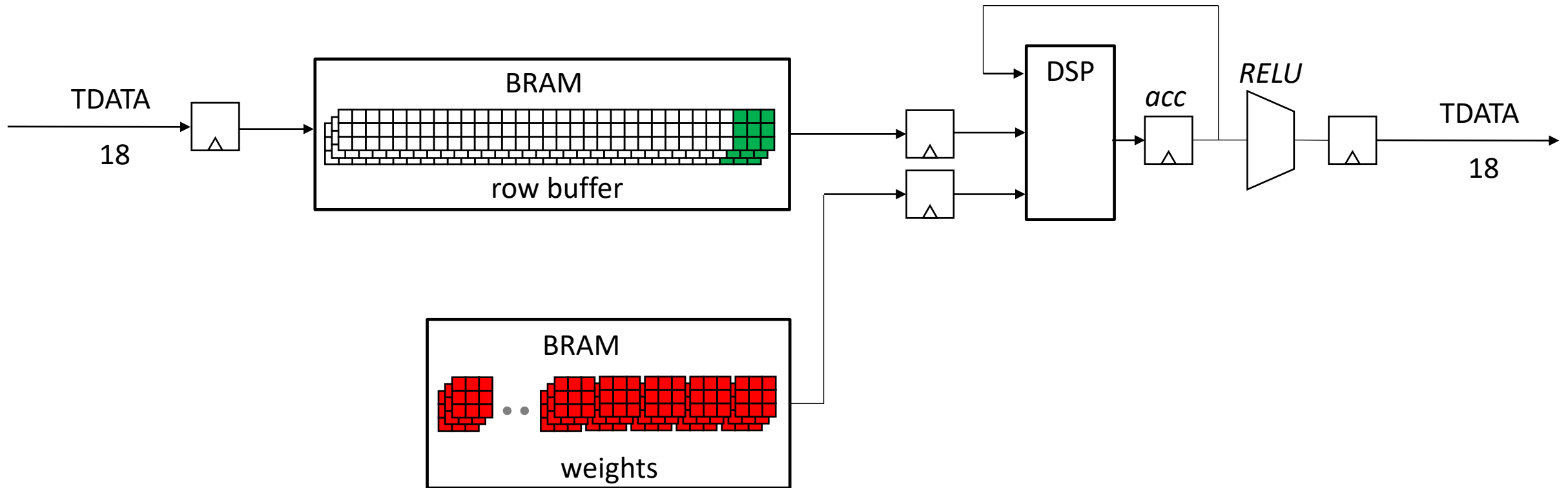
CNN primitive operation



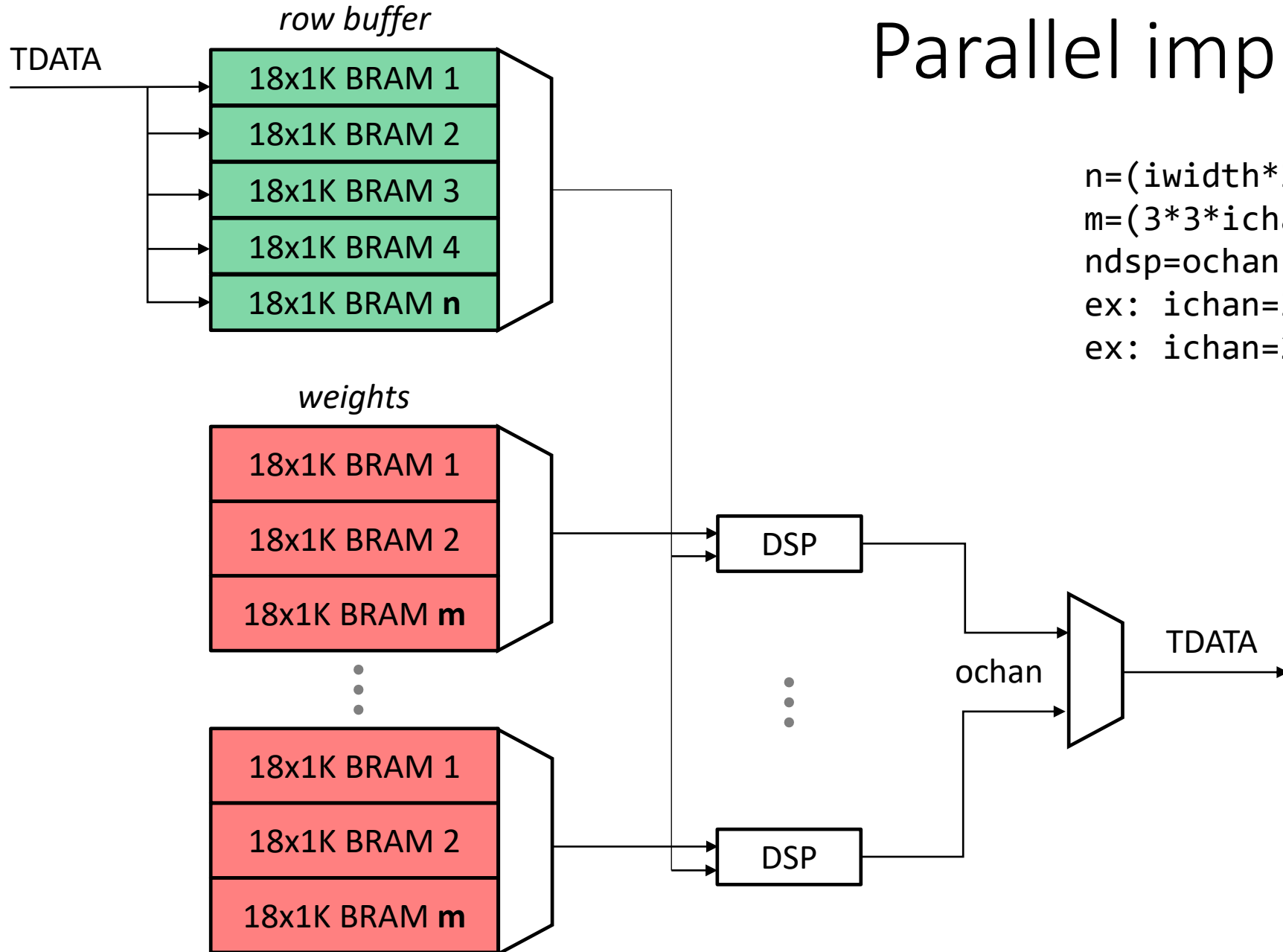
CNN dot product using FPGA



Sequential implementation



Parallel implementation



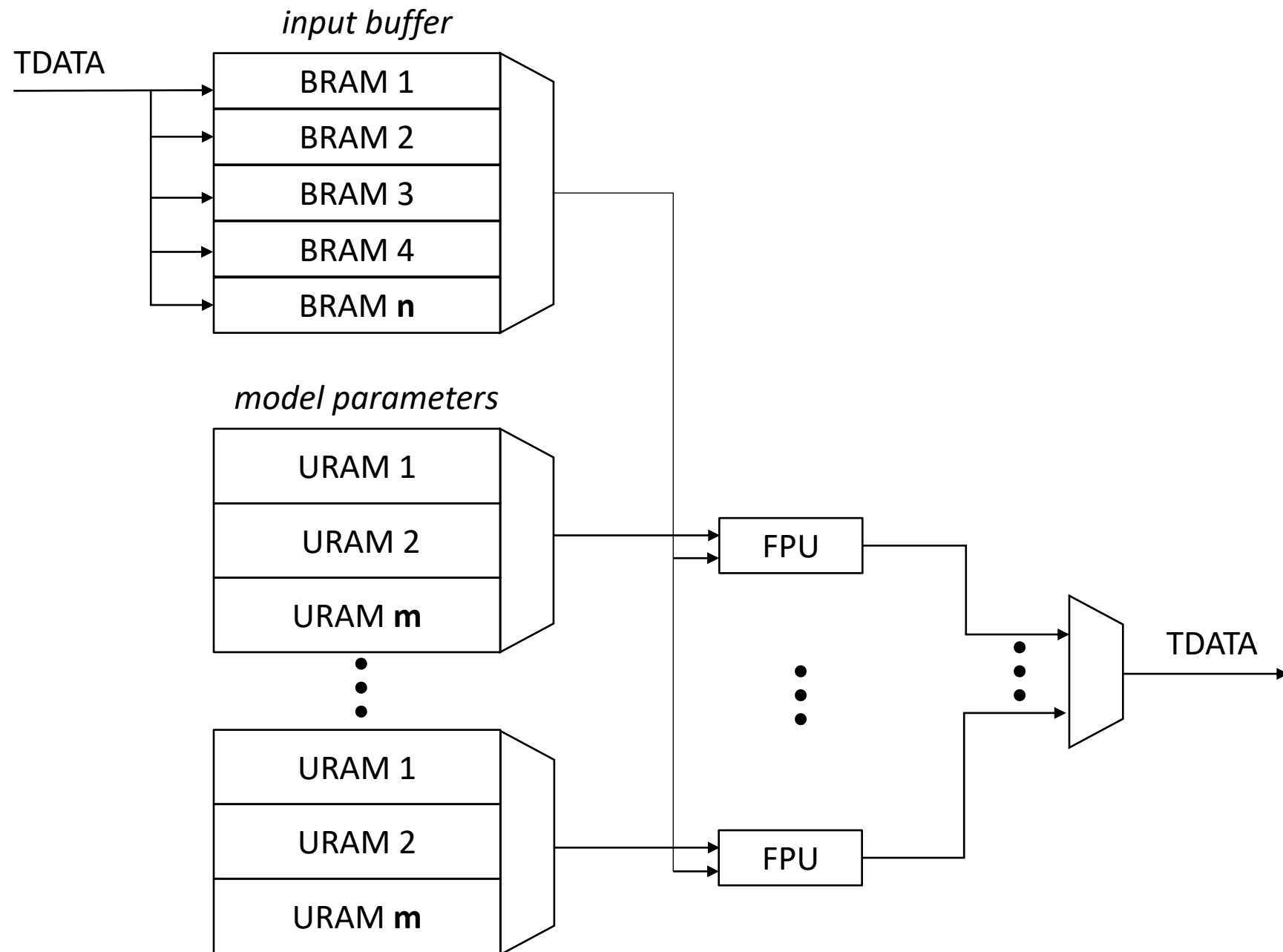
$$n = (\text{iwidth} * 3 * \text{ichan}) / 1K$$

$$m = (3 * 3 * \text{ichan}) / 1K$$

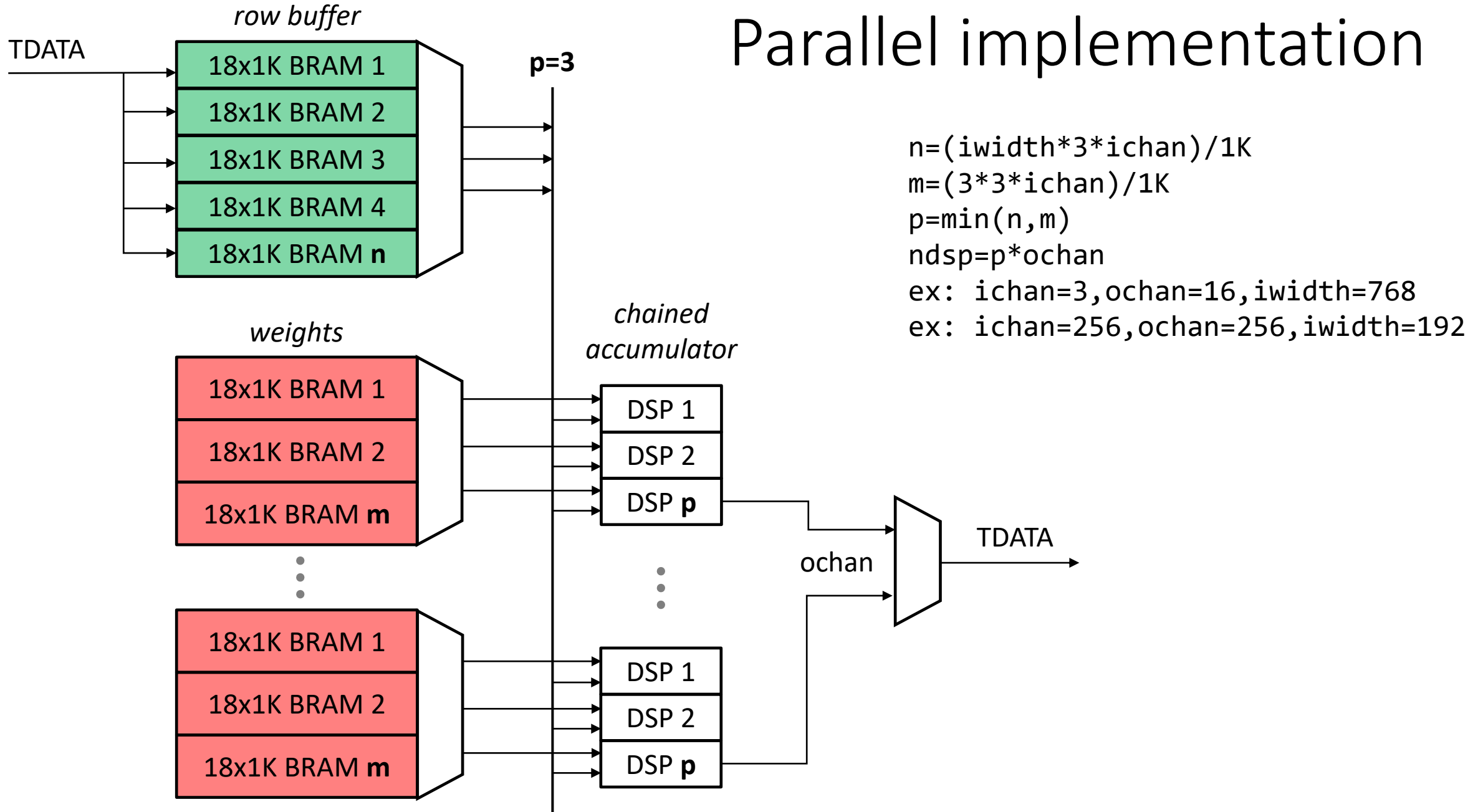
$$\text{ndsp} = \text{ochan}$$

$$\text{ex: } \text{ichan}=3, \text{ochan}=16, \text{iwidth}=768$$

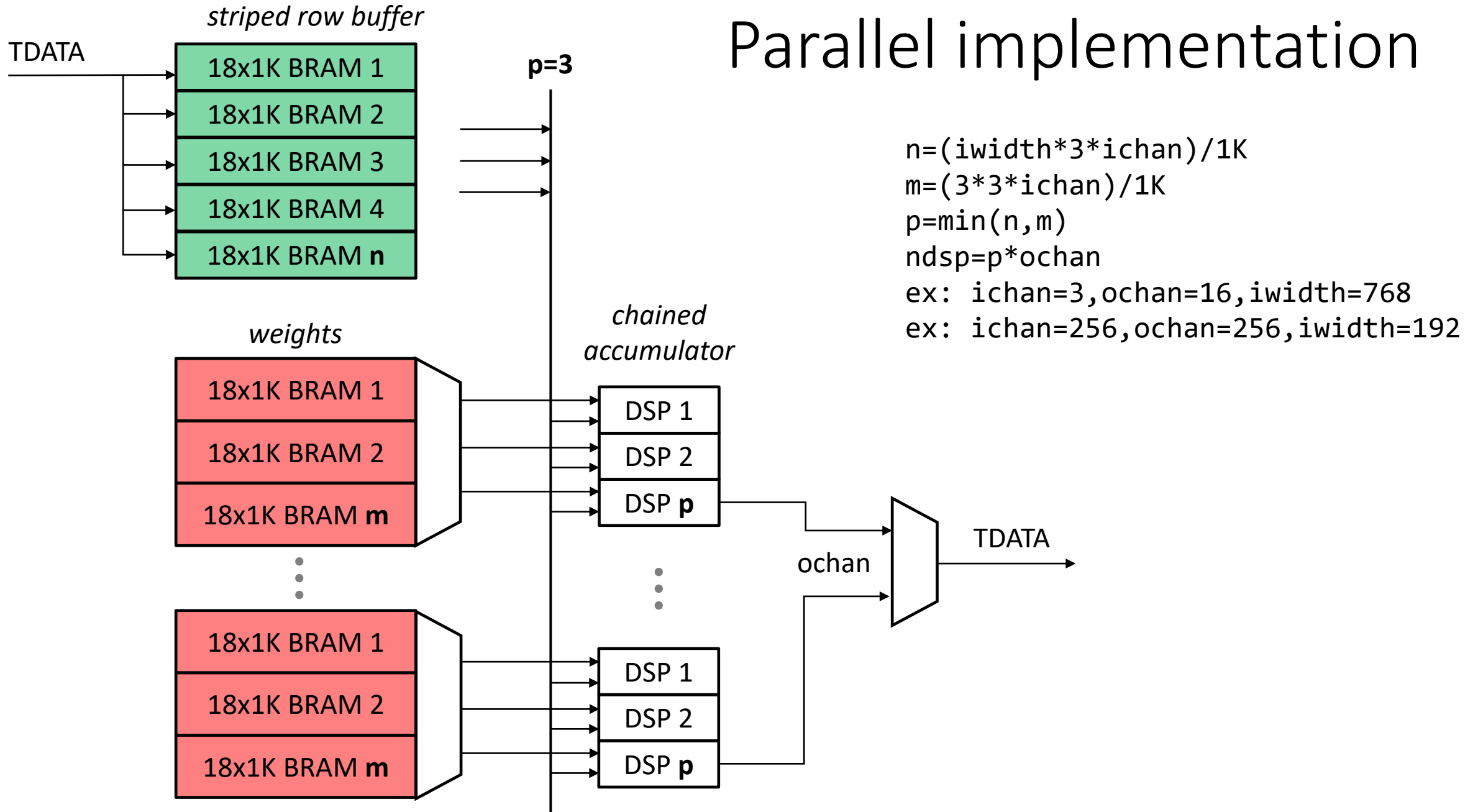
$$\text{ex: } \text{ichan}=256, \text{ochan}=256, \text{iwidth}=192$$

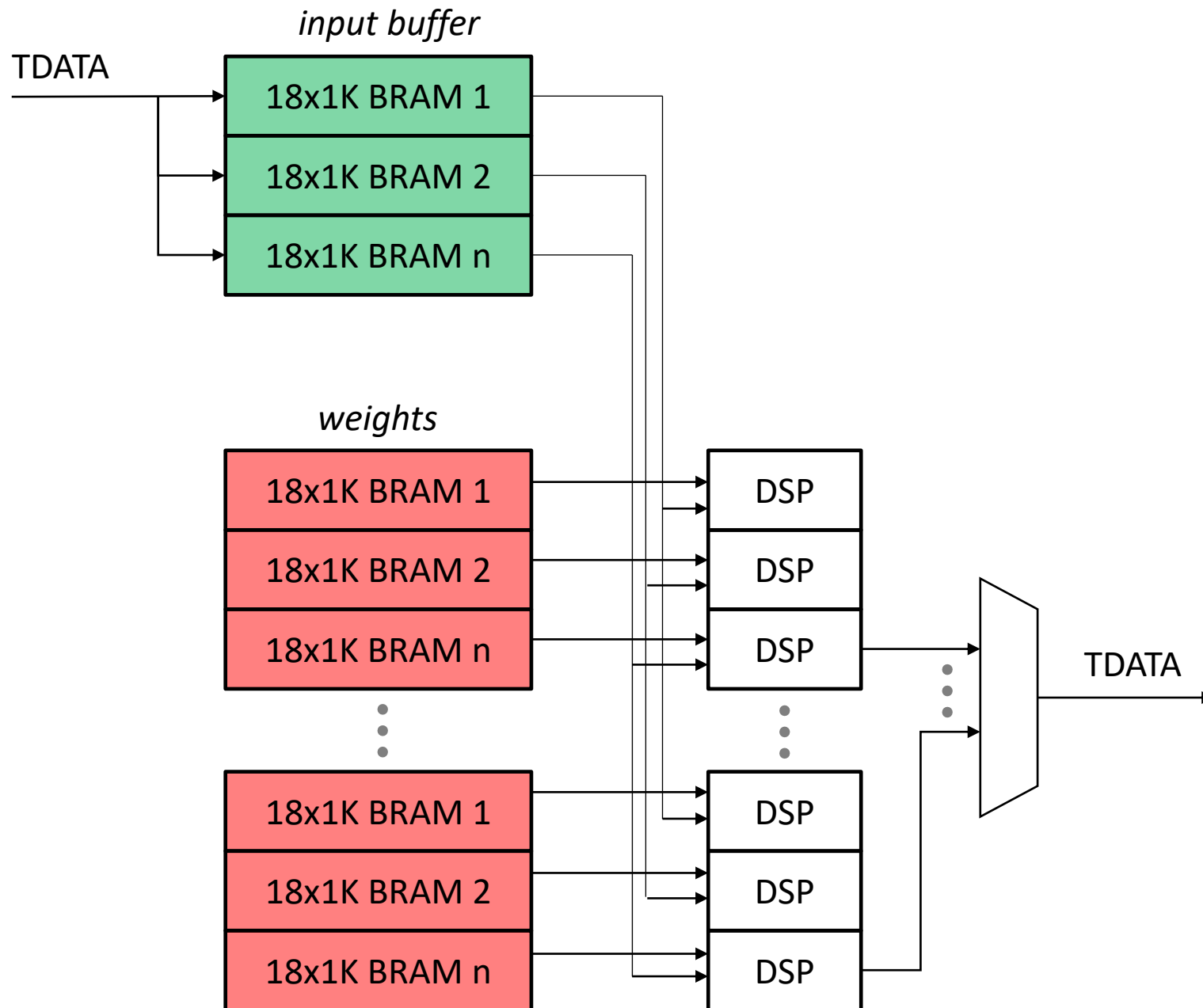


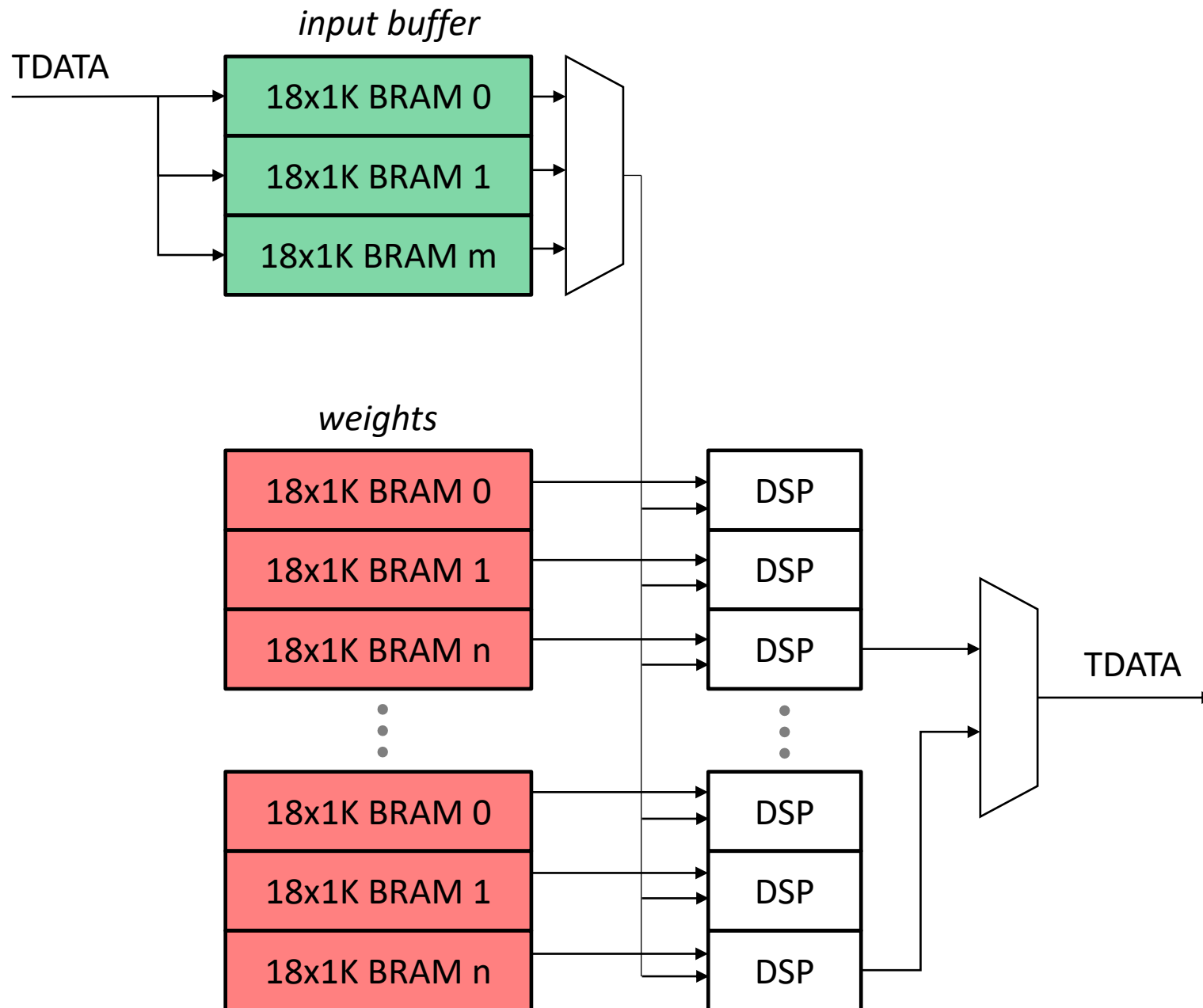
Parallel implementation

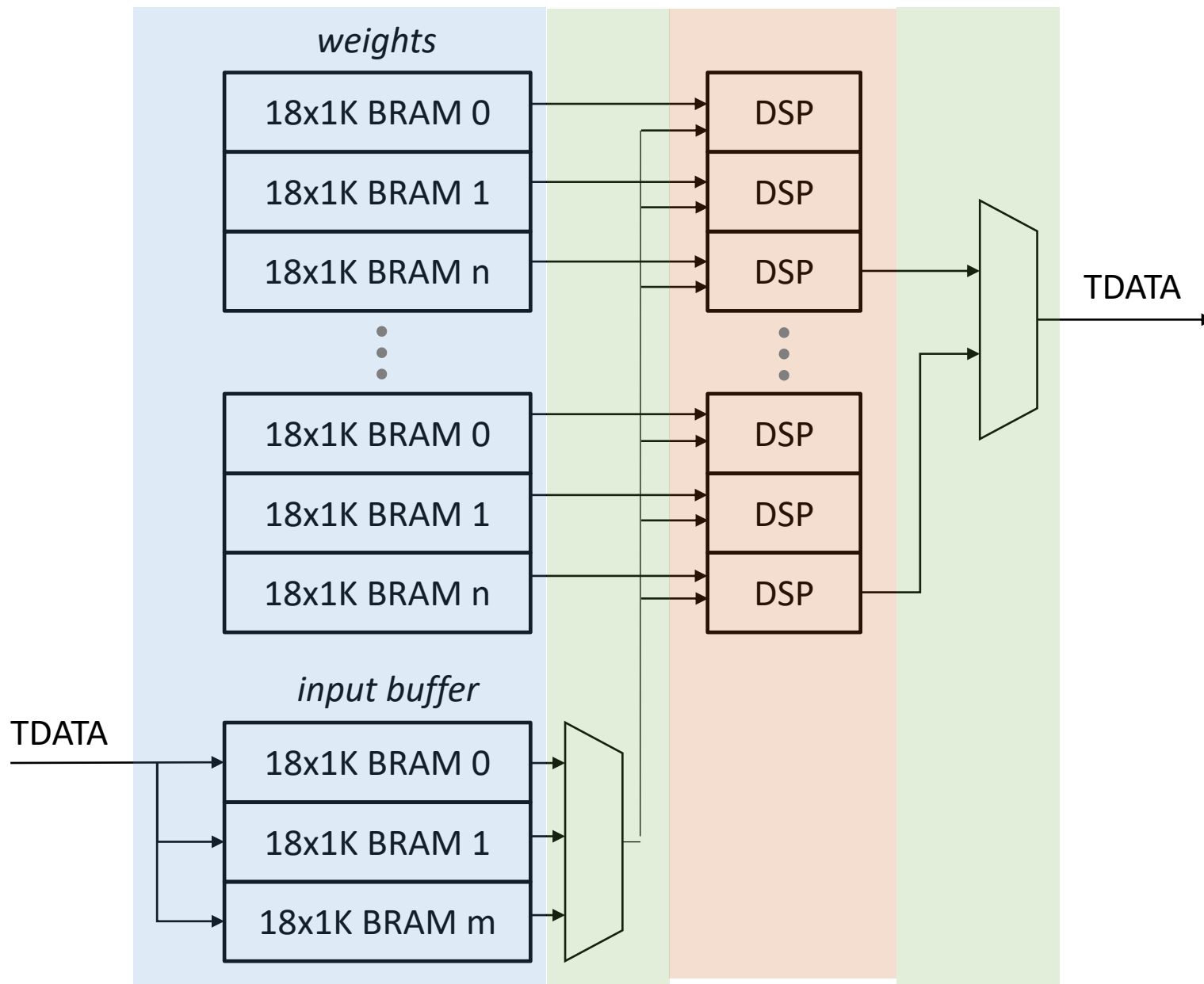


Parallel implementation





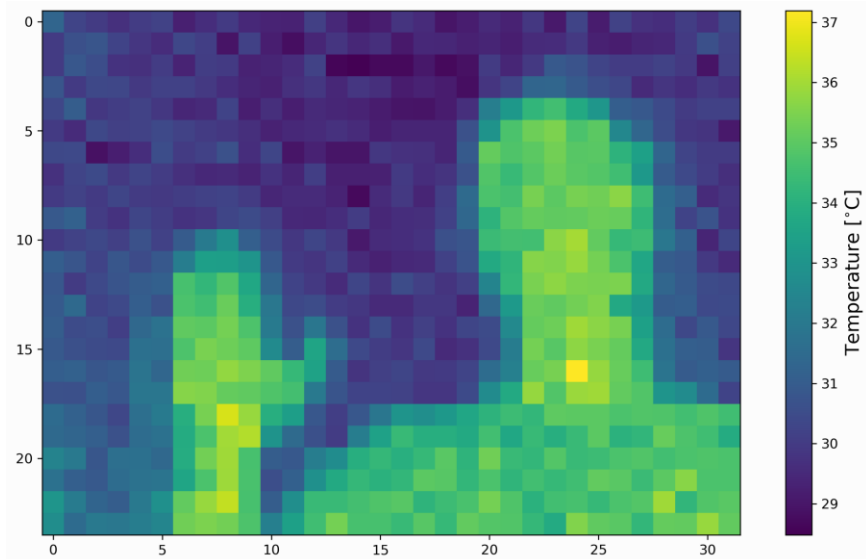




AI application concept – thermal imaging CNN



MLX90640 IR Array Thermal Imaging Camera
Module 32×24 Pixels 55° Field of View Camera
with I2C Interface for Raspberry Pi/Arduino



32x24 calibrated FIR pixels, 14 bit precision, 32 fps

High precision non-contact temperature measurements

Intrusion / Movement detection

Presence detection / Person localization

Temperature sensing element for intelligent HVAC

Thermal Comfort sensor in automotive air conditioning control systems

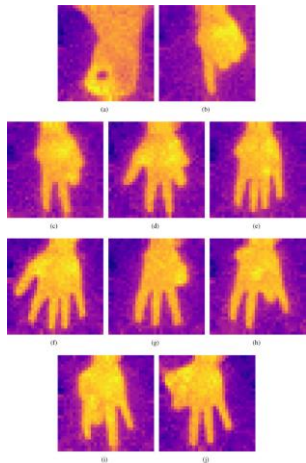
Microwave ovens

Industrial temperature control of moving parts

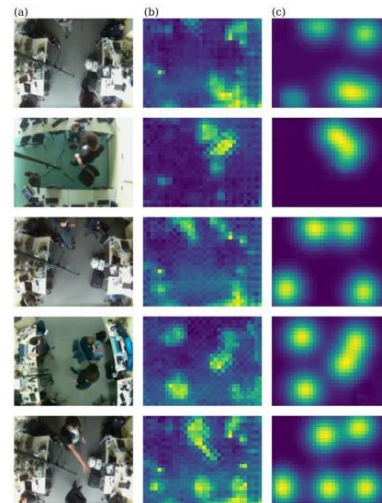
Visual IR thermometers

Example CNN tasks and training data

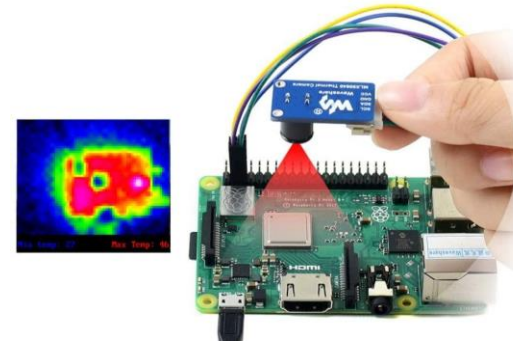
Gesture Recognition
Sign language



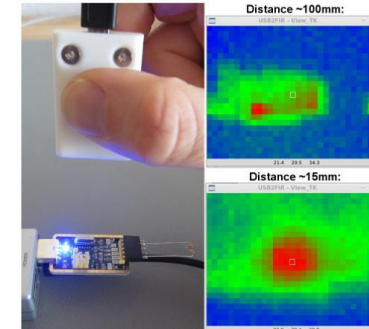
Occupancy Counting
Privacy preserving
Pose detection



Industrial Inspection
Anomaly Detection
PCB hot spot analysis
Animal health screening

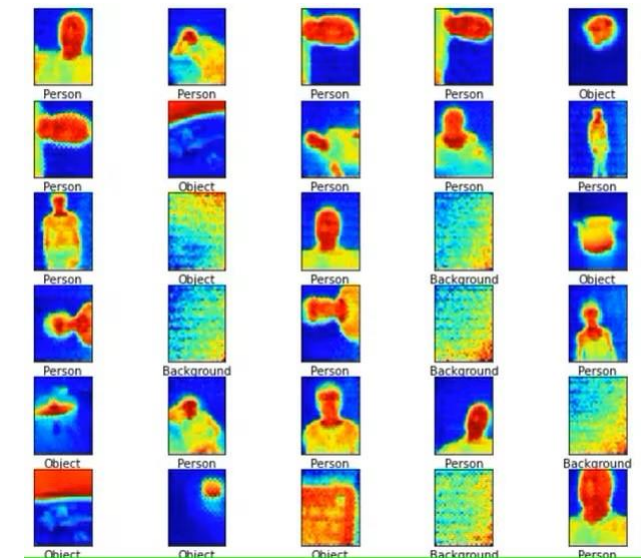


Thermography with USB2FIR
Find and evaluate hotspots on printed circuit boards
Distance ~100mm:

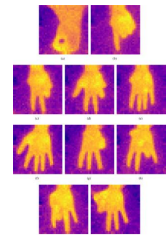
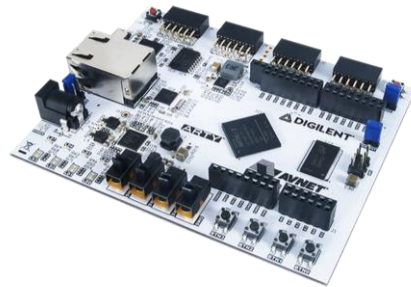
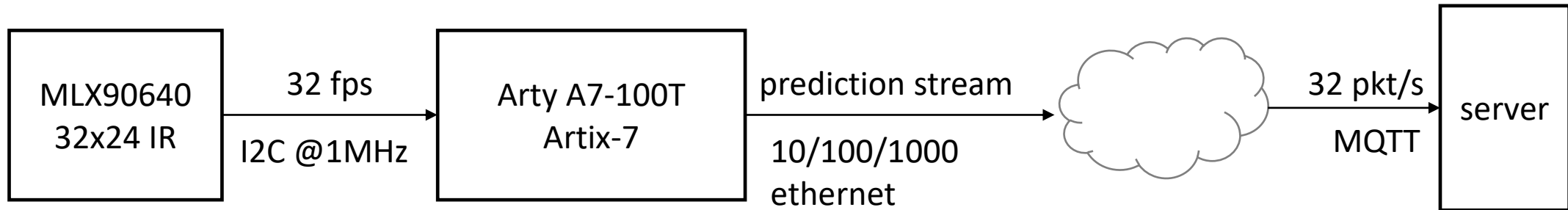


Hotspot in this example (USB-to-serial board with load resistor) is the voltage regulator (SOT-23) with ~39°C.

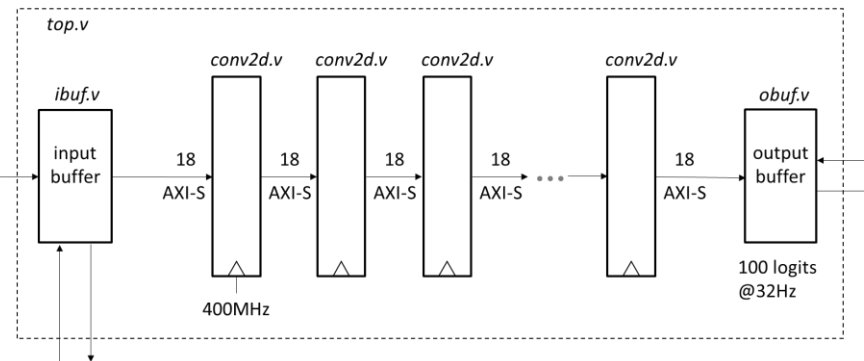
Action Recognition
Human fall detection

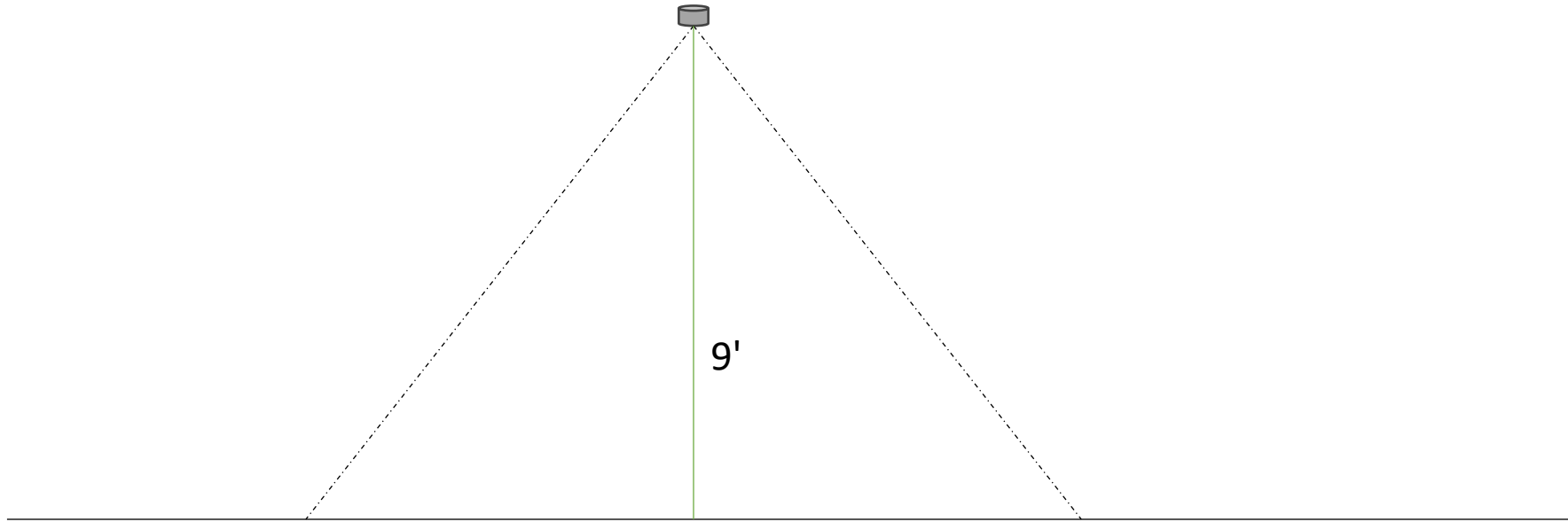


AI application prototype



"accelerator" vs. hard real time
All parameters on chip using BRAM
Model evaluated entirely in FPGA fabric
Real time streaming pipeline





110 deg : 24' diameter
55 deg : 10' diameter